



STANDARD

ANSI/ASHRAE Standard 62.1-2025

(Supersedes ANSI/ASHRAE Standard 62.1-2022)

Includes ANSI/ASHRAE addenda listed in Appendix Q

Ventilation and Acceptable Indoor Air Quality

See Appendix Q for approval dates by ASHRAE and the American National Standards Institute.

This Standard is under continuous maintenance by a Standing Standard Project Committee (SSPC) for which the Standards Committee has established a documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the Standard. Instructions for how to submit a change can be found on the ASHRAE® website (www.ashrae.org/continuous-maintenance).

The latest edition of an ASHRAE Standard may be purchased from the ASHRAE website (www.ashrae.org) or from ASHRAE Customer Service, 180 Technology Parkway, Peachtree Corners, GA 30092. E-mail: orders@ashrae.org. Fax: 678-539-2129. Telephone: 404-636-8400 (worldwide), or toll free 1-800-527-4723 (for orders in US and Canada). For reprint permission, go to www.ashrae.org/permissions.

© 2025 ASHRAE

ISSN 1041-2336



PDF includes hyperlinks for convenient navigation. Click on a reference to a section, table, figure, or equation to jump to its location. Return to the previous page via the bookmark menu.



Includes online access to Ventilation Rate Procedure (VRP) and Indoor Air Quality Procedure (IAQP) calculators and more.



ASHRAE Standing Standard Project Committee 62.1
Cognizant TC: 4.3, Ventilation Requirements and Infiltration
SPLS Liaison: Lawrence C. Markel • ASHRAE Staff Liaison: Mark Weber

Brendon J. Burley, <i>Chair</i>	James E. Dennison	Hao Li	Christopher P. Ruch
Wayne R. Thomann, <i>Co-Vice Chair</i>	William S. Dols	Guopeng Liu	Kevin A. Scarlett
Marwa Zaatari, <i>Co-Vice Chair</i>	Paul L. Doppel	Liang C. James Lo	Chandra Sekhar
Tony G. Aazan	Jonathan D. Douglas	Lauren MacGowens	Benjamin C. Seeley
Anthony M. Abate	Brett Duffy	Meghan K. McNulty	Terri Self
Sama Aghniaey	Taylor Duran	Robert E. Miller, Jr	Charles J. Seyffer
Nick H. Agopian	Rafael Da Costa Dutra	Asit Kumar Mishra	Michael S. Sherber
Hugo Aguilar	Christian Ebnetter	Christopher O. Muller	Paul H. Shipp
Hassan F. Ali Younes	E. Curtis Eichelberger, Jr.	Kashif Nawaz	Steven C. Sill
William P. Bahnfleth	Henry W. Ernst, Jr.	Nitin Naik	Peter Simmonds
Wane A. Baker	Sama Fakhimi	Victor G. Nino	Jeffrey K. Smith
Elizabeth C. Balke	Scott Forst	John Nelson, Jr.	Felipe Soberon
Scott D. Barr	Richard B. Fox	Lisa Chen Ng	Dennis A. Stanke
Charlene W. Bayer	Enrique Gonzalez	Brad North	Erica Stewart
Carl Bengtsson	Fred Grable	Michael D. Orcutt	Katlyn Stoker
Victoria L. Binz	Paul Grahovac	Faith Patrick	Drayton P. Stott
Jerry Bogert	Gregg Gress	Chirag D. Parikh	Eric Sturm
Hoy R. Bohanon, Jr.	Brian J. Hafendorfer	Andrew K. Persily	Christophe Suchy
Chakradhar Bonam	Gene Havard	Sara Persily	Eric Sun
Dean M. Bonessi	Roger L. Hedrick	Joseph J. Pessa	Richard Taft
Tina M. Brueckner	Benjamin L. Heyser	Ryan Peters	Steven T. Taylor
Anthony G. Buschur	Nathan L. Ho	Laura G. Petrillo-Groh	Dean T. Tompkins
Mark P. Buttner	Elliott Horner	Daniel C. Pettway	Stephen M. Trent
LaToya Carraway	Eli P. Howard, III	Heather Platt Gullede	Triantafyllos A. Triantafyllopoulos
Jordan D. Clark	Zalmie Hussein	Gary H. Pomerantz	Donald Weekes, Jr.
Michael Coomber	Jennifer A. Isenbeck	Janet Price	Jeffrey G. Whitelaw
Fabricio Dias da Costa Correa	Robert T. Jauch	Daniel Raheimanjato	Steven K. Winstead
Cara E. Cywinski	Ilona Johnson	Jonathan Rajala	Scott D. Williams
Leonard A. Damiano	Allen Johnston	Stephen Ray	Buzz Wright
Abdel K. Darwich	Gregory M. Kalinyak	Daniel J. Redmond	Junjing Yang
Mark Davidson	Jennifer Kane	Daniel J. Rice	Jeremy Zeedyk
Darryl W. DeAngelis	Manfred Kehrer	Tom Rice	

ASHRAE STANDARDS COMMITTEE 2025–2026

Adrienne G. Thomle, <i>Chair</i>	Susanne Dormann	Paul A. Lindahl, Jr.	Paolo M. Tronville
Jennifer A. Isenbeck-Pille, <i>Vice Chair</i>	Drake H. Erbe	Kenneth A. Monroe	Douglas K. Tucker
Anthony M. Abate	Marcus Hassen	Philip J. Naughton	Thomas E. Watson
Omar A. Abdelaziz	William M. Healy	Kathleen Owen	David P. Yuill
Charles S. Barnaby	Jaap Hogeling	Michael P. Patton	Patrick C. Marks, <i>BOD ExO</i>
Hoy R. Bohanon	Satish N. Iyengar	Karl L. Peterman	Devin A. Abellon, <i>CO</i>
Kelley P. Cramm	Phillip A. Johnson	Christopher J. Seeton	
Abdel K. Darwich	Tatsuro Kobayashi	Russell C. Tharp	

Ryan Shanley, *Senior Manager of Standards*

SPECIAL NOTE

This American National Standard (ANS) is a national voluntary consensus Standard developed under the auspices of ASHRAE. *Consensus* is defined by the American National Standards Institute (ANSI), of which ASHRAE is a member and which has approved this Standard as an ANS, as “substantial agreement reached by directly and materially affected interest categories. This signifies the concurrence of more than a simple majority, but not necessarily unanimity. Consensus requires that all views and objections be considered, and that an effort be made toward their resolution.” Compliance with this Standard is voluntary until and unless a legal jurisdiction makes compliance mandatory through legislation.

ASHRAE obtains consensus through participation of its national and international members, associated societies, and public review.

ASHRAE Standards are prepared by a Project Committee appointed specifically for the purpose of writing the Standard. The Project Committee Chair and Vice-Chair must be members of ASHRAE; while other committee members may or may not be ASHRAE members, all must be technically qualified in the subject area of the Standard. Every effort is made to balance the concerned interests on all Project Committees.

The Senior Manager of Standards of ASHRAE should be contacted for

- interpretation of the contents of this Standard,
- participation in the next review of the Standard,
- offering constructive criticism for improving the Standard, or
- permission to reprint portions of the Standard.

DISCLAIMER

ASHRAE uses its best efforts to promulgate Standards and Guidelines for the benefit of the public in light of available information and accepted industry practices. However, ASHRAE does not guarantee, certify, or assure the safety or performance of any products, components, or systems tested, installed, or operated in accordance with ASHRAE's Standards or Guidelines or that any tests conducted under its Standards or Guidelines will be nonhazardous or free from risk.

ASHRAE INDUSTRIAL ADVERTISING POLICY ON STANDARDS

ASHRAE Standards and Guidelines are established to assist industry and the public by offering a uniform method of testing for rating purposes, by suggesting safe practices in designing and installing equipment, by providing proper definitions of this equipment, and by providing other information that may serve to guide the industry. The creation of ASHRAE Standards and Guidelines is determined by the need for them, and conformance to them is completely voluntary.

In referring to this Standard or Guideline and in marking of equipment and in advertising, no claim shall be made, either stated or implied, that the product has been approved by ASHRAE.

CONTENTS
ANSI/ASHRAE Standard 62.1-2025
Ventilation and Acceptable Indoor Air Quality

SECTION	PAGE
Foreword	2
1 Purpose	2
2 Scope	2
3 Definitions	2
4 Outdoor Air Quality	7
5 Systems and Equipment	7
6 Procedures	16
7 Construction and System Start-Up	33
8 Operations and Maintenance	35
9 Normative References	40
Normative Appendix A: Multiple-Zone System Ventilation Efficiency: Alternative Procedure	43
Normative Appendix B: Separation of Exhaust Outlets and Outdoor Air Intakes	46
Normative Appendix C: Zone Air Distribution Effectiveness: Alternative Procedures	49
Normative Appendix D: Air Density Correction Factor	51
Normative Appendix E: Ventilation Rates for Outpatient Facilities not Covered by ASHRAE/ASHE Standard 170	52
Informative Appendix F: Information on Selected National Standards and Guidelines for PM10, PM2.5, and Ozone	53
Informative Appendix G: Acceptable Mass-Balance Equations for Use with the IAQ Procedure	55
Informative Appendix H: Simplified Ventilation Rate Calculation for Multiple-Zone Recirculating Systems Serving Only Specified Occupancy Categories in Existing Buildings	57
Informative Appendix I: Application	59
Informative Appendix J: Documentation	61
Informative Appendix K: Rate Rationale	65
Informative Appendix L: Information on Natural Ventilation	71
Informative Appendix M: Compliance	74
Informative Appendix N: Ventilation Rate Check Table	78
Informative Appendix O: Indoor Air Quality Procedure	82
Informative Appendix P: Informative References	86
Informative Appendix Q: Addenda Description	87
Standard 62.1-2025 Calculators: www.ashrae.org/pub9373452	

NOTE

Approved addenda, errata, or interpretations for this standard can be downloaded free of charge from the ASHRAE website at www.ashrae.org/technology.

© 2025 ASHRAE

180 Technology Parkway · Peachtree Corners, GA 30092 · www.ashrae.org · All rights reserved.
ASHRAE is a registered trademark of the American Society of Heating, Refrigerating and Air-Conditioning Engineers, Inc.
ANSI is a registered trademark of the American National Standards Institute.

(This foreword is not part of this standard. It is merely informative and does not contain requirements necessary for conformance to the standard. It has not been processed according to the ANSI requirements for a standard and may contain material that has not been subject to public review or a consensus process. Unresolved objections on informative material are not offered the right to appeal at ASHRAE or ANSI.)

FOREWORD

ASHRAE Standard 62.1 specifies minimum ventilation rates and other measures intended to provide indoor air quality (IAQ) that is acceptable to human occupants, minimizes adverse health effects, and protects the integrity of the building and its ventilation systems. A properly designed and maintained ventilation system reduces the risk of mold, mitigates against the spread of both nuisance and health hazards within the building, and contributes to occupant comfort and satisfaction.

The 2025 edition of the standard refines and expands the humidity control requirements, adds requirements for emergency ventilation controls to address atypical operating modes, and provides several new methods of calculation. Users of previous editions will find new methods for the calculation of separation distances between outdoor air intakes and exhausts; a new air density correction factor for all ventilation zones; a new method for calculating systems ventilation requirements when multiple standards are followed; a new demand control ventilation control sequence; requirements for air-cleaning system performance, including a calculation for end of useful life efficiency for certain contaminants; and revisions to the performance path for exhaust systems. Also included are updates to provide greater clarity of existing requirements and to align requirements with other codes and standards.

This standard is updated on a regular basis using ASHRAE's continuous maintenance procedures. Addenda are publicly reviewed, approved by ASHRAE and ANSI, and posted on the ASHRAE website. Change proposals can be submitted online at osr.ashrae.org. The project committee for Standard 62.1 takes formal action on all change proposals received.

1. PURPOSE

1.1 The purpose of this standard is to specify minimum ventilation rates and other measures intended to provide indoor air quality (IAQ) that is acceptable to human occupants and that minimizes adverse health effects.

1.2 This standard is intended for regulatory application to new buildings, additions to existing buildings, and those changes to existing buildings that are identified in the body of the standard.

1.3 This standard is intended to be used to guide the improvement of IAQ in existing buildings.

2. SCOPE

2.1 This standard applies to spaces intended for human occupancy within buildings except those within *dwelling units in residential occupancies* in which occupants are *nontransient*.

2.2 This standard defines requirements for *ventilation* and *air-cleaning system* design, installation, commissioning, and operations and maintenance.

2.3 In addition to *ventilation*, this standard contains requirements related to certain *contaminants* and *contaminant sources*, including *outdoor air*, construction processes, moisture, and biological growth.

2.4 This standard does not prescribe specific ventilation rate requirements for the following:

- a. Spaces that contain smoking or that do not meet the requirements in the standard for separation from spaces that contain smoking
- b. *Patient care areas* not listed in this standard
- c. Laboratories with *hazardous materials*

3. DEFINITIONS

3.1 Terminology (See Figure 3-1)

acceptable indoor air quality (IAQ): air in which there are no known *contaminants* at harmful *concentrations*, as determined by *cognizant authorities*, and with which a substantial majority (80% or more) of the people exposed do not express dissatisfaction.

air-cleaning system: a device or combination of devices applied to reduce the *concentration* of airborne *contaminants* such as *microorganisms*, dusts, fumes, respirable particles, other particulate matter, gases, vapors, or any combination thereof.

air conditioning: process of treating air to meet the requirements of a *conditioned space* by controlling its temperature, humidity, cleanliness, and distribution.

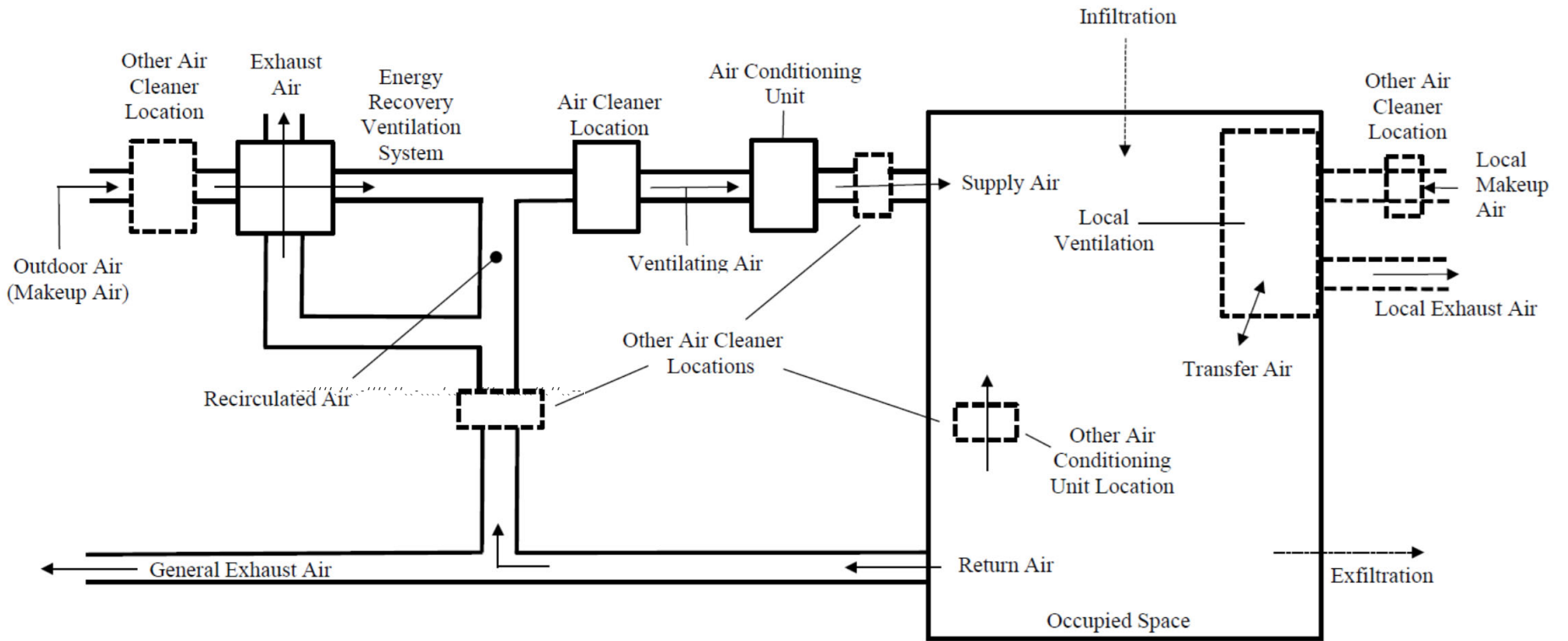


Figure 3-1 Ventilation system.

ambient air: air surrounding a building.

authority having jurisdiction (AHJ): the agency or agent responsible for enforcing a standard.

breathing zone: region within an occupied space between planes 3 and 72 in. (75 and 1800 mm) above the floor and more than 2 ft (600 mm) from the walls or fixed air-conditioning equipment.

ceiling return: air removed from the space more than 4.5 ft (1.4 m) above the floor.

ceiling supply: air supplied to the space more than 4.5 ft (1.4 m) above the floor.

Class 1 imaging room: imaging rooms that meet the criterion of Class 1 as per the Facility Guidelines Institute (FGI) *Guidelines for Design and Construction of Outpatient Facilities*,¹ Table 2.1-5.6.2.5.1.3.

classroom: a space for instruction in which the instructor regularly occupies and stores supplies in the space.

cognizant authority: an agency or organization that has the expertise and jurisdiction to establish and regulate *concentration* limits for *contaminants*, or an agency or organization that is recognized as authoritative and has the scope and expertise to establish guidelines, limit values, or *concentrations* levels for *contaminants*.

concentration: quantity of one constituent dispersed in a defined amount of another.

conditioned space: part of a building that is heated or cooled, or both, for the comfort of occupants.

contaminant: unwanted airborne constituent with the potential to reduce acceptability of the air.

contaminant mixture: two or more *contaminants* that target the same organ system.

cool air: air whose temperature is less than the average space temperature.

demand controlled ventilation (DCV): any means by which the *breathing zone* outdoor airflow (V_{bz}) can be varied to the occupied space or spaces based on the actual or estimated number of occupants, ventilation requirements of the occupied zone, or both.

design compounds (DCs): chemical compounds found in the indoor environment that have the potential to reduce acceptability of the air and are considered in designing to meet the requirements of the IAQ Procedure (IAQP).

dwelling unit: a single unit providing complete, independent living facilities for one or more persons, including permanent provisions for living, sleeping, eating, cooking, and sanitation.

energy recovery device: a device or combination of devices or system to transfer heat and/or water vapor between separate outdoor and exhaust airstreams.

environmental tobacco smoke (ETS): the combination of both side-stream smoke (smoke from the lit end of a cigarette or other tobacco product) and exhaled mainstream smoke (smoke that is exhaled by a smoker). (*Informative Note:* ETS is commonly referred to as “secondhand smoke.” This definition includes smoke produced from the combustion of cannabis and controlled substances and the emissions produced by electronic smoking devices.)

ETS area: spaces where smoking is permitted, as well as those spaces not separated from spaces where smoking is permitted in accordance with the requirements of Section 5 in this standard.

ETS-free area: an area where no smoking occurs and that is separated from *ETS areas* according to the requirements of this standard. (*Informative Note:* A no-smoking area is not necessarily an *ETS-free area*.)

evaporative heat rejection equipment: heat rejection devices, including open-circuit cooling towers, closed-circuit cooling towers, and evaporative condensers, with an open water circuit that reject heat through the evaporation of water.

exfiltration: uncontrolled outward air leakage from *conditioned spaces* through unintentional openings in ceilings, floors, and walls to unconditioned spaces or the outdoors caused by pressure differences across these openings due to wind, indoor-outdoor temperature differences (stack effect), and imbalances between outdoor and exhaust airflow rates.

exhaust air: air removed from a space and discharged to outside the building by means of mechanical or natural ventilation systems.

floor return: air removed from the space less than 4.5 ft (1.4 m) above the floor.

floor supply: air supplied to the space less than 4.5 ft (1.4 m) above the floor.

free area: ratio of area of the openings through a surface, such as openings between louver blades and undercuts, divided by the gross area (length times height) of the surface.

hazardous materials: any biological, chemical, radiological, or physical item or agent that has the potential to cause harm to humans, animals, or the environment, either by itself or through interaction with other factors. (*Informative Note:* Hazardous chemicals are any chemicals that are classified as a health hazard or simple asphyxiant in accordance with the *Hazard Communication Standard* [29 CFR 1910.1200],^{P1} and any

other particularly hazardous substances, including select carcinogens, reproductive toxins, and substances that have a high degree of acute toxicity. Hazardous biological agents are any pathogenic, allergenic, or toxicogenic *microorganisms*, including BSL2-4 agents as defined in the U.S. Department of Health and Human Service's *Biosafety in Microbiological and Biomedical Laboratories*.^{P2)}

hurricane-prone regions: areas vulnerable to hurricanes, defined as

- a. the U.S. Atlantic Ocean and Gulf of Mexico coasts where the basic design wind speed, V , for Risk Category II buildings is greater than 115 mph (51.4 m/s);
- b. Hawaii, Puerto Rico, Guam, Virgin Islands, and American Samoa; or
- c. as specified by the *authority having jurisdiction (AHJ)*.

indoor air: air in an *occupiable space*.

industrial space: an indoor environment where the primary activity is production or manufacturing processes.

infiltration: uncontrolled inward air leakage to *conditioned spaces* through unintentional openings in ceilings, floors, and walls from unconditioned spaces or the outdoors caused by the same pressure differences that induce *exfiltration*.

makeup air: any combination of *outdoor air* and *transfer air* intended to replace *exhaust air* and *exfiltration*.

mechanical ventilation: *ventilation* provided by mechanically powered equipment such as motor-driven fans and blowers but not by devices such as wind-driven turbine ventilators and mechanically operated windows.

microorganism: a microscopic organism, especially a bacterium, fungus, or protozoan.

modular room: an occupiable, prefabricated structure that is designed and used for activities other than sleeping.

natural ventilation: *ventilation* provided by thermal, wind, or diffusion effects through doors, windows, or other intentional openings in the building.

net occupiable area: floor area of an *occupiable space* defined by the inside surfaces of its walls but excluding shafts, column enclosures, and other permanently enclosed, inaccessible, and unoccupiable areas. (*Informative Note:* Obstructions in the space, such as furnishings, display or storage racks, and other obstructions, whether temporary or permanent, are considered to be part of the *net occupiable area*.)

nontransient: occupying a *dwelling unit* or *sleeping unit* for more than 30 days.

occupant sensor: a device such as a motion detector or a captive key system that detects the presence of one or more persons within a space.

occupiable space: an enclosed space intended for human activities, excluding spaces that are intended to be occupied occasionally and for short periods of time, such as storage rooms, equipment rooms, and emergency exitways.

occupied mode: when a zone is scheduled to be occupied.

occupied standby mode: when a zone is scheduled to be occupied and an *occupant sensor* indicates zero population within the zone.

odor: a quality of gases, liquids, or particles that stimulates the olfactory organ.

openable area: *net free area* of an opening.

outdoor air: *ambient air* and *ambient air* that enters a building through a ventilation system, through intentional openings for *natural ventilation*, or by *infiltration*.

particulate matter 2.5 (PM_{2.5}): aerosol particles with an aerodynamic diameter less than or equal to a nominal 2.5 μm .

patient care area: an area used primarily for the provision of clinical care to patients. Such care includes monitoring, evaluation, and treatment services.

primary air: air supplied to the *ventilation zone* prior to mixing with any locally *recirculated air*.

readily accessible: capable of being reached quickly for operation without requiring personnel to climb over or remove obstacles or to resort to the use of unsafe climbing aids such as tables or chairs.

recirculated air: air removed from a space, or treated within a space, that is delivered to a space.

residential occupancies: occupancies that are not classified as institutional by the *AHJ* and that contain permanent provisions for sleeping.

return air: air removed from a space to be recirculated or exhausted.

sleeping unit: a room or space not within a *dwelling unit* in which people sleep that includes permanent provisions for living, eating, and either sanitation or kitchen facilities but not both.

sleep pod: an occupiable, prefabricated structure that is designed and used for sleeping purposes.

space volume: total volume of an *occupiable space* enclosed by the building envelope, plus that of any spaces permanently open to the *occupiable space*, such as a ceiling attic used as a ceiling return plenum.

stratified air distribution system: a device or combination of devices applied to provide a stratified thermal and pollutant distribution within a zone.

supply air: air delivered by *mechanical ventilation* or *natural ventilation* to a space and composed of any combination of *outdoor air*, *recirculated air*, or *transfer air*.

transfer air: air moved from one indoor space to another.

unoccupied mode: when a zone is not scheduled to be occupied.

unusual source: an item or activity not usually expected within an occupancy category and that has the potential to create *contaminants*. (**Informative Note:** Informative Appendix K contains information on sources and *contaminants* expected in certain occupancy categories.)

ventilation: the process of supplying air to or removing air from a space for the purpose of controlling air *contaminant* levels, humidity, or temperature within the space.

ventilation air: portion of *supply air* that is *outdoor air* plus any *recirculated air* that has been treated for the purpose of maintaining *acceptable IAQ*.

ventilation zone: any indoor area that requires *ventilation* and comprises one or more spaces with the same occupancy category (see Table 6-2), occupant density, *zone air distribution effectiveness* (see Section 6.2.1.2), and design zone primary airflow (see Section 6.2.4.3.2 and Normative Appendix A) per unit area. (**Informative Note:** A *ventilation zone* is not necessarily an independent thermal control zone; however, spaces that can be combined for load calculation purposes can often be combined into a single zone for ventilation calculations purposes.)

warm air: air whose temperature is greater than the average space temperature.

zone air distribution effectiveness: ratio of the change of *contaminant concentration* between the air supply and air exhaust to the change of *contaminant concentration* between the air supply and the *breathing zone*.

3.2 Initialisms, Abbreviations, and Acronyms

AHJ	authority having jurisdiction
ACGIH	American Conference of Governmental Industrial Hygienists
AgBB	Committee for Health Related Evaluation of Building Products (German)
AHRI	Air Conditioning, Heating, and Refrigeration Institute
AMCA	Air Movement and Control Association
ANSI	American National Standards Institute
ASHE	American Society for Health Care Engineering
ASSP	American Society of Safety Professionals
BSL	biological safety level
Cal	California
CFR	Code of Federal Regulations
CIBSE	Chartered Institution of Building Services Engineers
CREL	chronic reference exposure levels
DC	design compound
DCV	demand controlled ventilation
DL	design limit
EHS	environmental health and safety
EPA	Environmental Protection Agency
ETS	environmental tobacco smoke
HVAC	heating, ventilation, and air conditioning
IAQ	indoor air quality
IAQP	Indoor Air Quality Procedure
I-P	inch-pound
ISO	International Organization for Standardization
LCI	lowest <i>concentration</i> of interest

MERV	minimum efficiency reporting value
NAAQS	National Ambient Air Quality Standards
NFPA	National Fire Protection Association
NSF	National Sanitation Foundation
O&M	operations and maintenance
OS	occupied standby
PM	particulate matter
SI	International System of Units
SMACNA	Sheet Metal and Air Conditioning Contractors National Association
UL	Underwriters Laboratories
VAV	variable air volume
VRP	Ventilation Rate Procedure

4. OUTDOOR AIR QUALITY

Outdoor air quality shall be investigated in accordance with Sections 4.1 and 4.2 prior to completion of *ventilation* system design. The results of this investigation shall be documented in accordance with Section 4.3.

4.1 Regional Air Quality. The status of compliance with ambient air quality standards shall be determined for the geographic area of the building site.

4.1.1 In the United States, compliance status shall be either in “attainment” or “nonattainment” with the NAAQS.² In the United States, areas with no U.S. Environmental Protection Agency (U.S. EPA) compliance status designation shall be considered “attainment” areas.

Informative Notes:

1. The NAAQS are shown in Informative Appendix F, Table F-1.
2. The U.S. EPA list of nonattainment areas can be found at www.epa.gov/green-book.
3. Air quality data collected at outdoor monitors across the U.S. can be found at www.epa.gov/outdoor-air-quality-data.
4. Internet links to detailed information on the ambient air quality standards and *contaminant* levels for other select counties and regions can be found in Informative Appendix F.

4.2 Local Air Quality. An observational survey of the building site and its immediate surroundings shall be conducted during hours the building is expected to be normally occupied to identify local *contaminants* from surrounding facilities that will be of concern if allowed to enter the building.

4.3 Documentation. Documentation of the outdoor air quality investigation shall be reviewed with building owners or their representative and shall include the following as a minimum:

- a. Regional air quality compliance status
- b. Local survey information
 1. Date of observations
 2. Time of observations
 3. Site description
 4. Description of facilities on site and on adjoining properties
 5. Observation of *odors* or irritants
 6. Observation of visible plumes or visible air *contaminants*
 7. Description of sources of vehicle exhaust on site and on adjoining properties
 8. Identification of potential *contaminant* sources on the site and from adjoining properties, including any that operate only seasonally
- c. Conclusion regarding the acceptability of outdoor air quality and the information supporting the conclusion

5. SYSTEMS AND EQUIPMENT

5.1 Building Envelope and Interior Surfaces. The building envelope and interior surfaces within the building envelope shall be designed in accordance with the following subsections.

5.1.1 Building Envelope. The building envelope, including roofs, walls, fenestration systems, and foundations, shall comply with the following:

- a. A weather barrier or other means shall be provided to prevent liquid-water penetration into the envelope.
Exception to (a): When the envelope is engineered to allow incidental water penetration to occur without resulting in damage to the envelope construction.

- b. An appropriately placed vapor retarder or other means shall be provided to limit water-vapor diffusion to prevent condensation on cold surfaces within the envelope.

Exception to (b): When the envelope is engineered to manage incidental condensation without resulting in damage to the envelope construction.

- c. Exterior joints, seams, or penetrations in the building envelope that are pathways for air leakage shall be caulked, gasketed, weather stripped, provided with a continuous air barrier, or otherwise sealed to limit infiltration through the envelope to reduce uncontrolled entry of outdoor-air moisture and pollutants.

Informative Note: In localities where soils contain high concentrations of radon or other soil-gas contaminants, the AHJ might impose additional measures such as subslab depressurization.

5.1.2 Condensation on Interior Surfaces. Pipes, ducts, and other surfaces within the building whose surface temperatures are expected to fall below the surrounding dew-point temperature shall be insulated. The insulation system thermal resistance and material characteristics shall prevent condensate from forming on the exposed surface and within the insulating material.

Exception to 5.1.2: Where condensate will wet only surfaces that will be managed to prevent or control mold growth. A management plan must be submitted along with the design specifying design assumptions and limits of the plan. The plan must be provided to the owner.

5.2 Buildings with Attached Parking Garages. In order to limit the entry of vehicular exhaust into *occupiable spaces*, buildings with attached parking garages shall be designed to

- a. maintain the garage pressure at or below the pressure of the adjacent *occupiable spaces*,
- b. use a vestibule to provide an airlock between the garage and the adjacent *occupiable spaces*, or
- c. otherwise limit migration of air from the attached parking garage into the adjacent *occupiable spaces* of the building in a manner acceptable to the *authority having jurisdiction (AHJ)*.

5.3 Requirements for Buildings Containing Environmental Tobacco Smoke (ETS) Areas and ETS-Free Areas. The requirements of this section must be met when a building contains both *ETS areas* and *ETS-free areas*. Such buildings shall be constructed and operated in accordance with Sections 5.3.1 through 5.3.8. This section does not purport to achieve *acceptable indoor air quality (IAQ)* in *ETS areas*.

5.3.1 Classification. All spaces shall be classified as either *ETS-free areas* or *ETS areas*.

5.3.2 Pressurization. *ETS-free areas* shall be at a positive pressure with respect to any adjacent or connected *ETS areas*.

Exceptions to 5.3.2:

1. *Dwelling units*, including hotel and motel guestrooms, and adjacent properties under different ownership with separation walls that are structurally independent and that contain no openings. This exception shall apply only when
 - a. the separation walls are constructed as smoke barriers in accordance with the requirements of applicable standards;
 - b. the separation walls include an air barrier consisting of a continuous membrane or surface treatment in the separation wall that has documented resistance to air leakage—continuity of the barrier shall be maintained at openings for pipes, ducts, and other conduits and at points where the barrier meets the outside walls and other barriers; and
 - c. interior corridors common to *ETS areas* and *ETS-free areas* are mechanically supplied with *outdoor air* at the rate of 0.1 cfm/ft² (0.5 L/s·m²).
2. Adjacent spaces otherwise required to be held at negative pressure and posted with signs due to the presence of hazardous or flammable materials or vapors.

Informative Note: Examples of methods for demonstrating relative pressure include engineering analysis, pressure differential measurement, and airflow measurement.

5.3.3 Separation. Solid walls, floors, ceilings, and doors equipped with automatic closing mechanisms shall separate *ETS areas* from *ETS-free areas*.

Exception to 5.3.3: Openings without doors are permitted in the separation where engineered systems are designed to provide airflow from *ETS-free areas* into *ETS areas*, notwithstanding eddies that may occur in the immediate vicinity of the boundary between the *ETS areas* and *ETS-free areas* and reverse flow that may occur due to short-term conditions such as wind gusts.

Informative Note: Examples of methods for demonstrating air motion are engineering analysis and the use of a directional airflow indicator at representative locations in the opening, such as on 1 ft (0.3 m) centers or at locations required for duct traverses in standard testing and balancing procedures, such as those described in ANSI/ASHRAE Standard 111.³

5.3.4 Transfer Air. When air is transferred from *ETS-free areas* to *ETS areas*, the transfer airflow rate shall be maintained regardless of whether operable doors or windows between *ETS-free areas* and *ETS areas* are opened or closed. Acceptable means of doing so include fixed openings in doors, walls, or floors, transfer grilles, transfer ducts, or unducted air plenums with air pressure differentials in compliance with Section 5.3.2.

5.3.5 Recirculation. Air-handling and natural ventilation systems shall not recirculate or transfer air from an *ETS area* to an *ETS-free area*.

5.3.6 Exhaust Systems. Exhaust or relief air from an *ETS area* shall be discharged such that none of the air is recirculated back into any *ETS-free area*.

5.3.7 Signage. A sign shall be posted outside each entrance to each *ETS area*. The sign shall state, as a minimum, "This Area May Contain Environmental Tobacco Smoke," in letters at least 1 in. (25 mm) high or otherwise in compliance with accessibility guidelines.

Exception to 5.3.7: Instead of the specified sign, equivalent notification means acceptable to the *AHJ* may be used.

Informative Note: Based on the definition of "*ETS area*," such a sign might be posted outside a larger *ETS area* that includes the area where smoking is permitted.

5.3.8 Reclassification. An area that was previously an *ETS area* but now meets the requirements of an *ETS-free area* shall be permitted to be classified as such where smoke exposure has stopped and *odor* and irritation from residual *ETS contaminants* are not apparent.

5.4 Outdoor Air Intakes. Ventilation system *outdoor air* intakes shall be designed in accordance with the following subsections.

5.4.1 Location. *Outdoor air* intakes (including openings that are required as part of a natural ventilation system) shall be located such that the shortest distance from the intake to any specific potential outdoor *contaminant* source listed in Table 5-1 shall be equal to or greater than

- a. the separation distance in Table 5-1 or
- b. the calculation methods in Normative Appendix B

and shall comply with all other requirements of this section.

5.4.1.1 Exhaust/Relief Outlets. Separation criteria for Class 2 and Class 3 exhaust/relief outlets apply to the distance from the *outdoor air* intakes for one ventilation system to the exhaust outlets and relief outlets for any other ventilation system.

5.4.1.2 Fuel-Burning Equipment. The minimum distances relative to fuel-fired appliances shall be as required by ANSI Z223.1/NFPA 54⁴ for fuel-gas-burning appliances and equipment, NFPA 31⁵ for oil-burning appliances and equipment, and NFPA 211⁶ for other combustion appliances and equipment.

5.4.1.3 Roof, Landscaped Grade, or Another Surface Directly Below Intake. Where snow accumulation is expected, the surface of the snow at the expected average snow depth shall be considered to be a surface directly below an intake.

Exception to 5.4.1.3: The minimum separation distance in Table 5-1 shall not apply where outdoor surfaces below the air intake are sloped more than 45 degrees from horizontal or where such surfaces are less than 1 in. (30 mm) in width.

5.4.1.4 Laboratory Exhaust. Separation criteria for fume hood exhaust shall be in compliance with ANSI/ASSP Z9.5.⁷

5.4.2 Rain Entrainment. Outdoor air intakes that are part of the mechanical ventilation system shall be designed to manage rain entrainment in accordance with one or more of the following:

- a. Limit water penetration through the intake to 0.07 oz/ft²·h (21.5 g/m²·h) of inlet area when tested using the rain test apparatus described in UL 1995⁸ or IEC 60355-2-40.⁹
- b. Select louvers that limit water penetration to a maximum of 0.01 oz/ft² (3 g/m²) of louver free area at the maximum free area intake velocity through the louver. This water penetration rate shall be determined for a minimum 15-minute test duration when subjected to a water flow rate of 0.25 gal/min (16 mL/s) as described under the water penetration test in AMCA 500-L¹⁰ or equivalent. Manage the water that penetrates the louver by providing a drainage area or moisture removal devices. For buildings located within *hurricane-prone regions*, select louvers that also comply with Section 5.4.2.1.
- c. Select louvers that are Class A when rated according to AMCA Publication 511¹¹ and tested per the AMCA 500-L¹⁰ wind-driven rain test when subjected to a simulated rainfall of 3 in. (75 mm) per hour and a 29 mph (13 m/s) wind velocity. The maximum design core area velocity through the louver shall

Table 5-1 Air Intake Minimum Separation Distance

Object	Minimum Distance, ft (m)
Class 2 air exhaust/relief outlet	10 (3)
Class 3 air exhaust/relief outlet	15 (5)
Class 4 air exhaust/relief outlet	30 (10)
Evaporative heat-rejection equipment exhaust	25 (7.5)
Evaporative heat-rejection equipment intake or basin	15 (5)
Driveway, street, or parking place	5 (1.5)
Garage entry, automobile loading area, or drive-in queue	15 (5)
Garbage storage/pick-up area, dumpsters	15 (5)
Plumbing vents terminating at least 3 ft (1 m) above the level of the <i>outdoor air</i> intake	3 (1)
Plumbing vents terminating less than 3 ft (1 m) above the level of the <i>outdoor air</i> intake	10 (3)
Roof, landscaped grade, or other surface directly below intake	1 (0.30)
Thoroughfare with high traffic volume	25 (7.5)
Truck loading area or dock, bus parking/idling area	25 (7.5)
Vents, chimneys, and flues from combustion appliances and equipment	15 (5)

correlate to a Class A rating. For buildings located within *hurricane-prone regions*, select louvers that also comply with Section 5.4.2.1.

- d. Use rain hoods sized for no more than 500 fpm (2.5 m/s) face velocity with a downward-facing intake, such that all intake air passes upward through a horizontal plane that intersects the solid surfaces of the hood before entering the system.
- e. Manage the water that penetrates the intake opening by providing a drainage area or moisture removal devices.

5.4.2.1 Louvers Located within Hurricane-Prone Regions. Louvers that protect air intake or exhaust openings in buildings located in *hurricane-prone regions* shall comply with the requirements of AMCA Standard 550.¹²

Exception to 5.4.2.1: Louvers integral to equipment.

5.4.3 Rain Intrusion. Air-handling and distribution equipment mounted outdoors shall be designed to prevent rain intrusion into the airstream when tested at design airflow and with no airflow, using the rain test apparatus described in UL 1995,⁸ Section 58.

5.4.4 Snow Entrainment. Where climate dictates, outdoor air intakes that are part of the mechanical ventilation system shall be designed as follows to manage water from snow that is blown or drawn into the system:

- a. Access doors to permit cleaning of wetted surfaces shall be provided.
- b. Outdoor air ductwork or plenums shall pitch to drains designed in accordance with the requirements of Section 5.7.

5.4.5 Bird Screens. Outdoor air intakes shall include a screening device designed to prevent penetration by a 0.5 in. (13 mm) diameter probe. The screening device material shall be corrosion resistant. The screening device shall be located, or other measures shall be taken, to prevent bird nesting within the outdoor air intake.

Informative Note: Any horizontal surface may be subject to bird nesting.

5.5 Particulate Matter Removal. Particulate matter filters or air cleaners having either

- a. a MERV of not less than 8 where rated in accordance with ANSI/ASHRAE Standard 52.2¹³ or
- b. the minimum efficiency within ISO ePM10 where rated in accordance with ISO 16890-1¹⁴

shall be provided upstream of all cooling coils or other devices with wetted surfaces through which air is supplied to an *occupiable space*.

Exception to 5.5: Cooling coils that are designed, controlled, and operated to provide sensible cooling only.

5.6 Finned-Tube Coils and Heat Exchangers

5.6.1 Drain Pans. A drain pan, in accordance with Section 5.7, shall be provided beneath all dehumidifying cooling-coil assemblies and all condensate-producing heat exchangers.

5.6.2 Finned-Tube-Coil Selection for Cleaning. Individual finned-tube coils or multiple finned-tube coils in series without intervening access spaces of at least 18 in. (457 mm) shall be selected to result in no more than 0.75 in. of water (187 Pa) combined dry-coil pressure drop at 500 fpm (2.54 m/s) face velocity.

5.7 Drain Pans. Drain pans, including their outlets and seals, shall be designed and constructed in accordance with this section.

5.7.1 Drain Pan Slope. Pans intended to collect and drain liquid water shall be sloped at least 0.125 in./ft (10 mm/m) from the horizontal toward the drain outlet or shall be otherwise designed such that water drains freely from the pan whether the fan is on or off.

5.7.2 Drain Outlet. The drain pan outlet shall be located at the lowest point(s) of the drain pan and shall be sized to preclude drain pan overflow under any normally expected operating condition.

5.7.3 Drain Seal. For configurations that result in negative static pressure at the drain pan relative to the drain outlet (such as a draw-through unit), the drain line shall include a P-trap or other sealing device designed to maintain a seal against ingestion of ambient air, while allowing complete drainage of the drain pan under any normally expected operating condition, whether the fan is on or off.

5.7.4 Pan Size. The drain pan shall be located under the water-producing device. Drain pan width shall be sized to collect water droplets across the entire width of the water-producing device or assembly. For horizontal airflow configurations, the drain pan length shall begin at the leading face or edge of the water-producing device or assembly and extend downstream from the leaving face or edge to a distance of either

- one half of the installed vertical dimension of the water-producing device or assembly or
- as necessary to limit water droplet carryover beyond the drain pan to 0.0044 oz/ft² (1.5 mL/m²) of face area per hour under peak sensible and peak dew-point design conditions, accounting for both latent load and coil face velocity.

5.8 Humidifiers and Water Spray Systems. Steam and direct-evaporative humidifiers, air washers, direct-evaporative coolers, and other water spray systems shall be designed in accordance with this section.

5.8.1 Water Quality. Water purity shall meet or exceed potable water standards at the point where it enters the ventilation system, space, or water vapor generator. Water vapor generated shall contain no chemical additives other than those chemicals in a potable water system.

Exceptions to 5.8.1:

- Water spray systems that use chemical additives that meet NSF/ANSI/CAN Standard 60.¹⁵
- Boiler water additives that meet the requirements of 21 CFR 173.310¹⁶ and include automated dosing devices.

5.8.2 Obstructions. Air cleaners or ductwork obstructions, such as turning vanes, volume dampers, and duct offsets greater than 15 degrees, that are installed downstream of humidifiers or water spray systems shall be located a distance equal to or greater than the absorption distance recommended by the humidifier or water spray system manufacturer.

Exception to 5.8.2: Equipment such as eliminators, coils, or evaporative media shall be permitted to be located within the absorption distance recommended by the manufacturer, provided a drain pan complying with the requirements of Section 5.7 is used to capture and remove any water that drops out of the airstream due to impingement on these obstructions.

5.9 Ozone-Generating Devices. The use of ozone-generating devices shall comply with Sections 5.9.1 and 5.9.2.

Exception to 5.9: Electronic devices used exclusively for the operation of HVAC equipment and controls.

Informative Note: Ozone generation is expected from ozone generators, corona discharge technology, some ultraviolet lights, electronic devices that create chemical reactions within the system, and some devices using a high voltage (>480 V). Motors and relays are examples of electronic devices that would be exempt.

5.9.1 Air-Cleaning Devices. Air-cleaning devices shall be listed and labeled in accordance with UL 2998.¹⁷

Informative Note: The use of devices not intended for air cleaning with the potential to generate ozone should be avoided.

5.9.2 Ultraviolet Devices. Ultraviolet-generating devices in *supply air* or spaces shall not transmit 185 nm wavelengths.

Informative Note: Ultraviolet devices used in treatment of closed water systems may produce 185 nm wavelengths, which may generate ozone.

5.10 Ventilation Air Distribution. Ventilating systems shall be designed in accordance with the requirements of Sections 5.10.1 through 5.10.3.

5.10.1 Designing for Air Balancing. Ventilation air distribution systems shall be provided that allow field verification of *outdoor air* intake flow (V_{ot}) during operation.

5.10.1.1 Designing for Varying Loads and Operating Conditions. The ventilation air distribution system for variable-air-volume (VAV) and multispeed constant-air-volume applications shall be provided with means to adjust the system to achieve at least the minimum ventilation airflow as required by Section 6 under any load condition or dynamic reset condition.

5.10.2 Plenum Systems. When the ceiling or floor plenum is used both to recirculate *return air* and to distribute *ventilation air* to ceiling-mounted or floor-mounted terminal units, the system shall be engineered such that each space is provided with its required minimum ventilation airflow.

Informative Note: Systems with direct connection of ventilation air ducts to terminal units, for example, comply with this requirement.

5.10.3 Documentation. The design documents shall specify minimum requirements for air balance testing or reference applicable standards for measuring and balancing airflow. The design documentation shall state assumptions that were made in the design with respect to ventilation rates and air distribution.

5.11 Airstream Surfaces. All airstream surfaces in equipment and ducts in the HVAC system shall be designed and constructed in accordance with the requirements of Sections 5.11.1 and 5.11.2.

5.11.1 Resistance to Mold Growth. Material surfaces shall be determined to be resistant to mold growth in accordance with a standardized test method.

Informative Note: Test methods include the mold growth and humidity test in UL 181,¹⁸ ASTM C1338,¹⁹ or ASTM D3273.²⁰

Exception to 5.11.1: Sheet-metal surfaces and metal fasteners.

Informative Note: Even with this resistance, any airstream surface that is continuously wetted is still subject to microbial growth.

5.11.2 Resistance to Erosion. Airstream surface materials shall be evaluated in accordance with the erosion test in UL 181¹⁸ and shall not break away, crack, peel, flake off, or show evidence of delamination or continued erosion under test conditions.

Exception to 5.11.2: Sheet metal surfaces and metal fasteners.

5.12 Mechanical and Indirect Evaporative Cooling Systems. Systems that cool by mechanical means or indirect evaporation shall be designed in accordance with Sections 5.12.1 and 5.12.2.

Exceptions to 5.12:

1. Systems in locations where the outdoor dew-point temperature is below 68°F (20°C) at the ASHRAE 2% annual dehumidification design condition.
2. Zones equipped with materials, assemblies, coatings, and furnishings that resist microbial growth and that are not damaged by continuously high indoor air humidity.
3. Indoor humidity shall be allowed to exceed the Section 5.12.1 humidity limit continuously for a period of up to 60 hours, provided the 30-day average humidity remains below the limit.

Informative Notes:

1. ASHRAE publishes design dehumidification conditions in the “Climatic Design Information” chapter of *ASHRAE Handbook—Fundamentals*.^{P3}
2. Examples of zones exempted by Exception 2 include shower rooms, swimming pool enclosures, kitchens, spa rooms, or semicooled warehouses that contain stored contents that are not damaged by continuously high indoor air humidity or microbial growth.

5.12.1 Humidity Limit. The indoor humidity shall not exceed a dew-point temperature of 60°F (15°C) during both occupied and unoccupied hours in any zone.

5.12.2 Controls. Devices and controls shall be provided to maintain the humidity at or below the limit defined in Section 5.12.1.

5.13 Direct Evaporatively Cooled Buildings. Systems that include cooling by direct evaporation into the *supply air* shall be designed in accordance with Sections 5.13.1 through 5.13.3.

5.13.1 Humidity Limits. Air in each HVAC zone shall be limited to a maximum relative humidity of 65% whenever evaporative cooling equipment is operating.

5.13.2 Analysis. The designer shall calculate the relative humidity in the HVAC zone at the outdoor cooling design condition. The HVAC zone design condition and resulting relative humidity and temperature

shall be included in the design documents to confirm that the design complies with the humidity limit defined in Section 5.13.1.

5.13.3 Controls. The design shall include at least one humidity sensor in each HVAC zone served by direct evaporative cooling equipment. Devices and controls shall be provided to maintain relative humidity of air in each HVAC zone at or below the limit defined in Section 5.13.1.

Exceptions to 5.13:

1. HVAC zones equipped with materials, assemblies, coatings, furnishings, and contents that resist microbial growth and that are not damaged by continuously high indoor air humidity.
2. Data centers, telephone closets, server rooms, and similar HVAC zones in mixed-use buildings.

Informative Notes:

1. Examples of HVAC zones exempted by Exception 1 include shower rooms, swimming pool enclosures, kitchens, spa rooms, or semicooled warehouses that contain stored contents that are not damaged by continuously high indoor air humidity or microbial growth.
2. Examples of HVAC zones potentially exempted by Exception 2 are those with installed equipment or machinery that generates a continuous sensible cooling load that is high enough to reduce relative humidity to less than 65% when calculated using the surface temperatures of walls, floor, and ceiling, rather than when calculated using the temperature of air in the HVAC zone. Surfaces that remain relatively warm are at relatively low risk for either condensation or moisture accumulation large enough to support health-relevant dampness and microbial growth.

5.14 Air Classification and Recirculation. Air shall be classified, and its recirculation shall be limited in accordance with Sections 5.14.1 through 5.14.4.

5.14.1 Classification. Air (*return air*, *transfer air*, or *exhaust air*) leaving each space or location shall be designated at an expected air-quality classification not less than that shown in Table 6-1, 6-2, or 6-3 or as approved by the AHJ. Air leaving spaces or locations that are not listed in Table 6-1, 6-2, or 6-3 shall be designated with the same classification as air from the most similar space or location listed in terms of occupant activities and building construction.

Exception to 5.14.1: Air from spaces where *ETS* is present.

Informative Notes: Classifications in Table 6-1, 6-2, or 6-3 are based on relative *contaminant concentration* using the following subjective criteria:

1. Class 1: Air with low *contaminant concentration*, low sensory-irritation intensity, and inoffensive *odor*.
2. Class 2: Air with moderate *contaminant concentration*, mild sensory-irritation intensity, or mildly offensive *odors*. (Class 2 air also includes air that is not necessarily harmful or objectionable but that is inappropriate for transfer or recirculation to spaces used for different purposes.)
3. Class 3: Air with significant *contaminant concentration*, significant sensory-irritation intensity, or offensive *odor*.
4. Class 4: Air with highly objectionable fumes or gases or with potentially dangerous particles, bioaerosols, or gases, at *concentrations* high enough to be considered as harmful.
5. Classification of air from *ETS areas* is not addressed. Spaces that are expected to include *ETS* do not have a classification listed in Table 6-1.

5.14.2 Redesignation

5.14.2.1 Air Cleaning. If air leaving a space or location passes through an *air-cleaning system*, redesignation of the cleaned air to a cleaner classification shall be permitted per the following requirements:

- a. Class 2 air where based on the subjective criteria in the informative note for Section 5.14.1 and where approved by the AHJ.
- b. Class 3 and Class 4 air when all requirements of Sections 6.3.1 through 6.3.6 are followed.

5.14.2.2 Transfer. A mixture of air that has been transferred through or returned from spaces or locations with different air classes shall be redesignated with the highest classification among the air classes mixed.

Informative Note: For example, mixed *return air* from a common system serving both a Class 1 space and a Class 2 space is designated as Class 2 air.

5.14.2.3 Ancillary Spaces. Redesignation of Class 1 air to Class 2 air shall be permitted for Class 1 spaces that are ancillary to Class 2 spaces.

Informative Note: For example, an office within a restaurant might be designated as a space ancillary to a Class 2 space, thus enabling the office to receive Class 2 air.

5.14.3 Recirculation Limitations. When the Ventilation Rate Procedure (VRP) of Section 6 is used to determine ventilation airflow values, recirculation of air shall be limited in accordance with the requirements of this section.

5.14.3.1 Class 1 Air. Recirculation or transfer of Class 1 air to any space shall be permitted.

5.14.3.2 Class 2 Air

5.14.3.2.1 Recirculation of Class 2 air within the space of origin shall be permitted.

5.14.3.2.2 Recirculation or transfer of Class 2 air to other Class 2 or Class 3 spaces shall be permitted, provided that the other spaces are used for the same or similar purpose or task and involve the same or similar pollutant sources as the Class 2 space.

5.14.3.2.3 Transfer of Class 2 air to toilet rooms shall be permitted.

5.14.3.2.4 Recirculation or transfer of Class 2 air to Class 4 spaces shall be permitted.

5.14.3.2.5 Class 2 air shall not be recirculated or transferred to Class 1 spaces.

Exception to 5.14.3.2.5: When using any *energy recovery device*, recirculation from leakage, carryover, or transfer from the exhaust side of the *energy recovery device* is permitted but shall not be counted as *outdoor air*. Exhaust air transfer ratio of Class 2 air shall not exceed 10% of the *outdoor air* intake flow at the design static pressure differential as defined in AHRI 1060.²¹

5.14.3.3 Class 3 Air

5.14.3.3.1 Recirculation of Class 3 air within the space of origin shall be permitted.

5.14.3.3.2 Class 3 air shall not be recirculated or transferred to any other space.

Exception to 5.14.3.3.2: When using any *energy recovery device*, recirculation from leakage, carryover, or transfer from the exhaust side of the *energy recovery device* is permitted but shall not be counted as *outdoor air*. Exhaust air transfer ratio of Class 3 air shall not exceed 5% of the *outdoor air* intake flow at the design static pressure differential as defined in AHRI 1060.²¹

5.14.3.4 Class 4 Air. Class 4 air shall not be recirculated or transferred to any space or recirculated within the space of origin.

5.14.4 Documentation. Design documentation shall indicate the justification for classification of air from any occupancy category, airstream, or location not listed in Table 6-1, 6-2, or 6-3.

5.15 Exhaust Duct Location

5.15.1 Exhaust ducts that convey Class 4 air shall be negatively pressurized relative to ducts, plenums, or *occupiable spaces* through which the ducts pass.

5.15.2 Exhaust ducts under positive pressure that convey Class 2 or Class 3 air shall not extend through ducts, plenums, or *occupiable spaces* other than the space from which the *exhaust air* is drawn.

Exception to 5.15.2: Exhaust ducts conveying Class 2 air and exhaust ducts conveying air from residential kitchen hoods that are sealed in accordance with Seal Class A as defined in ANSI/SMACNA 006.²²

5.16 Combustion Air. Fuel-burning appliances, both vented and unvented, shall be provided with air for combustion and removal of combustion products in accordance with manufacturer's instructions. Products of combustion from vented appliances shall be vented directly outdoors.

5.17 Local Capture of Contaminants. The discharge from noncombustion equipment that captures the *contaminants* generated by the equipment shall be ducted directly to the outdoors.

Exception to 5.17: Equipment specifically designed for discharge indoors in accordance with the manufacturer's recommendations.

5.18 Building Exfiltration. Ventilation systems for a building equipped with or served by mechanical cooling equipment shall be designed such that the total building outdoor air intake equals or exceeds the total building exhaust under all load and dynamic reset conditions.

Exceptions to 5.18:

1. Where an imbalance is required by process considerations and approved by the *AHJ*, such as in certain industrial facilities.
2. When *outdoor air* dry-bulb temperature is below the indoor space dew-point design temperature.

Informative Note: Although individual zones within a building may be neutral or negative with respect to outdoors or to other zones, net positive mechanical intake airflow for the building as a whole reduces *infiltration* of untreated *outdoor air*.

5.19 Ventilation System Controls. Mechanical ventilation systems shall include controls in accordance with Sections 5.19.1 and 5.19.2.

5.19.1 All systems shall be provided with manual or automatic controls to maintain not less than the outdoor air intake flow (V_{ot}) required by Section 6 under all load conditions or dynamic reset conditions.

5.19.2 Systems with fans supplying variable system primary airflow (V_{ps}) shall be provided with any combination of control equipment, methods, or devices to maintain no less than the outdoor air intake flow (V_{ot}) required for compliance with Section 5.19.1.

5.20 Ventilation System Emergency Control Modes. The building control system shall include modes of operation that adjust ventilation rates for emergency conditions. The control system shall allow initiation of the following modes:

- a. Economizer shutdown: Disable economizer controls such that systems operate with minimum outdoor airflow only.
- b. Outdoor air intake increase: *Demand controlled ventilation (DCV)* shall be disabled. The outdoor air intake flow (V_{ot}) and the exhaust rates shall be set to the maximum rates designated by the designer for system operation in this mode.

Informative Note: An example emergency during which the economizer shutdown mode of operation may be used is a nearby wildfire causing poor outdoor air quality. An example emergency during which the ventilation increase mode of operation may be used is a temporary period of unusual indoor *contaminant* source.

5.20.1 Infection Risk Management Mode (IRMM). When compliance with ASHRAE Standard 241²³ requires a change to the ventilation system operation, that mode of operation shall be designated as a ventilation system emergency control mode.

5.20.2 Outdoor Pollution Mode. When compliance with a guideline or standard for outdoor air pollution requires a change to the ventilation system operation, that mode of operation shall be designated as a ventilation system emergency control mode.

Informative Note: Refer to ASHRAE Guideline 44^{P4} for wildfires. Refer to ASHRAE Guideline 36^{P5} for other outdoor pollution events.

5.20.3 Ventilation system emergency controls modes shall include a timer-based reset that automatically restores normal operation after a user-adjustable period that shall be limited to not more than 72 hours. The initiation of these modes shall trigger a control system notification.

5.21 Access for Inspection, Cleaning, and Maintenance

5.21.1 Equipment Clearance. Ventilation equipment shall be installed with working space that will allow for inspection and routine maintenance, including filter replacement and fan belt adjustment and replacement.

5.21.2 Ventilation Equipment Access. Access doors, panels, or other means shall be provided and sized to allow unobstructed access for inspection, maintenance, and calibration of all ventilation system components for which routine inspection, maintenance, or calibration is necessary. Ventilation system components include air-handling units, fan-coil units, water-source heat pumps, other terminal units, controllers, and sensors.

5.21.3 Air Distribution System. Access doors, panels, or other means shall be provided in ventilation equipment, ductwork, and plenums, located and sized to allow convenient and unobstructed access for inspection, cleaning, and routine maintenance of the following:

- a. Outdoor air intake areaways or plenums
- b. Mixed-air plenums
- c. Upstream surface of each heating, cooling, and heat-recovery coil or coil assembly having a total of four rows or fewer
- d. Both upstream and downstream surface of each heating, cooling, and heat-recovery coil having a total of more than four rows, and air washers, evaporative coolers, heat wheels, and other heat exchangers
- e. Air cleaners
- f. Drain pans and drain seals
- g. Fans
- h. Humidifiers

5.22 Legionella Risk. The building owner shall be provided with written documentation of the design information for all of the HVAC-related water systems and elements that are required to be addressed by ANSI/ASHRAE Standard 188.²⁴

Informative Note: Typical elements of the HVAC system addressed by Standard 188 include open- and closed-circuit cooling towers and evaporative condensers that provide cooling or refrigeration for the

HVAC system and humidification systems. This requirement is related to HVAC's influence on Legionnaires' Disease risk, but it is important to note that there are multiple sources of potential water-to-people transmission inside buildings that are potable-water-based, such as shower heads, some humidifiers, and ice machines, as well as nonpotable sources that are not associated with HVAC, such as decorative water features.

6. PROCEDURES

6.1 General. The Ventilation Rate Procedure (VRP), the Indoor Air Quality Procedure (IAQP), the Natural Ventilation Procedure, or a combination thereof shall be used to meet the requirements of this section. In addition, the requirements for *outdoor air* treatment in Section 6.1.4 and exhaust *ventilation* in Section 6.5 shall be met regardless of the method used to determine minimum outdoor airflow rates.

Informative Note: Although the intake airflow determined using each of these approaches may differ significantly because of assumptions about the design, any of these approaches is a valid basis for design.

6.1.1 Ventilation Rate Procedure. The prescriptive design procedure presented in Section 6.2, in which outdoor air intake rates are determined based on space type/application, occupancy level, and floor area, shall be permitted to be used for any zone or system.

6.1.2 Indoor Air Quality Procedure. The IAQP, presented in Section 6.3, is an alternative to the VRP used to determine the design rate of outdoor airflow to maintain *concentrations* of *design compounds* (DCs) and *PM_{2.5}* in the indoor environment to be less than design limits (DLs), based on indoor and outdoor sources, air cleaning, and other variables. These *outdoor air* requirements shall be calculated with mass-balance equations. Verification of occupant satisfaction and indoor *DC concentrations* shall be performed after the building is completed.

6.1.3 Natural Ventilation Procedure. The prescriptive or engineered system design procedure presented in Section 6.4, in which *outdoor air* is provided through openings to the outdoors, shall be permitted to be used for any zone or portion of a zone in conjunction with *mechanical ventilation* in accordance with Section 6.4.

6.1.4 Outdoor Air Treatment. Each ventilation system that provides *outdoor air* shall comply with Sections 6.1.4.1 through 6.1.4.4.

Exception to 6.1.4: Systems supplying air for enclosed parking garages, warehouses, storage rooms, janitor's closets, trash rooms, recycling areas, shipping/receiving/distribution areas.

Informative Note: Occupied spaces ventilated with *outdoor air* that is judged to be unacceptable are subject to reduced air quality when *outdoor air* is not cleaned prior to introduction to the occupied spaces.

6.1.4.1 Particulate Matter Smaller than 10 Micrometers (PM₁₀). In buildings located in an area where the national standard or guideline for PM₁₀ is exceeded, particle filters or air-cleaning devices shall be provided to clean the *outdoor air* at any location prior to its introduction to occupied spaces. Particulate matter filters or air cleaners shall have either

- a MERV of not less than 8 where rated in accordance with ANSI/ASHRAE Standard 52.2¹³ or
- the minimum efficiency within ISO ePM₁₀ where rated in accordance with ISO 16890-1.¹⁴

Informative Note: See Informative Appendix F for resources regarding selected PM₁₀ national standards and guidelines.

6.1.4.2 Particulate Matter Smaller than 2.5 Micrometers (PM_{2.5}). In buildings located in an area where the national standard or guideline for PM_{2.5} is exceeded, particle filters or air-cleaning devices shall be provided to clean the *outdoor air* at any location prior to its introduction to occupied spaces. Particulate matter filters or air cleaners shall have either

- a MERV of not less than 11 where rated in accordance with ANSI/ASHRAE Standard 52.2¹³ or
- the minimum efficiency within ISO ePM_{2.5} where rated in accordance with ISO 16890-1.¹⁴

Informative Note: See Informative Appendix F for resources regarding selected PM_{2.5} national standards and guidelines.

6.1.4.3 Ozone. Air-cleaning devices for ozone shall be provided when the most recent three-year average annual fourth-highest daily maximum eight-hour average ozone *concentration* exceeds 0.100 ppm (195 µg/m³).

Such air-cleaning devices shall have a volumetric ozone removal efficiency of not less than 40% where installed, operated, and maintained in accordance with the manufacturer's recommendations and shall be approved by the *authority having jurisdiction* (AHJ). Such devices shall be operated where the outdoor ozone levels are expected to exceed 0.100 ppm (195 µg/m³).

Exceptions to 6.1.4.3: Air cleaning for ozone shall not be required where

1. the system design *outdoor air* intake flow is 1.5 ach or less,
2. controls are provided that sense outdoor ozone level and reduce intake airflow to 1.5 ach or less while complying with the outdoor airflow requirements of Section 6, or
3. *outdoor air* is brought into the building and heated by direct-fired *makeup air* units.

Informative Note: In the U.S., a most recent three-year average annual fourth-highest daily maximum eight-hour average ozone *concentration* exceeding 0.100 ppm (195 µg/m³) equates to a U.S. EPA eight-hour ozone classification of “Serious” or higher (Severe 15, Severe 17, or Extreme).

6.1.4.4 Other Outdoor Contaminants. In buildings located in an area where the national standard for one or more *contaminants* not addressed in Section 6.1.4 is exceeded, any design assumptions and calculations related to the impact on IAQ shall be included in the design documents.

6.2 Ventilation Rate Procedure. The outdoor air intake flow (V_{ot}) for a ventilation system shall be determined in accordance with Section 6.1.4 and Sections 6.2.1 through 6.2.6.

Informative Note: Additional explanation of terms used below is contained in Normative Appendix A, along with a ventilation system schematic (Figure A-1).

6.2.1 Zone Calculations. Ventilation zone parameters shall be determined in accordance with Sections 6.2.1.1 through Section 6.2.1.4 for *ventilation zones* served by the ventilation system, except that the ventilation rates from ANSI/ASHRAE/ASHE Standard 170²⁵ shall be used for the occupancy categories within the scope of Standard 170. For outpatient facilities outside the scope of Standard 170, refer to Normative Appendix E.

Informative Note: The ventilation rates in ANSI/ASHRAE/ASHE Standard 170 are intended to achieve asepsis and control *odor* migration and might not be adequate to achieve *acceptable IAQ* as defined in Standard 62.1.

6.2.1.1 Breathing Zone Outdoor Airflow. The flow rate of *outdoor air* required in the *breathing zone* (V_{bz}) of the *occupiable space* or spaces in a *ventilation zone* shall be not less than the value determined in accordance with Equation 6-1.

$$V_{bz} = R_p \times P_z + R_a \times A_z \quad (6-1)$$

where

A_z = zone floor area, the net occupiable floor area of the *ventilation zone*, ft² (m²)

P_z = zone population, the number of people in the *ventilation zone* during use

R_p = flow rate of *outdoor air* required per person as determined from Table 6-1 (**Informative Note:** These values are based on adapted occupants.)

R_a = flow rate of *outdoor air* required per unit area as determined from Table 6-1

Informative Notes:

1. Equation 6-1 accounts for people-related sources and area-related sources independently in the determination of the flow rate of *outdoor air* required at the *breathing zone*. The use of Equation 6-1 in the context of this standard does not necessarily imply that simple addition of outdoor airflow rates for different sources can be applied to any other aspect of IAQ.
2. The rates in Table 6-1 are based on all other applicable requirements of this standard being met. If other requirements of the standard are not met, then the rates do not apply.

6.2.1.1.1 Unlisted Occupancy. Where the occupancy category for a proposed space or zone is not listed, the requirements for the listed occupancy category that is most similar in terms of occupant density, activities, and building construction shall be used.

6.2.1.1.2 Source Strengths. The VRP minimum rates are based on *contaminant* sources and source strengths that are typical for the listed occupancy categories. Where *unusual sources* are expected, the additional *ventilation* or air cleaning required shall be calculated using Section 6.3 or criteria established by the environmental health and safety (EHS) professional responsible to the owner.

Informative Notes:

1. Zones where emissions are expected from stored *hazardous materials* are not typical for any listed occupancy category.
2. Dry ice, theatrical smoke, and smoke-producing activities are not typical for any listed occupancy categories.

6.2.1.1.3 Dwelling and Sleeping Units. Air from one *dwelling unit* with transient occupancy, or air from one *sleeping unit*, shall not be recirculated or transferred to any other unit.

6.2.1.1.4 Laboratories. Laboratory spaces that comply with all requirements of ANSI/ASSP Z9.5⁷ are not required to comply with the rates in Table 6-1 and Table 6-2.

6.2.1.1.5 Animal Facilities. Animal facilities that have completed a risk evaluation performed by the EHS professional responsible to the owner or to the owner's designee are not required to comply with the rates in Table 6-1 and Table 6-2.

6.2.1.1.6 Modular Rooms and Sleep Pods. *Modular rooms* and *sleep pods* shall use the requirements of the listed occupancy category that is most similar in terms of space use.

6.2.1.1.7 Design Zone Population. Design zone population (P_z) shall equal the largest (peak) number of people expected to occupy the *ventilation zone* during typical use.

Exceptions to 6.2.1.1.7:

1. Where the number of people expected to occupy the *ventilation zone* fluctuates, zone population equal to the average number of people shall be permitted, provided such average is determined in accordance with Section 6.2.5.2.
2. Where the largest or average number of people expected to occupy the *ventilation zone* cannot be established for a specific design, an estimated value for zone population shall be permitted, provided such value is the product of the *net occupiable area* of the *ventilation zone* and the default occupant density listed in Table 6-1.

6.2.1.2 Zone Air Distribution Effectiveness. The *zone air distribution effectiveness* (E_z) shall be determined in accordance with Table 6-4 or Normative Appendix C.

Informative Notes:

1. For some configurations, the default value depends on space and supply air temperature.
2. Calculation of E_z using the procedures in Normative Appendix C may result in values greater than those listed in Table 6-4 for systems with the same description.

6.2.1.2.1 Stratified Air Distribution Systems. A *stratified air distribution system* shall be designed in accordance with the following subsections, or the *zone air distribution effectiveness* (E_z) shall be determined in accordance with Normative Appendix C.

6.2.1.2.1.1 Supply Air. *Cool air* shall be at least 4°F (2°C) less than the average room air temperature.

6.2.1.2.1.2 Return Air. The return air openings or pathways shall be located more than 9 ft (2.8 m) above the floor.

6.2.1.2.1.3 Stratification. The zone shall not contain any devices that mechanically mix the air and shall be protected from impinging airstreams from adjacent *ventilation zones*.

Informative Note: Ceiling fans, blowers, air curtains, aspirating diffusers without adequate draft separation, or other devices that disrupt the stratification cause the *zone air distribution effectiveness* to be similar to a well-mixed system.

6.2.1.2.2 Personalized Ventilation Systems. A personalized ventilation system shall be designed in accordance with the following subsections, or the *zone air distribution effectiveness* (E_z) shall be determined in accordance with Normative Appendix C.

Informative Note: A personalized ventilation system is primarily for exposure control and dilution of *contaminants* in the *breathing zone* and may provide some spot cooling. Personalized *ventilation* is used when the occupant spends most of their time in one occupied space. The ventilation outlet is usually incorporated into or mounted on the furniture. It is used in conjunction with another air distribution system that handles the area ventilation requirements and thermal loads in the space.

6.2.1.2.2.1 Personalized Air. The personalized air shall be distributed in the *breathing zone* and designed such that the velocity is equal to or less than 50 fpm (0.25 m/s) at the head/facial region of the occupant.

6.2.1.2.2.2 Return Air. The *return air* openings or pathways shall be located more than 9 ft (2.8 m) above the floor.

6.2.1.3 Air Density Correction Factor. The air density correction factor (E_p) shall be determined in accordance with Table 6-5 or Normative Appendix D. *Outdoor air* intake elevation above sea level shall be measured to the center of the louver.

Informative Note: Where multiple louvers or systems are present the designer may utilize the elevation of the highest louver in lieu of the elevation at each louver.

Table 6-1 Minimum Ventilation Rates in Breathing Zone

Occupancy Category	People Outdoor Air Rate R_p		Area Outdoor Air Rate R_a		Default Values		OS (6.2.6.2)	Maximum CO ₂ above Ambient, $\Delta C_{6.1}$
	cfm/ person	L/s· person	cfm/ft ²	L/s·m ²	Occupant Density			
					#/1000 ft ² or #/100 m ²	Air Class		
Animal Facilities								
Animal exam room (veterinary office)	10	5	0.12	0.6	20	2		NA
Animal imaging (MRI/CT/PET)	10	5	0.18	0.9	20	3		NA
Animal operating rooms	10	5	0.18	0.9	20	3		NA
Animal postoperative recovery room	10	5	0.18	0.9	20	3		NA
Animal preparation rooms	10	5	0.18	0.9	20	3		NA
Animal procedure room	10	5	0.18	0.9	20	3		NA
Animal surgery scrub	10	5	0.18	0.9	20	3		NA
Large-animal holding room	10	5	0.18	0.9	20	3		NA
Necropsy	10	5	0.18	0.9	20	3		NA
Small-animal-cage room (static cages)	10	5	0.18	0.9	20	3		NA
Small-animal-cage room (ventilated cages)	10	5	0.18	0.9	20	3		NA
Correctional Facilities								
Booking/waiting	7.5	3.8	0.06	0.3	50	2		1200
Cell	5	2.5	0.12	0.6	25	2		NA
Dayroom	5	2.5	0.06	0.3	30	1		1500
Guard stations	5	2.5	0.06	0.3	15	1		1200
Educational Facilities								
Art classroom	10	5	0.18	0.9	20	2		NA
Assembly without fixed seats—concentrated (chairs only—fixed)	7.5	3.8	0.06	0.3	65	1	✓	1200
Classrooms (ages 5 to 8)	10	5	0.12	0.6	25	1		600
Classrooms (age 9+)	10	5	0.12	0.6	35	1		600
Corridors (ages 5+)	—	—	0.12	0.6	—	1		NA
Daycare sickroom	10	5	0.18	0.9	25	3		NA
Daycare (through age 4)	10	5	0.18	0.9	25	2		NA
Information technology equipment facilities (occupiable)	10	5	0.12	0.6	25	1		600
Lecture hall (fixed seats)	7.5	3.8	0.06	0.3	143	1	✓	1200
Libraries	5	2.5	0.12	0.6	10			600
Media center	10	5	0.12	0.6	25	1		600
Multiuse assembly (tables and chairs)	7.5	3.8	0.06	0.3	67	1	✓	1200
Music/theater/dance	10	5	0.06	0.3	35	1	✓	2100
Science laboratories	10	5	0.18	0.9	25	2		NA
University/college laboratories	10	5	0.18	0.9	25	2		NA
Wood/metal shop	10	5	0.18	0.9	20	2		NA
Food and Beverage Service								
Bars, cocktail lounges	7.5	3.8	0.18	0.9	100	2		1200

Table 6-1 Minimum Ventilation Rates in Breathing Zone (Continued)

Occupancy Category	People Outdoor Air Rate R_p		Area Outdoor Air Rate R_a		Default Values		OS (6.2.6.2)	Maximum CO ₂ above Ambient, $\Delta C_{6,1}$
	cfm/ person	L/s· person	cfm/ft ²	L/s·m ²	Occupant Density			
					#/1000 ft ² or #/100 m ²	Air Class		
Cafeteria/fast-food dining	7.5	3.8	0.18	0.9	67	2		900
Kitchen (cooking)	7.5	3.8	0.12	0.6	5	2		NA
Restaurant dining rooms	7.5	3.8	0.18	0.9	67	2		1500
General								
Break rooms	5	2.5	0.06	0.3	25	1	✓	1500
Coffee stations	5	2.5	0.06	0.3	20	1	✓	1200
Conference/meeting	5	2.5	0.06	0.3	50	1	✓	1500
Corridors	—	—	0.06	0.3	—	1	✓	NA
Occupiable storage rooms for liquids or gels	5	2.5	0.12	0.6	2	2		NA
Hotels, Motels, Resorts, Dormitories								
Barracks sleeping areas	5	2.5	0.06	0.3	20	1	✓	900
Bedroom/living room	5	2.5	0.06	0.3	10	1	✓	600
Laundry rooms, central	5	2.5	0.12	0.6	10	2		NA
Laundry rooms within <i>dwelling units</i>	5	2.5	0.12	0.6	10	1		NA
Lobbies/prefunction	7.5	3.8	0.06	0.3	30	1	✓	1500
Multipurpose assembly	5	2.5	0.06	0.3	120	1	✓	1800
Miscellaneous Spaces								
Banks or bank lobbies	7.5	3.8	0.06	0.3	15	1	✓	900
Bank vaults/safe deposit	5	2.5	0.06	0.3	5	2	✓	600
Freezer and refrigerated spaces (<50°F [10°C])	10	5	0	0	0	2		NA
Information technology equipment facilities (occupiable)	5	2.5	0.06	0.3	3	1	✓	NA
Manufacturing where <i>hazardous materials</i> are not used	10	5.0	0.18	0.9	7	2		600
Manufacturing where <i>hazardous materials</i> are used (excludes heavy industrial and chemical processes)	10	5.0	0.18	0.9	7	3		NA
Pharmacy (prep. area)	5	2.5	0.18	0.9	10	2		900
Photo studios	5	2.5	0.12	0.6	10	1		NA
Shipping/receiving	10	5	0.12	0.6	2	2		700
Sorting, packing, light assembly	7.5	3.8	0.12	0.6	7	2		900
Telephone closets	—	—	0.00	0.0	—	1		NA
Transportation waiting	7.5	3.8	0.06	0.3	67	1	✓	1800
Warehouses	10	5	0.06	0.3	—	2		700
Office Buildings								
Main entry lobbies	5	2.5	0.06	0.3	10	1	✓	1200
Occupiable storage rooms for dry materials	5	2.5	0.06	0.3	2	1		700
Office space	5	2.5	0.06	0.3	5	1	✓	600

Table 6-1 Minimum Ventilation Rates in Breathing Zone (Continued)

Occupancy Category	People Outdoor Air Rate R_p		Area Outdoor Air Rate R_a		Default Values		Air Class	OS (6.2.6.2)	Maximum CO ₂ above Ambient, $\Delta C_{6,1}$
	cfm/ person	L/s· person	cfm/ft ²	L/s·m ²	Occupant Density				
					#/1000 ft ² or #/100 m ²				
Reception areas	5	2.5	0.06	0.3	30	1	✓	2100	
Telephone/data entry	5	2.5	0.06	0.3	60	1	✓	1800	
Public Assembly Spaces									
Auditorium seating area	5	2.5	0.06	0.3	143	1	✓	1800	
Courtrooms	5	2.5	0.06	0.3	70	1	✓	1500	
Legislative chambers	5	2.5	0.06	0.3	50	1	✓	1800	
Libraries	5	2.5	0.12	0.6	10	1		600	
Lobbies	5	2.5	0.06	0.3	150	1	✓	1800	
Museums (children’s)	7.5	3.8	0.12	0.6	40	1		1800	
Museums/galleries	7.5	3.8	0.06	0.3	33	1	✓	1500	
Places of religious worship	5	2.5	0.06	0.3	120	1	✓	1800	
Residential									
Common corridors	—	—	0.06	0.3		1	✓	NA	
Retail									
Sales (except as below)	7.5	3.8	0.12	0.6	15	2		900	
Barbershop	7.5	3.8	0.06	0.3	7	2	✓	NA	
Beauty and nail salons	20	10	0.12	0.6	7	2		NA	
Coin-operated laundries	7.5	3.8	0.12	0.6	7	2		NA	
Mall common areas	7.5	3.8	0.06	0.3	33	1	✓	2100	
Pet shops (animal areas)	7.5	3.8	0.18	0.9	10	2		NA	
Supermarket	7.5	3.8	0.06	0.3	8	1	✓	1500	
Sports and Entertainment									
Bowling alley (seating)	10	5	0.12	0.6	40	1		900	
Disco/dance floors	20	10	0.06	0.3	100	2	✓	1500	
Gambling casinos	7.5	3.8	0.18	0.9	91	1		1200	
Game arcades	7.5	3.8	0.18	0.9	20	1		900	
Gym, sports arena (play area)	20	10	0.18	0.9	7	2		900	
Health club/aerobics room	20	10	0.06	0.3	20	2		1200	
Health club/weight rooms	20	10	0.06	0.3	10	2		1500	
Spectator areas	7.5	3.8	0.06	0.3	143	1	✓	1500	
Stages, studios	10	5	0.06	0.3	67	1	✓	1500	
Swimming (pool and deck)	—	—	0.48	2.4	—	2		NA	

Table 6-2 Minimum Exhaust Rates

Occupancy Category	Exhaust Rate, cfm/unit	Exhaust Rate, cfm/ft ²	Exhaust Rate, L/s·unit	Exhaust Rate, L/s·m ²	Air Class
Animal Facilities					
Animal imaging (MRI/CT/PET)	—	0.90	—	4.5	3
Animal operating rooms	—	3.00	—	15	3
Animal postoperative recovery room	—	1.50	—	7.5	3
Animal preparation rooms	—	1.50	—	7.5	3
Animal procedure room	—	2.25	—	11.3	3
Animal surgery scrub	—	1.50	—	7.5	3
Large-animal holding room	—	2.25	—	11.3	3
Necropsy	—	2.25	—	11.3	3
Small-animal-cage room (static cages)	—	2.25	—	11.3	3
Small-animal-cage room (ventilated cages)	—	1.50	—	7.5	3
Arenas	—	0.50	—	—	1
Art classrooms	—	0.70	—	3.5	2
Auto repair rooms	—	1.50	—	7.5	2
Barber shops	—	0.50	—	2.5	2
Beauty and nail salons	—	0.60	—	3.0	2
Cells with toilet	—	1.00	—	5.0	2
Copy, printing rooms	—	0.50	—	2.5	2
Darkrooms	—	1.00	—	5.0	2
Dwelling-unit kitchens		—		—	2
Continuous operation	50		25		
Intermittent operation	100		50		
Janitor closets, trash rooms, recycling	—	1.00	—	5.0	3
Kitchenettes	—	0.30	—	1.5	2
Kitchens—commercial	—	0.70	—	3.5	2
Locker rooms for athletic, industrial, and health care facilities	—	0.50	—	2.5	2
All other locker rooms	—	0.25	—	1.25	2
Shower rooms per shower head					2
Continuous operation	20		10		
Intermittent operation	50		25		
Parking garages	—	0.75	—	3.7	2
Pet shops (animal areas)	—	0.90	—	4.5	2
Science laboratories	—	0.35	—	1.75	2
Soiled laundry storage rooms	—	1.00	—	5.0	3
Storage rooms, chemical	—	1.50	—	7.5	4
Toilets—private (one person)		—		—	2
Continuous operation	25		12.5		
Intermittent operation	50		25		
Toilets—public (>1 person) per fixture (water closet or urinal)		—		—	2
Continuous operation	50		25		
Intermittent operation	70		35		
Woodwork shop/classrooms	—	0.50	—	2.5	2

Table 6-3 Airstreams or Sources

Description	Air Class
Kitchen grease hoods	4
Kitchen hoods other than grease hoods	3
Diazo printing equipment discharge	4
Hydraulic elevator machine room	2
Laboratory hoods	4
Paint spray booths	4
Refrigerating machinery rooms	3

Table 6-4 Zone Air Distribution Effectiveness (E_z)

Air Distribution Configuration	E_z
Well-Mixed-Air Distribution Systems	
<i>Ceiling supply of cool air</i>	1.0
<i>Ceiling supply of warm air and floor return</i>	1.0
<i>Ceiling supply of warm air 15°F (8°C) or more above space temperature and ceiling return</i>	0.8
<i>Ceiling supply of warm air less than 15°F (8°C) above average space temperature where the supply air-jet velocity is less than 150 fpm (0.8 m/s) within 4.5 ft (1.4 m) of the floor and ceiling return</i>	0.8
<i>Ceiling supply of warm air less than 15°F (8°C) above average space temperature where the supply air-jet velocity is equal to or greater than 150 fpm (0.8 m/s) within 4.5 ft (1.4 m) of the floor and ceiling return</i>	1.0
<i>Floor supply of warm air and floor return</i>	1.0
<i>Floor supply of warm air and ceiling return</i>	0.7
Makeup supply outlet located more than half the length of the space from the exhaust, return, or both	0.8
Makeup supply outlet located less than half the length of the space from the exhaust, return, or both	0.5
Stratified-Air Distribution Systems (Section 6.2.1.2.1)	
<i>Floor supply of cool air where the vertical throw is greater than or equal to 60 fpm (0.25 m/s) at a height of 4.5 ft (1.4 m) above the floor and ceiling return at a height less than or equal to 18 ft (5.5 m) above the floor</i>	1.05
<i>Floor supply of cool air where the vertical throw is less than 60 fpm (0.25 m/s) at a height of 4.5 ft (1.4 m) above the floor and ceiling return at a height less than or equal to 18 ft (5.5 m) above the floor</i>	1.2
<i>Floor supply of cool air where the vertical throw is less than 60 fpm (0.25 m/s) at a height of 4.5 ft (1.4 m) above the floor and ceiling return at a height greater than 18 ft (5.5 m) above the floor</i>	1.5
Personalized Ventilation Systems (Section 6.2.1.2.2)	
Personalized air at a height of 4.5 ft (1.4 m) above the floor combined with <i>ceiling supply of cool air and ceiling return</i>	1.40
Personalized air at a height of 4.5 ft (1.4 m) above the floor combined with <i>ceiling supply of warm air and ceiling return</i>	1.40
Personalized air at a height of 4.5 ft (1.4 m) above the floor combined with a <i>stratified air distribution system</i> with nonaspirating <i>floor supply</i> devices and <i>ceiling return</i>	1.20
Personalized air at a height of 4.5 ft (1.4 m) above the floor combined with a <i>stratified air distribution system</i> with aspirating <i>floor supply</i> devices and <i>ceiling return</i>	1.50

Table 6-5 Air Density Correction Factor

Outdoor Air Intake Elevation above Sea Level, ft	Outdoor Air Intake Elevation above Sea Level, m	E_p
≤515	≤158	1.00
515 to 1855	158 to 566	1.05
1855 to 3120	566 to 951	1.10
3120 to 4320	951 to 1317	1.15
4320 to 5460	1317 to 1664	1.20
5460 to 6540	1664 to 1994	1.25
6540 to 7575	1994 to 2309	1.30
7575 to 8560	2309 to 2609	1.35
8560 to 9505	2609 to 2897	1.40
9505 to 10410	2897 to 3173	1.45
10410 to 11275	3173 to 3437	1.50
>11275	>3437	Use Normative Appendix D

6.2.1.4 Zone Outdoor Airflow. The zone outdoor airflow (V_{oz}) provided to the *ventilation zone* by the supply air distribution system shall be determined in accordance with Equation 6-2.

$$V_{oz} = V_{bz}/E_z \times E_p \quad (6-2)$$

6.2.2 Single-Zone Systems. For ventilation systems wherein one or more air handler supplies a mixture of *outdoor air* and *recirculated air* to only one *ventilation zone*, the outdoor air intake flow (V_{ot}) shall be determined in accordance with Equation 6-3.

$$V_{ot} = V_{oz} \quad (6-3)$$

6.2.3 100% Outdoor Air Systems. For ventilation systems wherein one or more air handler supplies only *outdoor air* to one or more *ventilation zone*, the outdoor air intake flow (V_{ot}) shall be determined in accordance with Equation 6-4.

$$V_{ot} = \sum_{all\ zones} V_{oz} \quad (6-4)$$

6.2.4 Multiple-Zone Recirculating Systems. For ventilation systems wherein one or more air handler supplies a mixture of *outdoor air* and *recirculated air* to more than one *ventilation zone*, the outdoor air intake flow (V_{ot}) shall be determined in accordance with Sections 6.2.4.1 through 6.2.4.4.

6.2.4.1 Uncorrected Outdoor Air Intake. The uncorrected outdoor air intake flow (V_{ou}) shall be determined in accordance with Equation 6-5.

$$V_{ou} = D \sum_{all\ zones} (R_p \times P_z) + \sum_{all\ zones} (R_a \times A_z) \quad (6-5)$$

6.2.4.1.1 Occupant Diversity. The occupant diversity ratio (D) shall be determined in accordance with Equation 6-6 to account for variations in population within the *ventilation zones* served by the system.

$$D = P_s / \sum_{all\ zones} P_z \quad (6-6)$$

where the system population (P_s) is the total population in the area served by the system.

Exception to 6.2.4.1.1: Alternative methods to account for occupant diversity shall be permitted, provided the resulting V_{ou} value is not less than that determined using Equation 6-5.

6.2.4.1.2 Design System Population. Design system population (P_s) shall equal the largest (peak) number of people expected to occupy all *ventilation zones* served by the ventilation system during use.

Informative Note: Design system population is always equal to or less than the sum of design zone population for all zones in the area served by the system because all zones may not be simultaneously occupied at design population.

6.2.4.1.3 Other Ventilation Requirements. When a zone ventilation rate is obtained from criteria other than this standard, the ventilation rate shall be converted to cubic feet per minute (cfm) or liters per second (L/s) and the value added to V_{ou} for use in system design calculations.

Exception to 6.2.4.1.3: For systems serving spaces in the scope of both this standard and ANSI/ASHRAE/ASHE Standard 170,²⁵ the total design outdoor air intake flow shall be calculated per Standard 170 but shall not be less than the design outdoor air intake flow (V_{ot}) required by this standard.

6.2.4.2 System Ventilation Efficiency. The system ventilation efficiency (E_v) shall be determined in accordance with Section 6.2.4.3 for the Simplified Procedure or Normative Appendix A for the Alternative Procedure.

Informative Note: These procedures also establish zone minimum primary airflow rates for variable-air-volume systems.

6.2.4.3 Simplified Procedure

6.2.4.3.1 System Ventilation Efficiency. System ventilation efficiency (E_v) shall be determined in accordance with Equation 6-7 or 6-8.

$$E_v = 0.88 \times D + 0.22 \text{ for } D < 0.60 \quad (6-7)$$

$$E_v = 0.75 \text{ for } D \geq 0.60 \quad (6-8)$$

6.2.4.3.2 Zone Minimum Primary Airflow. For each zone, the minimum primary airflow (V_{pz-min}) shall be determined in accordance with Equation 6-9.

$$V_{pz-min} = V_{oz} \times 1.5 \quad (6-9)$$

6.2.4.4 Outdoor Air Intake. The design outdoor air intake flow (V_{ot}) shall be determined in accordance with Equation 6-10.

$$V_{ot} = V_{ou}/E_v \quad (6-10)$$

6.2.5 Design for Varying Operating Conditions

6.2.5.1 Variable Load Conditions. Ventilation systems shall be designed to be capable of providing not less than the minimum ventilation rates required in the *breathing zone* where the zones served by the system are occupied, including all full- and part-load conditions.

Informative Note: The minimum outdoor air intake flow may be less than the design value at part-load conditions.

6.2.5.2 Short-Term Conditions. Where it is known that peak occupancy will be of short duration, *ventilation* will be varied or interrupted for a short period of time, or both, the design shall be permitted to be based on the average conditions over a time period (T) determined by Equation 6-11 (I-P) or 6-12 (SI).

$$T = 3v/V_{bz} \quad (6-11)$$

$$T = 50v/V_{bz} \quad (6-12)$$

where

T = averaging time period, min

v = the volume of the *ventilation zone* where averaging is being applied, ft³ (m³)

V_{bz} = the flow rate of *outdoor air* required in the *breathing zone* calculated using Equation 6-1 and the design value of the zone population (P_z), cfm (L/s)

Acceptable design adjustments based on this optional provision include the following:

- Zones with fluctuating occupancy: The zone population (P_z) shall be permitted to be averaged over time (T).
- Zones with intermittent interruption of *supply air*: The average flow rate of *outdoor air* supplied to the *breathing zone* over time (T) shall be not less than the *breathing zone* outdoor airflow (V_{bz}) calculated using Equation 6-1.
- Systems with intermittent closure of the outdoor air intake: The average outdoor air intake flow over time (T) shall be not less than the minimum outdoor air intake flow (V_{ot}) calculated using Equation 6-3, 6-4, or 6-5 as appropriate.

6.2.6 Dynamic Reset. The system shall be permitted to be designed to reset the outdoor air intake flow (V_{ot}), the space or zone outdoor airflow (V_{oz}) as operating conditions change, or both.

6.2.6.1 Demand Controlled Ventilation (DCV). *DCV* shall be permitted as an optional means of dynamic reset.

6.2.6.1.1 For *DCV ventilation zones* in the *occupied mode*, breathing zone outdoor airflow (V_{bz}) shall be reset in response to current population. Current population estimates or indicators used in *DCV* control calculations shall not result in ventilation rates that are less than those required by the actual population when averaged over the time period determined in accordance with Section 6.2.5.2, or one hour, whichever is less.

6.2.6.1.2 For *DCV ventilation zones* in the *occupied mode*, breathing zone outdoor airflow (V_{bz}) shall be not less than the building component ($R_a \times A_z$) for the zone.

6.2.6.1.3 CO₂ sensors shall be permitted to be used for *DCV* in accordance with the following subsections.

Exception to 6.2.6.1.3: CO₂-based *DCV* shall not be applied in zones with indoor sources of CO₂ other than occupants, or with CO₂ removal mechanisms such as gaseous air cleaners.

6.2.6.1.3.1 CO₂ sensors shall be located in the space between 3 ft (0.9 m) and 6 ft (1.8 m) above the floor. There shall be at least one CO₂ sensor per *ventilation zone* and at least one per 5000 ft² (460 m²) of net occupiable floor area. Where *DCV ventilation zones* comprise more than one room, each room shall have a CO₂ sensor, and *ventilation* shall be controlled to the room requiring the most zone outdoor airflow.

Exception to 6.2.6.1.3.1: Other locations for CO₂ sensors are permitted if the locations are demonstrated to be accurate in measuring average CO₂ *concentrations* in the space *breathing zone*.

6.2.6.1.3.2 Demand ventilation controls shall maintain zone CO₂ *concentration* less than or equal to C_{max} determined in accordance with Equation 6-13.

$$C_{max} = C_{amb} + \Delta C \quad (6-13)$$

where

C_{amb} = outdoor air CO₂ *concentration* determined in accordance with Section 6.2.6.1.3.3

ΔC = maximum CO₂ *concentration* above ambient *concentration*. It shall be equal to $\Delta C_{6.1}$ listed in Table 6-1 if the design occupancy used to calculate the zone outdoor airflow per Section 6.2 is equal to the default occupant density listed in Table 6-1.

Where the design occupancy used to calculate the zone outdoor airflow per Section 6.2 differs from the default occupant density listed in Table 6-1, ΔC in Equation 6-13 shall be the adjusted value determined in accordance with Equation 6-14:

$$\Delta C_{adj} = \Delta C_{6.1} \frac{R_p + \frac{R_a}{PD_{6.1}}}{R_p + \frac{R_a}{PD_{des}}} \quad (6-14)$$

where

ΔC_{adj} = adjusted CO₂ maximum above ambient CO₂ *concentration*, ppm, rounded to the nearest 100 ppm

$\Delta C_{6.1}$ = maximum CO₂ above ambient CO₂ *concentration* from Table 6-1, ppm

R_p = flow rate of *outdoor air* required per person from Table 6-1, cfm/p (L/s/p)

R_a = flow rate of *outdoor air* required per unit area from Table 6-1, cfm/ft² (L/s/m²)

$PD_{6.1}$ = default occupant density, number of people per unit area from Table 6-1, #/1000 ft²/1000 (#/100 m²/100)

PD_{des} = design occupant density, number of people per unit area, equal to P_z/A_z where P_z and A_z are determined in accordance with Section 6.2.1.1

Where “NA” is listed in Table 6-1, CO₂ *DCV* shall not be used.

Informative Note: Steady-state CO₂ *concentrations* are not only dependent on occupancy density, but also on activity level, gender, body mass, and age. The CO₂ values in Table 6-1 are based on assumptions of these values that are typical of the occupancy type. The activity level (met) has the greatest impact on the steady-state CO₂ values. It is therefore possible that the differential CO₂ values in the space may exceed the *concentrations* listed in Table 6-1, even if the occupancy density and the ventilation rate meet the values listed in Table 6-1.

DCV controls shall maintain the zone ventilation rate at no less than the building component ($R_a \times A_z$) when zone CO₂ *concentration* is at C_{amb} . Upon a rise in measured CO₂ levels above C_{amb} , *ventilation* shall be increased proportional to the difference between zone CO₂ *concentration* and C_{amb} . When the zone CO₂ *concentration* is equal to C_{max} , the minimum zone ventilation rate shall be equal to its design ventilation rate determined in accordance with Section 6.2 at design occupancy.

Exceptions to 6.2.6.1.3.2:

1. Zone and system ventilation rates are not required to be larger than the design ventilation rate required by Section 6.2 at design occupancy regardless of CO₂ *concentration*.
2. Other *DCV* control logic shall be permitted to be used where it can be demonstrated to comply with Section 6.2.6.1.1 under all expected operating conditions.

Informative Note: The CO₂ values in Table 6-1 are only for the purposes of implementing CO₂ *DCV*. They are not intended to be, and should not be used as, indicators of IAQ.

- a. CO₂ *concentration* shall be assumed to be 400 ppm without any direct measurement.
- b. CO₂ *concentration* shall be dynamically measured using a CO₂ sensor located within 4 ft (1.2 m) of the outdoor air intake.

6.2.6.1.3.5 CO₂ sensors that sequentially sample air from multiple spaces shall sample each space no less than once every five minutes.

6.2.6.3 Ventilation Efficiency. Variations in the efficiency with which *outdoor air* is distributed to the occupants under different ventilation system airflows and temperatures shall be permitted as an optional basis of dynamic reset.

6.3 Indoor Air Quality Procedure (IAQP). Breathing zone outdoor airflow (V_{bz}) shall be determined in accordance with Sections 6.3.1 through 6.3.6.

1. Indoor emission rate information for some compounds is provided in Informative Appendix O.
2. Outdoor *concentrations* can be determined from Section 4.

$$E_m = \frac{C_1}{DL_1} + \frac{C_2}{DL_2} + \frac{C_i}{DL_i} \quad (6-15)$$

1. Benzene, phenol, and tetrachloroethylene shall not be included in the mixture calculation for upper respiratory tract irritation, eye irritation, and central nervous system depression. Outside the U.S., if the

Table 6-6 Design Compounds, PM2.5, and Their Design Limits

Compound or PM2.5	Cognizant Authority	Design Limit
Acetaldehyde	OEHHA CREL ²⁶	140 µg/m ³
Acetone	AgBB LCI ²⁷	120,000 µg/m ³
Benzene	OEHHA 8REL ²⁶	3 µg/m ³
Dichloromethane	OEHHA CREL	400 µg/m ³
Formaldehyde	CARB ²⁸	33 µg/m ³
Naphthalene	OEHHA CREL	9 µg/m ³
Phenol	AgBB LCI	70 µg/m ³
Tetrachloroethylene	OEHHA CREL	35 µg/m ³
Toluene	OEHHA CREL	420 µg/m ³
Xylene, total	AgBB LCI	500 µg/m ³
Carbon monoxide	40 CFR 50 ²	9 ppm
PM2.5	40 CFR 50 (annual mean)	9 µg/m ³
Ozone	40 CFR 50	70 ppb
Ammonia	OEHHA CREL	200 µg/m ³

Table 6-7 Mixtures of Compounds

Upper Respiratory Tract Irritation	Eye Irritation	Central Nervous System
Acetone	Acetone	Acetone
Xylene, total	Xylene, total	Xylene, total
Acetaldehyde	Acetaldehyde	Dichloromethane
Ozone	Ozone	Toluene
	Formaldehyde	

Source: American Conference of Governmental Industrial Hygienists (ACGIH)^{P6}

outdoor *concentrations* of carbon monoxide, *PM2.5*, or ozone exceed the DL, the DL shall be the applicable *ambient air* standards where the project is located or as accepted by the *AHJ*.

2. Ammonia shall be included only for spaces that include nonhuman animals.

6.3.3 Design Approach. Zone and system outdoor airflow rates shall be the larger of those determined in accordance with Sections 6.3.3.1 and 6.3.3.2, based on emission rates, *concentration* limits, and other relevant design parameters.

6.3.3.1 Mass-Balance Analysis. Using a steady-state or dynamic mass-balance analysis, the minimum outdoor airflow rates required to achieve the *concentration* limits specified in Section 6.3.2 shall be determined for each *DC*, mixture of *DCs*, and *PM2.5* within each zone served by the system.

Informative Note: Informative Appendix G includes steady-state mass-balance equations that describe the impact of air cleaning on *outdoor air* and recirculation rates for ventilation systems serving a single zone.

6.3.3.2 Perceived Indoor Air Quality. Zone outdoor airflow rates shall be sufficient to ensure that 80% or more of the people exposed do not express dissatisfaction with the quality of the air when tested as required in Section 7.3.2.

6.3.4 Air Cleaning. Where particulate matter or gas-phase air cleaning is included in the design, the removal efficiencies shall be specified as follows. Particulate matter filters shall report a minimum efficiency reporting value (MERV) in accordance with ANSI/ASHRAE Standard 52.2¹³ or a particle capture efficiency classification in accordance with ISO 16890-1.¹⁴ Gas-phase air cleaners shall report an efficiency test for all compounds included in the design in accordance with any of the following:

- a. ANSI/ASHRAE Standard 145.2²⁹
- b. ISO 10121-2³⁰

- c. Tested by methods in Sections 6.1.2, 10.4, and 10.5 of ANSI/ASHRAE Standard 145.2²⁹ and reported as required in Section 11 of Standard 145.2
- d. Tested to a national consensus standard approved by the *authority having jurisdiction*
- e. For technologies not covered by any of the above, tests developed to demonstrate the removal efficiency shall be performed by a third party. The custom efficiency test shall be conducted for all compounds included in the design and shall comply with the following:
 1. Test of the background *concentration* without the air cleaning in operation
 2. Test of the output *concentration* with the air cleaning in operation
 3. Be conducted under air-cleaning operating conditions that match the IAQP design operating conditions (**Informative Note:** Air-cleaning operating conditions include fan voltage, flow rate, and other settings that are consistent with the manufacturer's operating specifications.)
 4. Be conducted using the relevant laboratory methods for analysis and quantification as specified in Table 7-1. Inorganic compounds and *PM2.5* may be measured instead using direct-read instruments that are calibrated in accordance with the device manufacturer's recommendations, capable of measuring below the design limit, and consistent with the performance requirements specified in Table 7-2.

Any custom efficiency test description covering points (1) through (4) above and challenge test *concentration* shall be documented and approved by the *AHJ*. All test results, along with relevant equipment settings, shall be provided upon request.

6.3.4.1 End-of-Useful-Life Efficiency. Gas-phase filters and air cleaners that capture *contaminants* shall report an end-of-useful-life removal efficiency (E_{EOL}) for formaldehyde and at least one nonpolar volatile organic compound (VOC) from Table 6-6 that accounts for the expected accumulated mass of captured *contaminant* (M_{ACC}) at the specified replacement period. The expected M_{ACC} at the specified replacement period shall be determined in accordance with Equation 6-16 (I-P) or Equation 6-17 (SI). The E_{EOL} shall then be determined by repeating the test from Section 6.3.4 on a gas-phase filter that has been preloaded to the expected accumulated mass value. In Informative Appendix G, the E_{EOL} shall be used in place of E_f when calculating mass-balance removal efficiency for formaldehyde. All inputs used in Equation 6-16 or Equation 6-17, along with third-party test reports, shall be provided upon request.

$$M_{ACC} = T_y P C_{bz} V_c E_f / 266,974,172 \quad (6-16)$$

$$M_{ACC} = T_y P C_{bz} V_c E_f / 1,000,000 \quad (6-17)$$

where

M_{ACC} = accumulated mass of *contaminants*, lb (g)

T_y = annual building operating hours (assume 3744 hours based on 72 hours per week, unless otherwise specified), h/year

P = replacement period of the gas-phase filter or air cleaner as specified by the manufacturer for design purposes, years

C_{bz} = design limit for target *contaminant* per Table 6-6, $\mu\text{g}/\text{m}^3$

V_c = airflow rate through the gas-phase filter or air cleaner, cfm (m^3/h)

E_f = initial *contaminant* removal efficiency of the gas-phase filter or air cleaner

For technologies other than gas-phase filters and air cleaners, the E_{EOL} shall be depreciated from the initial removal efficiency to the manufacturer-certified E_{EOL} to account for degradation over the service life of the air cleaner. This depreciation shall be determined in accordance with the manufacturer's instructions and shall be tested and verified by a third party. Any custom efficiency test procedure or test description shall be documented and approved by the *AHJ*. All test results, including relevant equipment settings, shall be provided upon request.

6.3.5 Documentation. Design documentation shall include the list of *PM2.5*, *DCs*, and *DLs* and mixtures thereof; outdoor source data; emission rates, including citations; *cognizant authorities* for any additional *DCs*; mass-balance calculations for each zone; and specifications for verification required by Section 7.3.

6.3.6 Combined IAQ Procedure and Ventilation Rate Procedure. The IAQP in conjunction with the VRP shall be permitted to be applied to a zone or system. In this case, the VRP shall be used to determine the required zone minimum outdoor airflow, and the IAQP shall be used to determine the additional *outdoor air* or *air cleaning necessary* to achieve the *concentration* limits of the *contaminants* and *contaminant mixtures* of concern.

Informative Note: The improvement of IAQ through the use of air cleaning or provision of additional *outdoor air* in conjunction with minimum ventilation rates may be quantified using the IAQP.

6.4 Natural Ventilation Procedure. Natural ventilation systems shall comply with the requirements of either Section 6.4.1 or 6.4.2. Designers shall provide interior air barriers, insulation, or other means that separate naturally ventilated spaces from mechanically cooled spaces to prevent high-dew-point *outdoor air* from coming into contact with mechanically cooled surfaces.

6.4.1 Prescriptive Compliance Path. Any zone designed for *natural ventilation* shall include *mechanical ventilation* designed in accordance with Section 6.2, Section 6.3, or both.

Exceptions to 6.4.1:

1. Zones in buildings that have all of the following:
 - a. Natural ventilation openings that comply with the requirements of Section 6.4.1.
 - b. Controls that prevent the natural ventilation openings from being closed during periods of expected occupancy, or natural ventilation openings that are permanently open.
2. Zones that are not served by heating or cooling equipment.

6.4.1.1 Ceiling Height. For ceilings that are parallel to the floor, the ceiling height (H) to be used in Sections 6.4.1.3 through 6.4.1.5 shall be the minimum ceiling height in the zone.

For zones wherein ceiling height increases as distance from the natural ventilation openings increases, the ceiling height shall be the average height of the ceiling determined over a distance not greater than 20 ft (6 m) from the openings.

6.4.1.2 Floor Area to Be Ventilated. The naturally ventilated area in zones or portions of zones shall extend from the openings to a distance determined by Sections 6.4.1.3, 6.4.1.4, or 6.4.1.5. Openings shall meet the requirements of Section 6.4.1.6. For zones where ceilings are not parallel to the floor, the ceiling height shall be determined in accordance with Section 6.4.1.1.

6.4.1.3 Single Side Opening. For zones with openings on only one side of the zone, the naturally ventilated area shall extend to a distance not greater than two times the height of the ceiling from the openings.

6.4.1.4 Double Side Opening. For zones with openings on two opposite sides of the zone, the naturally ventilated area shall extend between the openings separated by a distance not greater than five times the height of the ceiling.

6.4.1.5 Corner Openings. For zones with openings on two adjacent sides of a zone, the naturally ventilated area shall extend to a distance not greater than five times the height of the ceiling along a line drawn between the outside edges of the two openings that are the farthest apart. Floor area outside that line shall comply with Section 6.4.1.3 as a zone having openings on only one side of the zone.

Informative Note: “Floor area outside that line” refers to the remaining area of the zone that is not bounded by the walls that have the openings and the line drawn between the openings.

6.4.1.6 Location and Size of Openings. Zones or portions of zones to be naturally ventilated shall have a permanently open airflow path to openings directly connected to the outdoors. The minimum flow rate to the zone shall be determined in accordance with Section 6.2.1.1. This flow rate shall be used to determine the required *openable area* of openings, accounting only for buoyancy-driven flow. Wind-driven flow shall be used only where it can be demonstrated that the minimum flow rate is provided during all occupied hours. Openings shall be sized in accordance with Section 6.4.1.6.1 (Path A) or Section 6.4.1.6.2 (Path B).

Informative Note: “Permanently open airflow path” refers to pathways that would allow airflow unimpeded by partitions, walls, furnishings, etc.

6.4.1.6.1 Sizing Openings—Path A. Where the zone is ventilated using a single opening or multiple single openings located at the same elevation, the *openable area* as a percent of the net occupiable floor area shall be greater than or equal to the value indicated in Table 6-8. Where the zone is ventilated using two openings located at different elevations or multiple pairs of such openings, the *openable area* as a percent of the net occupiable floor area shall be greater than or equal to the value indicated in Table 6-9.

Where openings are obstructed by louvers or screens, the *openable area* shall be based on the net *free area* of the opening. Where interior zones, or portions of zones, without direct openings to the outdoors are ventilated through adjoining zones, the opening between zones shall be permanently unobstructed and have a *free area* of not less than twice the percent of occupiable floor area used to determine the opening size of adjacent exterior zones, or 25 ft² (2.3 m²), whichever is greater.

Informative Note: Tables 6-8 and 6-9 are based solely on buoyancy-driven flow and have not been created to address thermal comfort.

6.4.1.6.2 Sizing Openings—Path B. The required *openable area* for a single zone shall be calculated using CIBSE AM10, Section 4.3.³¹

Table 6-8 Minimum Openable Areas: Single Openings ^a

$V_{bz}/A_z \leq$, (L/s)/m ²	$V_{bz}/A_z \leq$, cfm/ft ²	Total Openable Areas in Zone as a Percentage of A_z		
		$H_S/W_S \leq 0.1$	$0.1 < H_S/W_S \leq 1$	$H_S/W_S > 1$
1.0	0.2	4.0	2.9	2.5
2.0	0.4	6.9	5.0	4.4
3.0	0.6	9.5	6.9	6.0
4.0	0.8	12.0	8.7	7.6
5.5	1.1	15.5	11.2	9.8

where

V_{bz} = breathing zone outdoor airflow, per Section 6.2.1.1.

A_z = zone floor area, the net occupiable floor area of the *ventilation zone*.

W_S = aggregated width of all single outdoor openings located at the same elevation

H_S = vertical dimension of the single opening or the least vertical dimension of the openings where there are multiple openings

a. Volumetric airflow rates used to estimate required *openable area* are based on the following:

- Dry-air density of 0.075 lbda/ft³ (1.2 kgda/m³) at a barometric pressure of 1 atm (101.3 kPa) and an air temperature of 70°F (21°C)
- Temperature difference between indoors and outdoors of 1.8°F (1°C)
- Gravity constant of 32.2 ft/s² (9.81 m/s²)
- Window discharge coefficient of 0.6

Table 6-9 Minimum Openable Areas: Two Vertically Spaced Openings ^a

$V_{bz}/A_z \leq$, (L/s)/m ²	$V_{bz}/A_z \leq$, cfm/ft ²	Total Openable Areas in Zone as a Percentage of A_z					
		$H_{vs} \leq 8.2 \text{ ft (2.5 m)}$		$8.2 \text{ ft (2.5m)} < H_{vs} \leq 16.4 \text{ ft (5 m)}$		$16.4 \text{ ft (5 m)} < H_{vs}$	
		$A_s/A_l \leq 0.5$	$A_s/A_l > 0.5$	$A_s/A_l \leq 0.5$	$A_s/A_l > 0.5$	$A_s/A_l \leq 0.5$	$A_s/A_l > 0.5$
1.0	0.2	2.0	1.3	1.3	0.8	0.9	0.6
2.0	0.4	4.0	2.6	2.5	1.6	1.8	1.2
3.0	0.6	6.0	3.9	3.8	2.5	2.7	1.7
4.0	0.8	8.0	5.2	5.0	3.3	3.6	2.3
5.5	1.1	11.0	7.1	6.9	4.5	4.9	3.2

where

V_{bz} = breathing zone outdoor airflow, per Section 6.2.1.1.

A_z = zone floor area, the net occupiable floor area of the *ventilation zone*

H_{vs} = vertical separation between the center of the top and bottom openings' free operable area; in case of multiple horizontally spaced pairs of openings, use shortest distance encountered

A_s = *openable area* of smallest opening (top or bottom); in case of multiple horizontally spaced pairs of top-and-bottom openings, use aggregated areas

A_l = *openable area* of largest opening (top or bottom); in case of multiple horizontally spaced pairs of top-and-bottom openings, use aggregated areas

a. Volumetric airflow rates used to estimate required operable area are based on the following:

- Dry-air density of 0.075 lbda/ft³ (1.2 kgda/m³) at a barometric pressure of 1 atm (101.3 kPa) and an air temperature of 70°F (21°C)
- Temperature difference between indoors and outdoors of 1.8°F (1°C)
- Gravity constant of 32.2 ft/s² (9.81 m/s²)
- Window discharge coefficient of 0.6

6.4.2 Engineered System Compliance Path. For engineered *natural ventilation*, the designer shall

- determine hourly environmental conditions, including *outdoor air* dry-bulb temperature; dew-point temperature; outdoor *concentration of contaminants*, including *PM2.5*, *PM10*, and ozone where data are available; wind speed and direction; and internal heat gains during expected hours of natural ventilation operation.
- determine the effect of pressure losses along natural ventilation airflow paths on the resulting flow rates, including inlet openings, air transfer grills, ventilation stacks, and outlet openings during representative conditions of expected natural ventilation system use.
- quantify natural ventilation airflow rates of identified airflow paths accounting for wind-induced and thermally induced driving pressures during representative conditions of expected natural ventilation system use.
- design to provide *outdoor air* in quantities sufficient to result in *acceptable IAQ* as established under Section 6.2.1.1 or 6.3 during representative conditions of expected natural ventilation system use.

6.4.3 Control and Accessibility. The means to open required openings shall be *readily accessible* to building occupants whenever the space is occupied. Controls shall be designed to coordinate operation of the natural and mechanical ventilation systems.

6.4.4 Documentation. Where the Natural Ventilation Procedure is used, the designer shall document the values and calculations that demonstrate conformance with the compliance path and the controls systems and sequences required for operation of the natural ventilation system, including coordination with mechanical ventilation systems. Where the Prescriptive Compliance Path in Section 6.5.1 is used for buildings located in an area where the national standard for one or more *contaminants* is exceeded, any design assumptions and calculations related to the impact on IAQ shall be included in the design documents.

6.5 Exhaust Ventilation. The Prescriptive Compliance Path in Section 6.5.1 or the Performance Compliance Path in Section 6.5.2 shall be used to meet the requirements of this section. Exhaust *makeup air* shall be permitted to be any combination of *outdoor air*, *recirculated air*, or *transfer air*.

6.5.1 Prescriptive Compliance Path. The design exhaust airflow shall be determined in accordance with the requirements in Tables 6-2 and 6-3.

Exceptions to 6.5.1:

1. Laboratory spaces that comply with all requirements of ANSI/ASSP Z9.5⁷ or determined otherwise by the EHS professional responsible to the owner.
2. Parking garages where two or more sides comprise walls that are at least 50% open to the outside.
3. *Exhaust air* from toilets that has been cleaned to meet Class 1 criteria per Section 5.14.1 shall be permitted to be recirculated.

6.5.1.1 Source Strengths. The minimum exhaust rates in Table 6-2 are based on *contaminant* sources and source strengths that are typical for the listed occupancy categories. Where *unusual sources* are expected within a listed occupancy category, the additional exhaust required shall be calculated using Section 6.5.2.

6.5.1.2 Combustion Equipment. For specific occupancy categories, the exhaust requirements in Sections 6.5.1.2.1 through 6.5.1.2.4 shall apply whenever combustion equipment is operated. For occupancy categories where combustion equipment is operated, the exhaust requirements in Section 6.5.2 shall apply.

6.5.1.2.1 Auto Repair Rooms. Stands where engines are run in auto repair rooms shall have exhaust systems that directly connect to the engine exhaust and prevent escape of fumes.

6.5.1.2.2 Parking Garages. Systems in parking garages shall be designed to operate continuously whenever the garage is in use.

Exception to 6.5.1.2.2: Intermittent operation shall be allowed, provided the system is designed in accordance with the procedure in Section 6.5.2.

6.5.1.2.3 Kitchens. Kitchen exhaust shall comply with requirements of ANSI/ASHRAE Standard 154.³²

6.5.1.2.4 Laboratories. Laboratory spaces with combustion equipment shall comply with all requirements of ANSI/ASSP Z9.5⁷ or with the requirements provided by the EHS professional responsible to the owner.

6.5.1.3 Laboratory Hoods. Exhaust from laboratory hoods shall be Air Class 4 unless determined otherwise by the EHS professional responsible to the owner or to the owner's designee.

6.5.1.4 Pressure Requirements. While the required exhaust systems are operating, the exhaust airflow of zones listed in Table 6-2 shall be larger than their respective supply airflow. If zones listed in Table 6-2 are adjacent, the difference between the exhaust and the supply airflow shall be larger for the zone with the higher number class of air.

Exception to 6.5.1.4: Where airflow offset requirements are established by the EHS professional responsible to the owner or owner's designee.

Informative Notes:

1. Exhaust systems are required for any occupancy category listed in Table 6-2.
2. Where intermittent operation is allowed in Table 6-2, exhaust equipment is intended to be operated when the space is in use.

6.5.2 Performance Compliance Path. The exhaust airflow shall be determined in accordance with Sections 6.5.2.1 through 6.5.2.5.

6.5.2.1 Design Compounds and PM_{2.5} Sources. The exhaust system design shall be based on the *DCs* and *PM_{2.5}* specified in Table 6-6 at a minimum. Additional compounds from outdoor sources identified in accordance with Section 4 or from *unusual sources* shall be determined and shall be added to the *DC* list if a design limit from a *cognizant authority* exists. For each *DC* and *PM_{2.5}*, the emission rates from indoor sources from people, building materials, furnishings, equipment, and other sources and the rate of *contaminant* influx into the building (mass per unit time) shall be determined.

Informative Note: See Section 6.3.

6.5.2.2 Design Compounds and PM2.5 Concentration. The *concentration* limits, referred to as “design limits,” shall be as specified in Table 6-6. Design exhaust shall be such that the calculated *concentration* of each *DC*, mixture of *DCs*, and *PM2.5* does not exceed its limit. For any compounds added to the minimum *DC* list in Table 6-6, data from *cognizant authorities* shall be used to determine if the compound causes the effects listed in Table 6-7 (Section 6.3.2) and compounds having one or more of the mixture effects will be added to the mixture list for that effect. For each mixture, the mixed exposure sum (E_m) as determined by Equation 6-15 shall be less than 1.0.

6.5.2.3 Objective Evaluation. Perform *DC* and *PM2.5* measurement in the completed and occupied exhaust zone to verify that design limits are met. The peak *concentration* over an eight-hour occupied period shall not exceed the design limit for carbon monoxide. For ozone and *PM2.5*, the average *concentration* measured over an eight-hour occupied period shall be below the design limit.

For all other compounds, the *concentration* measured over the maximum period allowed by the test method up to eight hours shall be below the design limit for each *DC*. For *DC* mixtures, the mixture calculation shall be less than 1.0. The *concentrations* shall be measured using the relevant laboratory methods specified in Table 7-1. Inorganic compounds and *PM2.5* may be measured instead using direct-read instruments that are calibrated in accordance with the device manufacturer’s recommendations, are capable of measuring below the design limit, and that follow the performance requirements specified in Table 7-2.

6.5.2.4 Subjective Evaluation. Using a subjective occupant evaluation conducted in the completed building, the minimum exhaust airflow rates required to achieve acceptability of 80% or more shall be determined within each zone served by the exhaust system.

Exception to 6.5.2.4: The minimum exhaust airflow rates shall be not less than those found in accordance with Section 6.5.2.4 for a substantially similar zone.

6.5.2.5 Dynamic Reset. The system shall be permitted to be designed to reset the exhaust flow as operating conditions change. Monitoring and control systems shall be provided to automatically detect *DC* or *PM2.5* levels and modulate exhaust airflow such that *DC* or *PM2.5* levels are maintained at not greater than the specified *DC* or *PM2.5* compound *concentration* limits.

6.6 Design Documentation Procedures. Design criteria and assumptions shall be documented and made available for operation of the system after installation. See Sections 4.3, 5.10.3, 5.14.4, 6.2.6.1.4, 6.3.5, and 6.4.4 regarding assumptions to be detailed in the documentation.

7. CONSTRUCTION AND SYSTEM START-UP

Compliance with Sections 7.1 and 7.2 is required for all buildings. Section 7.3 is required for buildings designed under the Indoor Air Quality Procedure (IAQP) in Section 6.3.

7.1 Construction Phase

7.1.1 Application. The requirements of this section apply to ventilation systems and the spaces they serve in new buildings and additions to or alterations in existing buildings.

7.1.2 Filters. Systems designed with particle filters shall not be operated without filters in place.

7.1.3 Protection of Materials. When recommended by the manufacturer, building materials shall be protected from rain and other sources of moisture by appropriate in-transit and on-site procedures. Porous materials with visible microbial growth shall not be installed. Nonporous materials with visible microbial growth shall be decontaminated.

7.1.4 Protection of Occupied Areas

7.1.4.1 Application. The requirements of Section 7.1.4 apply when construction requires a building permit and entails sanding, cutting, grinding, or other activities that generate significant amounts of *contaminants*.

7.1.4.2 Protective Measures. Measures shall be employed to reduce the migration of construction generated *contaminants* to occupied areas.

Informative Note: Examples of acceptable measures include, but are not limited to, sealing the construction area using temporary walls or plastic sheathing, exhausting the construction area, or pressurizing contiguous occupied areas.

7.1.5 Air Duct System Construction. Air duct systems shall be constructed in accordance with the following standards, as applicable:

- a. The following sections of ANSI/SMACNA 006, *HVAC Duct Construction Standards—Metal and Flexible*:²²
 - Section S1.9 of Section 1.3.1, “Duct Construction and Installation Standards”

Table 7-1 Allowed Laboratory Test Methods

Compound	Allowed Test Methods
VOCs except formaldehyde, acetaldehyde and acetone	EPA IP-1; ³⁶ EPA TO-17; ³⁷ ISO 16000-6; ³⁸ ISO 16017-1; ³⁹ ISO 16017-2 ⁴⁰
Formaldehyde	ASTM D5197; ⁴¹ EPA IP-6; ⁴² EPA TO-11A; ISO 16000-3; ⁴³ or testing method that is compliant with the California Air Resources Board's (CARB) § 93120
Acetaldehyde and acetone	ASTM D5197; EPA IP-6; EPA TO-11A; ⁴⁴ EPA TO-17; ISO 16000-3
Carbon monoxide	EPA IP-3 ⁴⁵ ; ISO 4224 ⁴⁶

Table 7-2 Direct Reading Instruments Minimum Specifications

	Ozone	PM _{2.5}	Carbon Monoxide	Formaldehyde ^a
Accuracy (±)	5 ppb	Greater of 5 µg /m ³ or 20% of reading	Greater of 3 ppm or 20% of reading	2% full scale within calibrated linearity range
Resolution (±)	1 ppb	5 µg/m ³	1 ppm	0.1 ppb

a. Include the "calibrated linearity range" in all reports.

Table 7-3 Number of Measurements Points

Total Occupied Floor Area, ft ² (m ²)	Number of Measurements
≤25,000 (2500)	1
>25,000 (2500) and ≤50,000 (5000)	2
>50,000 (5000) and ≤100,000 (10,000)	4
>100,000 (10,000)	6

- Section 7.4, "Installation Standards for Rectangular Ducts Using Flexible Liner"
 - Section 3.5, "Flexible Duct Installation Standards"
 - Section 3.6, "Specification for Joining and Attaching Flexible Duct"
 - Section 3.7, "Specification for Supporting Flexible Duct"
 - Sections S6.1, S6.3, S6.4, and S6.5 of Section 9.1, "Casing and Plenum Construction Standards"
- b. All sections of SMACNA's *Fibrous Glass Duct Construction Standards*³³
- c. NFPA 90A, *Standard for the Installation of Air-Conditioning and Ventilating Systems*³⁴
- d. NFPA 90B, *Standard for the Installation of Warm Air Heating and Air-Conditioning Systems*³⁵

7.2 System Start-Up

7.2.1 Application. The requirements of this section apply to the following ventilation systems:

- a. Newly installed air-handling systems
- b. Existing air-handling systems undergoing *supply air* or outdoor airflow reduction (Only the requirements of Section 7.2.2 shall apply to these altered systems.)
- c. Existing air-handling distribution systems undergoing alterations affecting more than 25% of the floor area served by the systems (Only the requirements of Section 7.2.2 shall apply to these altered systems.)

7.2.2 Air Balancing and Verification of Outdoor Air Performance. Ventilation systems shall be balanced in accordance with ANSI/ASHRAE Standard 111³ or another applicable national standard so as to verify conformance with the total outdoor airflow requirements of this standard (V_{ot}).

7.2.3 Testing of Drain Pans. To minimize conditions of water stagnation that may result in microbial growth, drain pans shall be field tested under operating conditions that are the most restrictive to condensate flow to demonstrate proper drainage.

Exception to 7.2.3: Field testing of drain pans is not required if units with factory-installed drain pans have been certified (attested in writing) by the manufacturer for proper drainage when installed as recommended.

Informative Note: Above conditions usually occur at full fan airflow for draw-through fans and minimum fan airflow for blow-through fans.

7.2.4 Ventilation System Start-Up. Ventilation air distribution systems shall be clean of dirt and debris.

7.2.5 Outdoor Air Dampers. Prior to occupancy, each ventilation system shall be tested to demonstrate that outdoor air dampers operate in accordance with the system design.

7.2.6 Documentation. The following ventilation system documentation shall be provided to the building owner or owner's designee, retained within the building, and made available to the building operating personnel:

- a. An operations and maintenance (O&M) manual describing basic data relating to the O&M of ventilation systems and equipment as installed
- b. HVAC controls information consisting of diagrams, schematics, control sequence narratives, and maintenance and/or calibration information
- c. An air-balance report documenting the work performed for Section 7.2.2
- d. Construction drawings of record, control drawings, and final design drawings
- e. Design criteria and assumptions

7.3 Indoor Air Quality Procedure Verification

7.3.1 Objective Evaluation. Perform *design compound (DC)* and *PM_{2.5}* measurement in the completed building to verify that design limits (DLs) are met.

For carbon monoxide, ozone, and *PM_{2.5}*, the average *concentration* measured over an eight-hour occupied period shall not exceed the DL.

For all other compounds, the *concentration* measured over the maximum period allowed by the test method up to eight hours shall not exceed the DL for each *DC*. For *DC* mixtures, the mixture calculation shall be less than 1.0. The *concentrations* shall be measured using the relevant laboratory methods specified in Table 7-1. Inorganic compounds and *PM_{2.5}* may be measured instead using direct-read instruments that are calibrated in accordance with the device manufacturer's recommendations, are capable of measuring below the DL, and that follow the performance requirements specified in Table 7-2.

Exception to 7.3.1: Objective and subjective evaluation are not required for every substantially similar zone. The minimum outdoor airflow rates shall be not less than those found in accordance with Sections 7.3.1 and 7.3.2 for a substantially similar zone.

7.3.1.1 Design Compounds and PM_{2.5} Measurement Test. The measurement equipment shall be positioned in the *breathing zone*. The measurement shall be conducted with the HVAC system in normal operation and lowest outdoor air intake setting expected during the year. The number of measurement points shall be specified according to Table 7-3.

7.3.2 Subjective Evaluation. Using a subjective occupant evaluation conducted in the completed building, the survey test results shall demonstrate occupant level of acceptability of 80% or more within each zone served by the system.

Informative Note: Informative Appendix O presents one approach to subjective occupant evaluation.

Exception to 7.3.2: Objective and subjective evaluation are not required for every substantially similar zone. The minimum outdoor airflow rates shall be not less than those found in accordance with Sections 7.3.1 and 7.3.2 for a substantially similar zone.

Informative Note: For example, in a building with 100 single-person private offices of the same area, it is not necessary to perform an objective and subjective evaluation in each office as long as the minimum outdoor airflow for each office is greater than or equal to the airflow in the office that is tested according to Section 7.3.2. A *ventilation zone* is similar by definition if it has the same occupancy category (see Table 6-1), occupant density, *zone air distribution effectiveness* (see Section 6.2.1.2), and design zone primary airflow (see Section 6.2.4.3.2 and Normative Appendix A) per unit area.

7.3.3 Documentation. Documentation shall include methodology of testing, test results and any adjustments to design *outdoor air* from the values determined in Section 6.3, if applicable.

8. OPERATIONS AND MAINTENANCE

8.1 General

8.1.1 Application. The requirements of this section apply to buildings and their ventilation systems and their components after they are placed into service.

8.1.2 Building Alterations or Change of Use. Where buildings are altered or where changes in building use, occupant category, significant change in occupant density, or other changes inconsistent with system design assumptions are made, the ventilation system design, operation, and maintenance shall be reevaluated and the operations and maintenance (O&M) manual updated as necessary.

Table 8-1 Minimum Inspection, Verification, and Validation Activity and Frequency for Ventilation Systems and Associated Components

Task No.*	Related Section	Inspection, Verification, Validation Task	Frequency
1a	4.1 6.1.4	Verify that the building is located in an area not designated as a nonattainment area. If the area has been designated as a nonattainment area, prepare and implement a plan to comply with the requirements of Section 6.1.4.	Annually
2a	5.1.2	Inspect piping and ductwork insulation system in accessible areas for integrity and signs of moisture or biological growth and repair as needed.	Annually
3a	5.4.1	Verify that no modifications to the HVAC system or the facility have been made that impact separation distances. If deficiencies are identified, document the deficiencies and undertake and document corrective actions.	Annually
4a	5.4.3 5.4.4 5.4.5	Inspect outdoor air intakes and exhaust openings, bird screens, louvers, dampers, and other attached components and ducting for rain intrusion, snow entrainment, pest intrusion, and physical condition, including the following: a. Indication of biological growth b. Indication of corrosion c. Buildup of dirt and debris d. Indication or presence of birds, insects, or other animals e. Integrity Clean and repair as needed; determine root cause and corrective actions.	Semiannually
5a	5.5 5.9	Verify air filtration fit, function, and performance. a. Verify efficiency and particulate matter rating meets or exceeds minimum requirements. b. Verify filters are correct thickness, fit, and size for housing assembly. c. Verify seal integrity so that airflow cannot bypass the filter(s). d. Verify filter scheduled replacement frequency per the O&M manual. e. Replace filters that exceed the maximum pressure drop of the filter per the O&M manual. f. Verify electronic air cleaning devices are operating in accordance with manufacturer's instructions. Repair, replace, or clean filters or assembly components as needed.	Quarterly or at scheduled replacement if sooner
6a	5.7.1 5.7.2 5.7.3 5.7.4 7.2.3	Inspect drain pans, seals, traps, pumping systems, drains and drain piping within air handling equipment, ducting, and plenums. a. Verify drain pan is positioned under water producing devices. Informative Note: The operator may check sizing per Sections 5.7.4 and 7.2.3. b. Verify drain pan slope is in direction of drain outlet. c. Inspect for biological growth, corrosion, or other debris that would prevent intended drain operation. d. Verify water drains freely out of pan and through attached drain piping. e. Verify traps, seals, and priming devices are wet and operational. f. Verify pumps are operational. g. Inspect for signs of overflow or water carryover. h. Validate shutoff devices or alarms function as intended. Clean and repair as needed; determine root cause and make corrective actions.	Semiannually

Notes:

- Task numbers labeled "a" are intended as verifications (see Note 2). Task numbers labeled "b" are intended as validations (see Note 3).
- "Verify" shall mean checking a condition through means such as visual inspection, review of documentation, reading gages, using telltales, viewing local displays, or analyzing live or trend data in a building automation system (BAS). These tasks are intended to be accomplished without special tools or specialized instrumentation and by persons without specialized training, knowledge, certifications, or licenses.
- "Validate" shall mean taking action that requires specific tools, calibrated instrumentation, adjustments, and changes in operation and that is performed by persons needing specialized training, knowledge, certifications, or licenses.
- If the desired results of a task cannot be verified, as described in Note 2, then the task shall be validated as described in Note 3, regardless of listed minimum frequency.

Table 8-1 Minimum Inspection, Verification, and Validation Activity and Frequency for Ventilation Systems and Associated Components (Continued)

Task No.*	Related Section	Inspection, Verification, Validation Task	Frequency
7a	5.8	Inspect humidifiers and dehumidifiers. a. Verify drain components function as intended. b. Verify water feed components function as intended. c. Inspect for biological growth, corrosion, or other debris on units or associated ducting as an indication of improper operation. d. Verify humidity, dew point, or other moisture measurements are within designed ranges and associated sensors are functional. e. Verify operation of desiccant dehumidifiers per manufacturer's instructions. Clean and repair as needed; determine root cause and make corrective actions.	Semiannually
7b	5.8	Validate that humidity, dew point, and other sensors that measure moisture are calibrated and function as intended.	Every 3 years
8a	5.9.1 5.9.2	Verify operation and condition of electrical air-cleaning devices and ultraviolet devices. a. Verify operation of air-cleaning devices per manufacturer's instructions. b. Inspect for indication of lamp malfunction or ineffectiveness and determine root cause and corrective action. c. Clean lamps in accordance with manufacturer's instructions. d. Replace lamps per manufacturer's schedule or upon lamp failure.	Quarterly
9a	5.10.1 5.21.1 5.21.2 5.21.3 7.2.2	Verify that space and access is provided and is maintained unobstructed for the following areas: a. All areas necessary for air balancing, verification, and measurement of <i>ventilation</i> by Section 7.2.2 b. All other areas that require routine maintenance and inspection by Section 8.2 c. All components and equipment that require verification by Section 8.5 d. All sensors, instruments, and ventilation system components, controls, and equipment e. All access doors and panels i. Verify access doors and panels are functional and correctly seal when not open for testing, inspection, or maintenance. Clean, repair, and provide access as needed.	Semiannually
10a	5.10.2 5.11.1 7.2.4	Inspect floor, ceiling, or mechanical room plenum systems and ductwork. a. Verify that there is no biological growth, corrosion, or indication of insects or other animals. b. Verify chemicals, cleaning products or equipment, or other materials are not kept in air-handling spaces or plenums. Clean and remove foreign objects; determine root cause and make corrective actions.	Annually

Notes:

- Task numbers labeled "a" are intended as verifications (see Note 2). Task numbers labeled "b" are intended as validations (see Note 3).
- "Verify" shall mean checking a condition through means such as visual inspection, review of documentation, reading gages, using telltales, viewing local displays, or analyzing live or trend data in a building automation system (BAS). These tasks are intended to be accomplished without special tools or specialized instrumentation and by persons without specialized training, knowledge, certifications, or licenses.
- "Validate" shall mean taking action that requires specific tools, calibrated instrumentation, adjustments, and changes in operation and that is performed by persons needing specialized training, knowledge, certifications, or licenses.
- If the desired results of a task cannot be verified, as described in Note 2, then the task shall be validated as described in Note 3, regardless of listed minimum frequency.

Table 8-1 Minimum Inspection, Verification, and Validation Activity and Frequency for Ventilation Systems and Associated Components (Continued)

Task No.*	Related Section	Inspection, Verification, Validation Task	Frequency
11a	5.10.3 7.2.2	Perform air balance verification. a. If occupancy or space utilization has changed, determine if airflow rates meet the requirements of this standard. i. Review design documents, most recent testing, adjusting, and balancing (TAB) report, and current requirements. ii. Determine if reported airflows meet design intent. b. When airflow is displayed: i. Verify outdoor airflows meet design requirements of this standard. ii. In spaces that require exhaust by Table 6-2, verify exhaust airflow is greater than supply airflow. c. Verify dynamic reset and outdoor airflow per Task No. 14a of this table. Verify CO₂ sensor calibration date and CO₂ DCV setup. a. When the time elapsed since the previous CO₂ sensor calibration date is beyond manufacturer's calibration frequency, have the sensor calibrated. b. Determine space type and design occupancy and verify maximum CO₂ limit is correctly set. c. Verify when the space is unoccupied that <i>ventilation</i> for the building component ventilation rate is provided or the space is in occupancy standby.	Annually
11b	5.10.3 5.19 6.2.5 6.2.6 7.2.2	Perform TAB air balance validation. a. Measure outdoor airflows of all units and adjust as necessary. i. Spot check space level airflows at outlets and inlets at a minimum of 20% of all zones. ii. Rebalance as necessary to achieve compliance with the design intent and this standard. iii. If occupancy or space utilization has changed and airflows no longer meets design intent or this standard, rebalance all zone and space level airflows. b. Rebalance exhaust airflows to maintain supply and exhaust relationship and pressurization requirements. c. Validate CO₂ sensor calibration date and CO₂ DCV setup. i. When CO₂ sensor calibration date is beyond manufacturer's calibration frequency, calibrate in accordance with manufacturer's instructions. d. Validate, maintain, and calibrate ventilation sensors. i. Calibrate static and differential pressure transducers used to control fan pressure, room pressure, VAV boxes, airflow, and filters. ii. Validate airflow rates of airflow sensors. Clean and calibrate airflow sensors per manufacturer's instructions.	Every 3 years
12a	5.12	Verify cooling systems maintain indoor conditions below humidity limits as referenced in Section 5.12. Make corrective actions.	Annually
13a	5.14 5.18 6.5.1.4	Verify directional airflow and building pressurization. a. Verify that zones are pressurized positively or negatively per design and that flow moves toward the exhaust. b. Verify operation of any pressure indication or measurement sensors.	Annually

Notes:

1. Task numbers labeled "a" are intended as verifications (see Note 2). Task numbers labeled "b" are intended as validations (see Note 3).
2. "Verify" shall mean checking a condition through means such as visual inspection, review of documentation, reading gages, using telltales, viewing local displays, or analyzing live or trend data in a building automation system (BAS). These tasks are intended to be accomplished without special tools or specialized instrumentation and by persons without specialized training, knowledge, certifications, or licenses.
3. "Validate" shall mean taking action that requires specific tools, calibrated instrumentation, adjustments, and changes in operation and that is performed by persons needing specialized training, knowledge, certifications, or licenses.
4. If the desired results of a task cannot be verified, as described in Note 2, then the task shall be validated as described in Note 3, regardless of listed minimum frequency.

Table 8-1 Minimum Inspection, Verification, and Validation Activity and Frequency for Ventilation Systems and Associated Components (Continued)

Task No.*	Related Section	Inspection, Verification, Validation Task	Frequency
14a	5.19 6.2.5 6.2.6	Verify ventilation system with variable load or dynamic reset controls. a. Inspect and test damper assemblies and adjust and repair as needed. i. Verify seals are intact, linkages and dampers operate smoothly, dampers operate full range, actuator clamps are tight on shaft, dampers close tight, and actuators modulate as intended and fail-safe on power loss. b. Inspect fan assemblies for the following. Clean, lubricate, adjust, and repair as needed. i. Belt tension and wear, sheave alignment, bearing operation ii. Fan balance, tightness, and cleanliness iii. Electrical connections, motor controllers, variable frequency drives, and other speed modulating devices c. For <i>ventilation zones</i> with airflow sensors, verify that zone level controls reset to changing conditions. i. Verify that zone air ventilation rates are maintained during supply fan turn-down and other reduced load conditions. ii. Verify that demand control <i>ventilation zones</i> respond to changes in occupancy. d. Verify that occupancy controls shut off the zone <i>ventilation</i> when the space is either unoccupied or in occupied standby.	Every 2 years
15a	5.22	Verify <i>Legionella</i> plan is implemented and maintained.	Annually
16a	6.3.1 6.3.3.2 7.3	For systems designed per the <i>Indoor Air Quality Procedure (IAQP)</i> , verify if changes to occupancy, changes to space utilization, new procedures, new components, refurbishments, or renovations have been made. If so, see Task No. 16b. Informative Notes: 1. New procedures or components in the space such as new furniture, printers, computers, cleaning chemicals, or other additions could increase the compounds listed in Table 6-5. 2. Refurbishment of carpets, paint, tiles, windows, or other additions could increase the compounds listed in Table 6-5.	Every 2 years
16b	6.3.1 6.3.3.2 7.3	For systems designed per the IAQP, validate that <i>concentrations</i> of <i>design compounds (DCs)</i> and <i>PM2.5</i> remain less than the design limits (DLs) per Section 6.3.1. a. Perform objective evaluation per Section 7.3.1 to verify DLs continue to be met. i. If any limit is exceeded, perform root cause analysis to determine if flow rates need adjustment or air-cleaning equipment needs repair or replacement. b. Conduct subjective evaluation of occupants per Section 7.3.2. i. Address concerns of occupants.	As required
17a	6.4.3	Verify <i>natural ventilation</i> controls and accessibility. a. Verify that occupant-operated natural ventilation openings are functional and accessible. b. Verify control sequences and instrumentation that automate natural ventilation openings operate during occupied periods. Validate that openings cannot be closed except during unoccupied periods or when mechanical ventilation system is active. c. Verify controls and sequences for <i>mechanical ventilation</i> operate when conditions for <i>natural ventilation</i> are inadequate.	Annually

Notes:

- Task numbers labeled “a” are intended as verifications (see Note 2). Task numbers labeled “b” are intended as validations (see Note 3).
- “Verify” shall mean checking a condition through means such as visual inspection, review of documentation, reading gages, using telltales, viewing local displays, or analyzing live or trend data in a building automation system (BAS). These tasks are intended to be accomplished without special tools or specialized instrumentation and by persons without specialized training, knowledge, certifications, or licenses.
- “Validate” shall mean taking action that requires specific tools, calibrated instrumentation, adjustments, and changes in operation and that is performed by persons needing specialized training, knowledge, certifications, or licenses.
- If the desired results of a task cannot be verified, as described in Note 2, then the task shall be validated as described in Note 3, regardless of listed minimum frequency.

Table 8-1 Minimum Inspection, Verification, and Validation Activity and Frequency for Ventilation Systems and Associated Components (Continued)

Task No.*	Related Section	Inspection, Verification, Validation Task	Frequency
18a	6.5	Verify exhaust <i>ventilation</i>. a. When following the Prescriptive Compliance Path (Section 6.5.1), review design documents and most recent TAB report exhaust flow rates. i. Determine if exhaust airflow rates are per Tables 6-2 and 6-3. ii. If occupancy or space utilization has changed, identify new exhaust rates. b. When following the Performance Compliance Path (Section 6.5.2), review design documents to determine that expected <i>contaminants</i> and mixtures of concern have not changed. i. Verify that exhaust monitoring and control system are detecting and maintaining <i>concentrations</i> below limits. For systems equipped with airflow sensors verify design airflow rates are maintained. ii. If occupancy or space utilization has changed identify new <i>contaminants</i> or mixtures of concern, <i>concentration</i> limits, and exhaust flow rates. c. If <i>concentration</i> levels, airflow rates, or pressurization requirements are not being maintained per design, perform the necessary actions described in Task No. 11b of this table.	Annually

Notes:

- Task numbers labeled “a” are intended as verifications (see Note 2). Task numbers labeled “b” are intended as validations (see Note 3).
- “Verify” shall mean checking a condition through means such as visual inspection, review of documentation, reading gages, using telltales, viewing local displays, or analyzing live or trend data in a building automation system (BAS). These tasks are intended to be accomplished without special tools or specialized instrumentation and by persons without specialized training, knowledge, certifications, or licenses.
- “Validate” shall mean taking action that requires specific tools, calibrated instrumentation, adjustments, and changes in operation and that is performed by persons needing specialized training, knowledge, certifications, or licenses.
- If the desired results of a task cannot be verified, as described in Note 2, then the task shall be validated as described in Note 3, regardless of listed minimum frequency.

8.2 Operations and Maintenance Manual. An O&M manual, either written or electronic, shall be developed and maintained on site or in a centrally accessible location for the working life of the applicable ventilation system equipment or components. This manual shall be updated as necessary. The manual shall include the O&M procedures, ventilation system operating schedules and any changes made thereto, final design drawings, maintenance schedules based on manufacturer’s instructions, and the inspection and maintenance requirements and frequencies provided in Table 8-1 and in ANSI/ASHRAE/ACCA Standard 180.⁴⁷

Informative Note: Standard 180, provides requirements for the development of a comprehensive maintenance plan for commercial HVAC systems.

8.3 Ventilation System Operation. Mechanical and natural ventilation systems shall be operated in a manner consistent with the O&M manual. Systems shall be operated such that spaces are ventilated in accordance with Section 6 during periods of expected occupancy.

8.4 Ventilation System Maintenance. The building ventilation system components shall be maintained in accordance with the O&M manual.

8.5 Ventilation System Inspection, Verification, and Validation. The building ventilation system and components shall be inspected and verified to continue to meet requirements as designed in accordance with this standard. The inspection and verification shall be performed per minimum frequencies provided in Table 8-1.

Informative Note: Tasks may need to be increased in frequency when operating in a mode that is beyond normal operation. Tasks may result in changes to the O&M manual or identification of ventilation system deficiencies that require corrective action. Failure to perform tasks, make repairs, or correct deficiencies may result in less than *acceptable IAQ*.

9. NORMATIVE REFERENCES

- FGI. 2022. *Guidelines for Design and Construction of Outpatient Facilities*. Florence, MA: Facility Guidelines Institute. (<https://fgiguideines.org>).
- GPO. 2024. 40 CFR 50, *National Primary and Secondary Ambient Air Quality Standards*. Washington, DC: U.S. Government Printing Office.
- ASHRAE. 2024. ANSI/ASHRAE Standard 111-2024, *Measurement, Testing, Adjusting, and Balancing of Building HVAC Systems*. Peachtree Corners, GA: ASHRAE.
- NFPA. 2024. ANSI Z223.1/NFPA 54. *National Fuel Gas Code*. Quincy, MA: National Fire Protection Association.

5. NFPA. 2024. NFPA 31. *Standard for the Installation of Oil-Burning Equipment*. Quincy, MA: National Fire Protection Association.
6. NFPA. 2024. NFPA 211. *Standard for Chimneys, Fireplaces, Vents, and Solid Fuel-Burning Appliances*. Quincy, MA: National Fire Protection Association.
7. ASSP. 2022. ANSI/ASSP Z9.5, *Standard for Laboratory Ventilation*. Park Ridge, IL: American Society of Safety Professionals.
8. UL. 2015. ANSI/UL 1995, *Heating and Cooling Equipment*, 5th Edition. Northbrook, IL: Underwriters Laboratories, LLC.
9. IEC. 2022. IEC 60355-2-40, Household and Similar Electrical Appliances—Safety—Part 2-40: Particular Requirements for Electrical Heat Pumps, Air-Conditioners and Dehumidifiers, 4th Edition. Geneva, Switzerland: International Electrotechnical Commission.
10. AMCA. 2023. AMCA 500-L-23, *Laboratory Methods of Testing Louvers for Rating*. Arlington Heights, IL: Air Movement and Control Association International, Inc.
11. AMCA. 2022. AMCA Publication 511-21 (Rev. 12-22), *Certified Ratings Program—Product Rating Manual for Air Control Devices*. Arlington Heights, IL: Air Movement and Control Association International, Inc.
12. AMCA. 2022. ANSI/AMCA Standard 550-22, *Test Method for High Velocity Wind Driven Rain Resistant Louvers*. Arlington Heights, IL: Air Movement and Control Association International, Inc.
13. ASHRAE. 2017. ANSI/ASHRAE Standard 52.2, *Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size*. Peachtree Corners, GA: ASHRAE.
14. ISO. 2016. ISO 16890-1, *Air Filters for General Ventilation*. Geneva, Switzerland: International Organization for Standardization.
15. NSF. 2024. NSF/ANSI/CAN Standard 60, *Drinking Water Treatment Chemicals—Health Effects*. Ann Arbor, MI: NSF International.
16. GPO. 21 CFR 173.310, *Secondary Direct Food Additives Permitted in Food for Human Consumption—Boiler Water Additives*. Washington, DC: U.S. Government Printing Office.
17. UL. 2020. ANSI/UL 2998, *Environmental Claim Validation Procedure (ECVP) for Zero Ozone Emissions from Air Cleaners*, 3rd Edition. Northbrook, IL: Underwriters Laboratories, LLC.
18. UL. 2013. ANSI/UL 181, *Factory-Made Air Ducts and Air Connectors*, 11th Edition. Northbrook, IL: Underwriters Laboratories, LLC.
19. ASTM. 2019. ASTM C1338, *Standard Test Method for Determining Fungi Resistance of Insulation Materials and Facings*. West Conshohocken, PA: ASTM International.
20. ASTM. 2021. ASTM D3273, *Standard Test Method for Resistance to Growth of Mold on the Surface of Interior Coatings in an Environmental Chamber*. West Conshohocken, PA: ASTM International.
21. AHRI. 2023. AHRI 1060, *Performance Rating of Air-to-Air Exchangers for Energy Recovery Ventilation Equipment*. Arlington, VA: Air Conditioning, Heating and Refrigeration Institute.
22. SMACNA. 2020. ANSI/SMACNA 006, *HVAC Duct Construction Standards—Metal and Flexible*, 4th Edition. Chantilly, VA: Sheet Metal and Air Conditioning Contractors National Association.
23. ASHRAE. 2023. ASHRAE Standard 241, *Control of Infectious Aerosols*. Peachtree Corners, GA: ASHRAE.
24. ASHRAE. 2021. ANSI/ASHRAE Standard 188, *Legionellosis: Risk Management for Building Water Systems*. Peachtree Corners, GA: ASHRAE.
25. ASHRAE. 2021. ANSI/ASHRAE/ASHE Standard 170, *Ventilation for Health Care Facilities*. Peachtree Corners, GA: ASHRAE.
26. CEPA. 2015. *The Air Toxics Hot Spots Program Guidance Manual for Preparation of Health Risk Assessments*. Sacramento, CA: California Environmental Protection Agency (Office of Environmental Health Hazard Assessment [OEHHA]).
27. AgBB. 2024. *Requirements for the Indoor Air Quality in Buildings: Health-Related Evaluation Procedure for Emissions of Volatile Organic Compounds (VVOC, VOC and SVOC) from Building Products*. Dessau-Roßlau, Germany: Ausschuss zur gesundheitlichen Bewertung von Bauprodukten.
28. CARB. 2004. *Indoor Air Quality Guideline No. 1, Formaldehyde in the Home*. Sacramento, CA: California Air Resources Board.
29. ASHRAE. 2016. ANSI/ASHRAE Standard 145.2, *Laboratory Test Method for Assessing the Performance of Gas-Phase Air-Cleaning Systems: Air-Cleaning Devices*. Peachtree Corners, GA: ASHRAE.

30. ISO. 2013. ISO 10121-2, *Test methods for assessing the performance of gas-phase air cleaning media and devices for general ventilation—Part 2: Gas-phase air cleaning devices (GPACD)*. Geneva, Switzerland: International Organization for Standardization.
31. CIBSE. 2005. CIBSE AM10, *Natural Ventilation in Non-Domestic Buildings*. London, UK: Chartered Institution of Building Services Engineers.
32. ASHRAE. 2016. ANSI/ASHRAE Standard 154, *Ventilation for Commercial Cooking Operations*. Peachtree Corners, GA: ASHRAE.
33. SMACNA. 2021. *Fibrous Glass Duct Construction Standards*, 8th Edition. Chantilly, VA: Sheet Metal and Air Conditioning Contractors National Association.
34. NFPA. 2024. NFPA 90A, *Standard for the Installation of Air-Conditioning and Ventilating Systems*. Quincy, MA: National Fire Protection Association.
35. NFPA. 2024. NFPA 90B, *Standard for the Installation of Warm Air Heating and Air-Conditioning Systems*. Quincy, MA: National Fire Protection Association.
36. EPA. 1990. EPA IP-1, *Determination of Volatile Organic Compounds (VOCs) in Indoor Air in Compendium of Methods for the Determination of Air Pollutants in Indoor Air*. Washington, DC: U.S. Environmental Protection Agency.
37. EPA. 1999. EPA TO-17, *Determination of Volatile Organic Compounds in Ambient Air Using Active Sampling Onto Sorbent Tubes in Compendium of Methods for the Determination of Toxic Organic Compounds in Ambient Air*, 2nd Ed. Washington, DC: U.S. Environmental Protection Agency.
38. ISO. 2021. ISO 16000-6, *Indoor air—Part 6: Determination of volatile organic compounds in indoor and test chamber air by active sampling on Tenax TA sorbent, thermal desorption and gas chromatography using MS or MS-FID*. Geneva, Switzerland: International Organization for Standardization.
39. ISO. 2000. ISO 16017-1, *Indoor, ambient and workplace air—Sampling and analysis of volatile organic compounds by sorbent tube/thermal desorption/capillary gas chromatography—Part 1: Pumped sampling*. Geneva, Switzerland: International Organization for Standardization.
40. ISO. 2003. ISO 16017-2, *Indoor, ambient and workplace air—Sampling and analysis of volatile organic compounds by sorbent tube/thermal desorption/capillary gas chromatography—Part 2: Diffusive sampling*. Geneva, Switzerland: International Organization for Standardization.
41. ASTM. 2021. ASTM D5197, *Standard Test Method for Determination of Formaldehyde and Other Carbonyl Compounds in Air (Active Sampler Methodology)*. West Conshohocken, PA: ASTM International.
42. EPA. 1990. EPA IP-6, *Determination of Formaldehyde or other Aldehydes in Indoor Air in Compendium of Methods for the Determination of Air Pollutants in Indoor Air*. Washington, DC: U.S. Environmental Protection Agency.
43. ISO. 2022. ISO 16000-3, *Indoor air—Part 3: Determination of formaldehyde and other carbonyl compounds in indoor air and test chamber air—Active sampling method*. Geneva, Switzerland: International Organization for Standardization.
44. EPA. 1999. EPA TO-11A, *Determination of Formaldehyde in Ambient Air Using Adsorbent Cartridge Followed by High Performance Liquid Chromatography (HPLC) [Active Sampling Methodology] in Compendium of Methods for the Determination of Toxic Organic Compounds in Ambient Air*, 2nd Edition. Washington, DC: U.S. Environmental Protection Agency.
45. EPA. 1990. EPA IP-3, *Determination of Carbon Monoxide (CO) or Carbon Dioxide (CO₂) in Indoor Air in Compendium of Methods for the Determination of Air Pollutants in Indoor Air*. Washington, DC: U.S. Environmental Protection Agency.
46. ISO. 2000. ISO 4224, *Ambient air—Determination of carbon monoxide—Non-dispersive infrared spectrometric method*. Geneva, Switzerland: International Organization for Standardization.
47. ASHRAE. 2018. ANSI/ASHRAE/ACCA Standard 180, *Standard Practice for Inspection and Maintenance of Commercial Building HVAC Systems*. Peachtree Corners, GA: ASHRAE.
48. NFPA. 2024. NFPA 45, *Standard on Fire Protection for Laboratories Using Chemicals*. Quincy, MA: National Fire Protection Association.

(This is a normative appendix and is part of the standard.)

NORMATIVE APPENDIX A MULTIPLE-ZONE SYSTEM VENTILATION EFFICIENCY: ALTERNATIVE PROCEDURE

This appendix presents an alternative procedure for calculating the system ventilation efficiency (E_v) for multiple-zone recirculating systems that must be used when Section 6.2.4.3 is not used. In this alternative procedure, E_v is equal to the lowest calculated value of the zone ventilation efficiency (E_{vz}) (see Equation A-2). Figure A-1 contains a ventilation system schematic depicting most of the quantities used in this appendix.

A1. SYSTEM VENTILATION EFFICIENCY

For any multiple-zone recirculating system, the system ventilation efficiency (E_v) shall be calculated in accordance with Sections A1.1 through A1.3.

A1.1 Average Outdoor Air Fraction. The average outdoor air fraction for the system (X_s) shall be determined in accordance with Equation A-1.

$$X_s = V_{out}/V_{ps} \quad (A-1)$$

where the uncorrected outdoor air intake (V_{out}) is found in accordance with Section 6.2.4.1, and the system primary airflow (V_{ps}) is found at the condition analyzed.

Informative Note: For variable-air-volume (VAV) system design purposes, V_{ps} is the highest expected system primary airflow at the design condition analyzed. System primary airflow at design is usually less than the sum of design zone primary airflow values, because primary airflow seldom peaks simultaneously in all VAV zones.

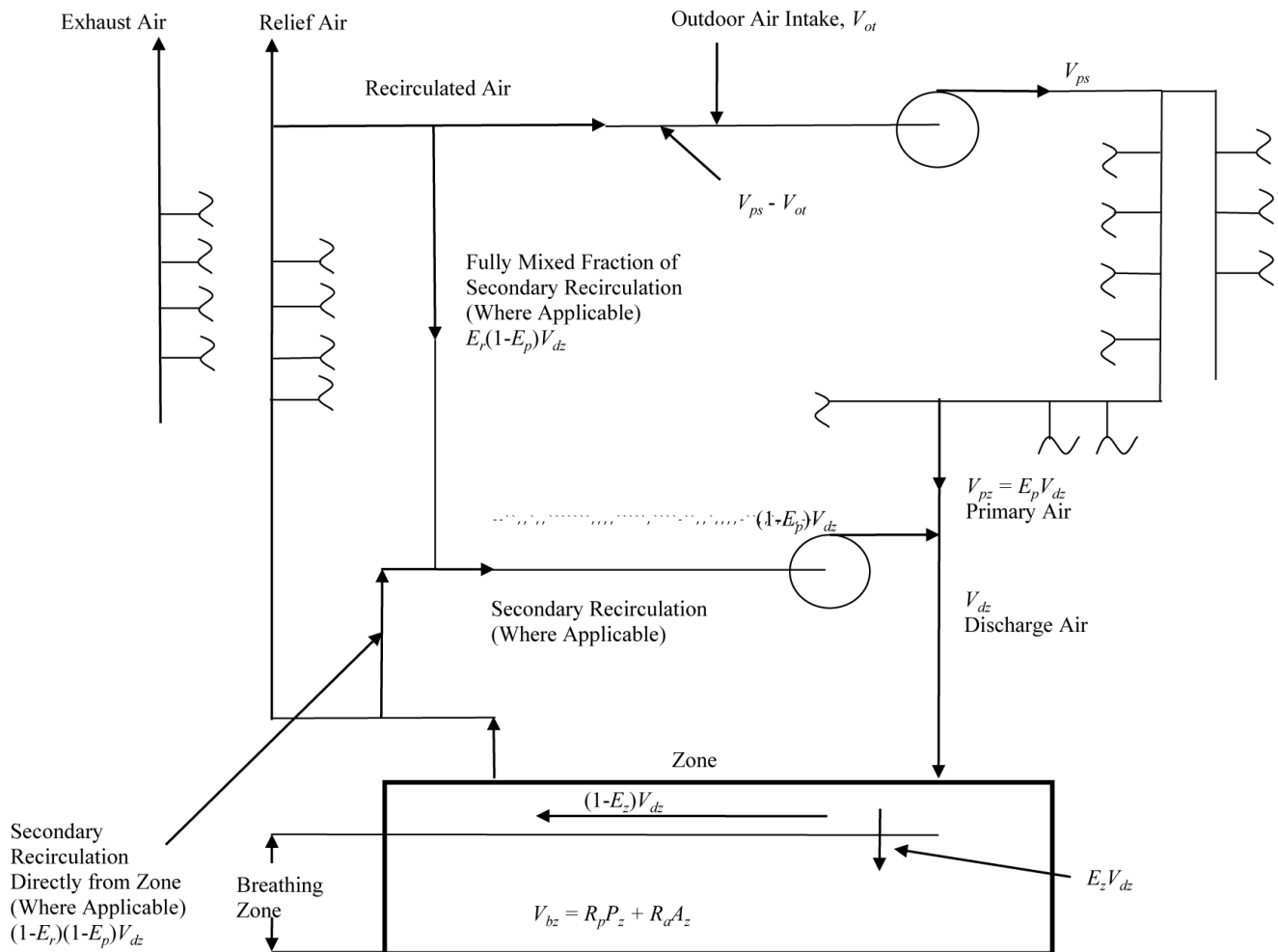


Figure A-1 Ventilation system schematic.

A1.2 Zone Ventilation Efficiency. The zone ventilation efficiency (E_{vz}) shall be determined in accordance with Sections A1.2.1 or A1.2.2.

A1.2.1 Single Supply Systems. For single supply systems, wherein all of the air supplied to each *ventilation zone* is a mixture of *outdoor air* and system-level *recirculated air*, zone ventilation efficiency (E_{vz}) shall be determined in accordance with Equation A-2. Examples of single supply systems include constant-air-volume reheat, single-duct VAV, single-fan dual-duct, and multiple-zone systems.

$$E_{vz} = 1 + X_s - Z_{pz} \quad (\text{A-2})$$

where the average outdoor air fraction for the system (X_s) is determined in accordance with Equation A-1, and the primary outdoor air fraction for the zone (Z_{pz}) is determined in accordance with Equation A-3.

$$Z_{pz} = V_{oz}/V_{pz} \quad (\text{A-3})$$

For VAV systems, V_{pz} is the lowest zone primary airflow value expected at the design condition analyzed.

A1.2.2 Secondary Recirculation Systems. For secondary recirculation systems wherein all or part of the *supply air* to each *ventilation zone* is *recirculated air* (air that has not been directly mixed with *outdoor air*) from other zones, zone ventilation efficiency (E_{vz}) shall be determined in accordance with Equation A-4. Examples of secondary recirculation systems include dual-fan dual-duct and fan-powered mixing-box systems and systems that include transfer fans for conference rooms.

$$E_{vz} = (F_a + X_s \times F_b - Z_{pz} \times E_p \times F_c)/F_a \quad (\text{A-4})$$

where system air fractions F_a , F_b , and F_c are determined in accordance with Equations A-5, A-6, and A-7, respectively.

$$F_a = E_p + (1 - E_p) \times E_r \quad (\text{A-5})$$

$$F_b = E_p \quad (\text{A-6})$$

$$F_c = 1 - (1 - E_z) \times (1 - E_r) \times (1 - E_p) \quad (\text{A-7})$$

Where the zone primary air fraction (E_p) is determined in accordance with Equation A-8, zone secondary recirculation fraction (E_r) is determined by the designer based on system configuration, and *zone air distribution effectiveness* (E_z) is determined in accordance with Section 6.2.1.2.

$$E_p = V_{pz}/V_{dz} \quad (\text{A-8})$$

where V_{dz} is zone discharge airflow.

Informative Notes:

1. For plenum return systems with secondary recirculation (e.g., fan-powered VAV with plenum return), E_r is usually less than 1.0, although values may range from 0.1 to 1.2 depending on the location of the *ventilation zone* relative to other zones and the air handler. For ducted return systems with secondary recirculation (e.g., fan-powered VAV with ducted return), E_r is typically 0.0, while for those with system-level recirculation (e.g., dual-fan dual-duct systems with ducted return), E_r is typically 1.0. For other system types, E_r is typically 0.75.
2. For single-zone and single-supply systems, E_p is 1.0.

A1.3 System Ventilation Efficiency. The system ventilation efficiency shall equal the lowest zone ventilation efficiency among all *ventilation zones* served by the air handler in accordance with Equation A-9.

$$E_v = \text{minimum}(E_{vz}) \quad (\text{A-9})$$

A2. DESIGN PROCESS

The system ventilation efficiency and, therefore, the outdoor air intake flow (V_{ot}) for the system determined as part of the design process are based on the design and minimum expected supply airflows to individual *ventilation zones* as well as the design zone outdoor airflow (V_{oz}) requirements to the zones. For VAV system design purposes, zone ventilation efficiency (E_{vz}) for each *ventilation zone* shall be found using the minimum expected zone primary airflow (V_{pz}) and using the highest expected system primary airflow (V_{ps}) at the design condition analyzed.

Informative Note: Increasing the zone supply airflow values during the design process, particularly to the critical zones requiring the highest fraction of *outdoor air*, reduces the system *outdoor air* intake flow requirement determined in the calculation.

A2.1 Selecting Zones for Calculation. Zone ventilation efficiency (E_{vz}) shall be calculated for all *ventilation zones*.

Exception to A2.1: Because system ventilation efficiency (E_v) is determined by the minimum value of the zone ventilation efficiency (E_{vz}) in accordance with Equation A-9, calculation of E_{vz} is not required for any *ventilation zone* that has an E_{vz} value that is equal to or larger than that of the *ventilation zone* for which a calculation has been made.

Informative Note: The value of E_{vz} for a *ventilation zone* will be equal to or larger than that for another *ventilation zone* if all of the following are true relative to the other *ventilation zone*:

- Floor area per occupant (A_z/P_z) is no lower.
- Minimum zone discharge airflow rate per unit area (V_{dz}/A_z) is no lower.
- Primary air fraction (E_p) is no lower.
- Zone air distribution effectiveness (E_z) is no lower.
- Flow rate of *outdoor air* required per unit area (R_a) is no higher.
- Flow rate of *outdoor air* required per person (R_p) is no higher.

A3. SYMBOLS

A_z	= zone floor area: net occupiable floor area of the <i>ventilation zone</i> , ft ² (m ²).
D	= occupant diversity: ratio of the system population to the sum of the zone populations.
E_p	= primary air fraction: fraction of <i>primary air</i> in the discharge air to the <i>ventilation zone</i> .
E_r	= secondary recirculation fraction: in systems with secondary recirculation of <i>return air</i> , the fraction of secondary <i>recirculated air</i> to the zone that is representative of average system <i>return air</i> rather than air directly recirculated from the zone.
E_v	= system ventilation efficiency: efficiency with which the system distributes air from the outdoor air intake to the <i>breathing zone</i> in the <i>ventilation-critical zone</i> , which requires the largest fraction of <i>outdoor air</i> in the primary airstream.
E_{vz}	= zone ventilation efficiency: efficiency with which the system distributes air from the outdoor air intake to the <i>breathing zone</i> in any particular <i>ventilation zone</i> .
E_z	= zone air distribution effectiveness: a measure of the effectiveness of <i>supply air</i> distribution to the <i>breathing zone</i> . E_z is determined in accordance with Section 6.2.1.2 or Normative Appendix C.
F_a	= supply air fraction: fraction of <i>supply air</i> to the <i>ventilation zone</i> that includes sources of air from outside the zone.
F_b	= mixed-air fraction: fraction of <i>supply air</i> to the <i>ventilation zone</i> from fully mixed <i>primary air</i> .
F_c	= outdoor air fraction: fraction of <i>outdoor air</i> to the <i>ventilation zone</i> that includes sources of air from outside the zone.
P_s	= system population: simultaneous number of occupants in the area served by the ventilation system.
P_z	= zone population: see Section 6.2.1.1.
R_a	= area outdoor air rate: see Section 6.2.1.1.
R_p	= people outdoor air rate: see Section 6.2.1.1.
$\dot{V}_{bz}$	= breathing zone outdoor airflow: see Section 6.2.1.1.
V_{dz}	= zone discharge airflow: expected discharge (supply) airflow to the zone that includes primary airflow and secondary recirculated airflow, cfm (L/s).
V_{ot}	= outdoor air intake flow: see Sections 6.2.2, 6.2.3, and 6.2.4.4.
V_{ou}	= uncorrected outdoor air intake: see Section 6.2.4.1.
V_{oz}	= zone outdoor airflow: see Section 6.2.1.4.
V_{ps}	= system primary airflow: total primary airflow supplied to all zones served by the system from the air-handling unit at which the outdoor air intake is located.
V_{pz}	= zone primary airflow: zone primary airflow to the <i>ventilation zone</i> , including <i>outdoor air</i> and <i>recirculated air</i> .
X_s	= average outdoor air fraction: at the primary air handler, the fraction of outdoor air intake flow in the system primary airflow.
Z_{pz}	= primary outdoor air fraction: <i>outdoor air</i> fraction required in the <i>primary air</i> supplied to the <i>ventilation zone</i> prior to the introduction of any secondary recirculation air.

(This is a normative appendix and is part of the standard.)

NORMATIVE APPENDIX B SEPARATION OF EXHAUST OUTLETS AND OUTDOOR AIR INTAKES

B1. GENERAL

This appendix presents an alternative procedure for determining separation distance between outdoor air intakes and *exhaust air* and vent outlets. This analytical method can be used instead of Table 5-1.

Exhaust air and vent outlets, as defined in Table 5-1, shall be located no closer to outdoor air intakes, or to operable windows, skylights, and doors, both those on the subject property and those on adjacent properties, than the minimum separation distance (L) specified in this section. The distance (L) is defined as the shortest “stretched string” distance measured from the closest point of the outlet opening to the closest point of the *outdoor air* intake opening, or to the operable window, skylight, or door opening, along a trajectory as if a string were stretched between them.

B1.1 Application. Exhaust outlets and outdoor air intakes or other openings shall be separated in accordance with Section B2.

Exceptions to B1.1:

1. Laboratory fume hood *exhaust air* outlets shall be in compliance with NFPA 45⁴⁸ and ANSI/ASSP Z9.5.⁷
2. Laboratory and industrial ventilation process exhausts are not addressed by this procedure.
3. Large, industrial-sized combustion flues and stacks are not addressed by this procedure.
4. Packaged units that have integral exhaust and intake locations are not addressed by this procedure.
5. *Evaporative heat rejection equipment* is not addressed by this procedure.
6. The minimum separation distances for heat rejection equipment shall conform to Table 5-1.
7. Separation distances do not apply when exhaust and outdoor air intake systems are controlled such that they cannot operate simultaneously.

B2. DETERMINING DISTANCE L

B2.1 General Equations. The minimum separation distance (L) shall be calculated using Equation B-1 through B-5b.

Informative Note: A spreadsheet that assists with Normative Appendix B calculations can be downloaded at www.ashrae.org/pub9373452.

$$F1 = 13.6 \frac{DF \times Q}{U_H} \quad (\text{B-1})$$

$$F2 = 33.37h_s^2 + 254.9\beta \frac{B_{fac} \times h_s \times Q}{d_e \times U_H} + 486.9\beta \left(\frac{B_{fac} \times Q}{d_e \times U_H} \right)^2 \quad (\text{B-2})$$

$$B_{fac} = \left\{ 1 + \left[\frac{15(T_s - T_a)T_s}{T_a^2 \times U_H \times V_e} \right] \right\}^{0.5} \quad (\text{I-P}) \quad (\text{B-3a})$$

$$B_{fac} = \left\{ 1 + \left[\frac{580,000(T_s - T_a)T_s}{T_a^2 \times U_H \times V_e} \right] \right\}^{0.5} \quad (\text{SI}) \quad (\text{B-3b})$$

$$V_e = Q/(\pi d_e^2/4) \quad (\text{B-4})$$

Find the maximum of $F1 - F2$ by varying U_H between 300 fpm (1.5 m/s) and the maximum wind speed. If local wind speed data is not available, U_H shall be varied from 300 fpm (1.5 m/s) to 2000 fpm (10 m/s).

$$\text{If the maximum of } [F1 - F2] > 0, \text{ then } L = [F1 - F2]^{0.5} \quad (\text{B-5a})$$

$$\text{If the maximum of } [F1 - F2] \leq 0, \text{ then } L = 0 \quad (\text{B-5b})$$

where:

Q = exhaust airflow rate, cfm (m^3/s). For variable flow, use the minimum airflow rate. For gravity vents, such as plumbing vents, use an exhaust rate of 150 cfm (0.0075 m^3/s). For flue vents from fuel-burning appliances, assume a value of 250 cfm per million Btu/h (0.00043 m^3/s per kW) of combustion input (or obtain actual rates from the combustion appliance manufacturer)

DF = dilution factor, which is the ratio of outdoor airflow to entrained exhaust airflow in the outdoor air intake; the minimum dilution factor shall be determined as a function of exhaust air class in Table B-1

L = minimum separation distance, ft (m) (stretched string as shown in Figure B-1)

Table B-1 Minimum Dilution Factors, DF

Exhaust Type	Minimum Dilution Factor, DF
Class 1 air exhaust/relief outlet	5
Class 2 air exhaust/relief outlet	10
Class 3 air exhaust/relief outlet	50
Class 4 air exhaust/relief—based on kitchen grease hoods	300
Wood burning kitchen exhaust	700
General boilers, natural gas and fuel oil—based on NO _x ppm factor ^a	$2.8 \times p$
Garage entry, automobile loading area, or drive-in queue (light-duty gasoline vehicles)	50
Diesel generators, diesel truck loading area or dock, diesel bus parking/idling area ^b	$2000 \times e$

a. p is the maximum NO_x ppm emitted from the boiler flue. If the maximum NO_x ppm is 10 ppm, $p = 10$ and $DF = 28$.

b. $e = 1$ – the VOC removal efficiency of the odor filter as defined by manufacturer or the cognizant authority (e.g. if the filter is 80% efficient, $e = 0.2$ and $DF = 400$).

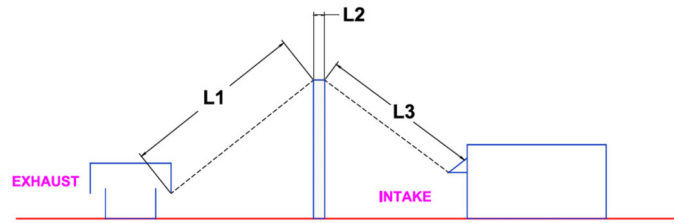


Figure B-1 Diagram showing how to calculate string distance, L . In the figure $L = L1 + L2 + L3$.

- U_H = wind speed at stack top, fpm (m/s)
 T_s = exhaust temperature, °R (K)
 T_a = ambient temperature, °R (K)
 h_s = stack height above or below the top of the air intake (ft [m]). Height above the top of the air intake is a positive number; height below the top of the air intake is a negative number.
 d_e = exhaust diameter (ft [m]); for rectangular exhaust (capped, horizontal, or vertical), an equivalent round stack diameter shall be calculated using the following equation:

$$d_{e,eff} = [\text{Exhaust Area} \times 4/\pi]^{0.5} \quad (\text{B-6})$$

for louvered round or louvered rectangular exhaust (capped, horizontal, or vertical), an equivalent round stack diameter should be calculated using the following equation:

$$d_{e,eff} = [\text{Exhaust Area} \times \text{Open Fraction} \times 4/\pi]^{0.5} \quad (\text{B-7})$$

for heated capped or heated horizontal exhaust (including louvered), the exhaust diameter prescribed in Section B2.2.5 applies

- β = 1 for uncapped stacks and 0 for capped or horizontal (including louvered) exhaust

Equation B-8a (I-P) or B-8b (SI) shall be used to determine whether the exhaust is considered heated:

$$T_c = 0.00626 \left(\frac{v_s^{1/3}}{d_{e,eff}^{2/3}} \right) T_s \quad (\text{I-P}) \quad (\text{B-8a})$$

$$T_c = 0.0297 \left(\frac{v_s^{1/3}}{d_{e,eff}^{2/3}} \right) T_s \quad (\text{SI}) \quad (\text{B-8b})$$

where

- T_c = the crossover temperature difference in (°R [K]); if $T_c < T_s - T_a$, the exhaust is considered to be heated
 v_s = exhaust air discharge velocity (fpm [m/s])

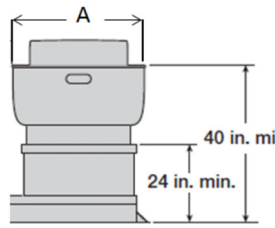


Figure B-2 Typical upblast exhaust.

$$v_s = \frac{Q_e}{d_{e,eff}}$$

If the exhaust is not considered heated, the exhaust temperature shall be equal to the ambient temperature. Both summer and winter design conditions shall be evaluated.

Informative Note: Equation B-8 is derived from the EPA’s User’s Guide for the Industrial Source Complex (ISC3) Dispersion Models,^{P7} Equation 1-10.

B2.2 Special Cases. When a special case occurs, the alternate compliance conditions and equations in the following sections are permitted to be used to determine input values instead of those prescribed in Section B2.1.

B2.2.1 Horizontal Exhaust Pointed Away from Intake. When an exhaust is pointed away from an intake and the wind is blowing toward the intake, the exhaust travels some direction upwind and then turns around.

Informative Note: The upwind distance traveled depends on the ratio of exhaust velocity to wind speed (velocity ratio). The plume is also diluted as it travels upwind. For small velocity ratios, the exhaust turns around quickly (within $0.5d_e$ for a velocity ratio of 0.5). For high velocity ratios, the plume travels upwind for a larger distance ($6d_e$ for a velocity ratio of 5).

B2.2.1.1 Pointed Away. “Pointed away” includes cases where the direction of the exhaust is oriented $180 \text{ degrees} \pm 45 \text{ degrees}$ away from the intake.

B2.2.1.2 Allowable Adjustments. Equations B-9 through B-11 shall be used to adjust the values in Section B2.1.

$$U_{H,pa} = V_e \quad (\text{B-9})$$

$$DF_{pa} = DF/1.7 \quad (\text{B-10})$$

$$L_{pa} = L/(1.75 \times d_e) \quad (\text{B-11})$$

B2.2.2 Upblast Exhaust. For upblast exhaust (typically used for kitchen exhaust), the effective exhaust velocity is computed using the dimension “A” for d_e (see Figure B-2) and the exhaust volume flow rate along with Equation B-4.

B2.2.3 Downblast Exhaust. Downblast exhaust (e.g. “mushroom” exhausters) are treated the same as a capped exhaust stack. The input exhaust diameter is “A,” as shown in Figure B-2.

Informative Note: If the downblast stack is heated, the method in Section B2.2.5 is permitted to be used.

B2.2.4 Hidden Intakes. A hidden intake is one that cannot be seen if standing at the exhaust location. A hidden intake must be at a similar elevation as the exhaust and on a building sidewall. For hidden intakes, the minimum dilution factor from Table B-1 shall be divided by 2, as shown in Equation B-12:

$$DF_h = DF/2 \quad (\text{B-12})$$

Informative Note: Typically, hidden intakes are located on building sidewalls or on the side of a large mechanical penthouse or unit.

B2.2.5 Capped Heated Exhaust. Capped stacks or horizontal louvered exhausts that are heated will have plume rise due to buoyancy effects. For capped heated exhaust, Equations B-13 and B-14 shall be used to calculate the value of $F2$ in Equation B-2.

$$d_e = d_{e,capheat} = 10 d_{e,eff} \quad (\text{B-13})$$

$$\beta = 1 \quad (\text{B-14})$$

(This is a normative appendix and is part of the standard.)

NORMATIVE APPENDIX C

ZONE AIR DISTRIBUTION EFFECTIVENESS: ALTERNATIVE PROCEDURES

This appendix provides a procedure for determining *zone air distribution effectiveness* (E_z) for all system types.

Informative Note: Table 6-4 provides default values of E_z that are permitted to be used for the air distribution configurations described in the table. The reference E_z value of 1 is typical of ideal mixing in the zone.

The strategy of removing *contaminants* or displacing *contaminants* from the *breathing zone* may result in an effective E_z value greater than unity, which is typical of stratified systems.

C1. ZONE AIR DISTRIBUTION EFFECTIVENESS

Zone air distribution effectiveness shall be calculated in accordance with Equation C-1:

$$E_z = (C_e - C_s)/(C - C_s) \quad (\text{C-1})$$

where

E_z = *zone air distribution effectiveness*

C = *average contaminant concentration at the breathing zone*

C_e = *average contaminant concentration at the exhaust*

C_s = *average contaminant concentration at the supply*

C1.1 Personalized Ventilation Systems. For the purpose of calculating *zone air distribution effectiveness* for personalized ventilation systems, the *breathing zone* shall be 9 ft² (0.8 m²) centered on each occupant with a height of 4.5 ft (1.4 m) from the floor.

C2. MODELED AIR DISTRIBUTION SYSTEM

C2.1 Computational Model. The computational fluid dynamics model for calculating *zone air distribution effectiveness* shall be in accordance with Sections C2.1.1 through C2.1.7.

C2.1.1 Computational Domain. The computational domain shall comprise all sensible heat sources, all major obstructions to airflow, and all air distribution devices. The calculation domain shall include all boundary walls.

C2.1.2 Solution Variables. Analysis shall include the solutions for fluid flow, heat transfer, and chemical species transport. The buoyancy (gravitational) effects shall be included in the calculation procedure.

C2.1.3 Boundary Conditions. Sensible heat sources shall be permitted to be modeled as volumetric heat sources to allow the air to pass through the source or as hollow blocks (no mesh inside) specified with either heat flux or constant temperature on the surfaces of the blocks. Boundary walls shall be modeled as adiabatic (zero heat flux), specified heat flux, or specified temperature boundary.

C2.1.4 Species Transport. The sources shall be modeled as volumetric source or a boundary flux with known generation rate with zero release velocity. The analysis shall be performed with a uniformly distributed source at the *breathing zone* level of the occupants. All the boundary walls shall be modeled as impermeable to the chemical species.

Informative Note: The species modeled should be a tracer gas, such as CO₂. Discretion is left to the modeler to determine the appropriate model depending on the *design compounds* in the zone.

C2.1.5 Turbulence Model. Reynolds (ensemble) averaging turbulence models shall be used.

Informative Note: Renormalization group and realizable k-ε models meet the requirements of this section.

C2.1.6 Computational Mesh. A fine mesh shall be generated near the sensible heat sources, such as occupants and computers, to resolve the thermal plume surrounding these sources. The fine mesh shall be generated on all *supply air* and *return air* locations.

C2.1.7 Solution Convergence. The solution convergence levels shall include the monitoring of relevant physical quantities, such as temperature or species *concentration*, at strategic locations. The globally scaled residuals shall be decreased to 10⁻³ for all equations except the energy and species equations, for which the residuals shall be decreased to 10⁻⁷. The mass and energy balance shall be calculated up to at least four (4) decimal places.

Informative Note: Refer to ASHRAE Research projects RP-1009^{P8} and RP-1373^{P9} for additional considerations related to computational models.

C2.2 Zone Air Distribution Effectiveness. *Zone air distribution effectiveness* (E_z) shall be computed in accordance with Equation C-1 for each computational cell in the *breathing zone*. The *zone air distribution effectiveness* of the system shall be the average value of the *zone air distribution effectiveness* of each com-

putational cell within the *breathing zone*. The analysis shall be performed for both summer cooling conditions and winter heating conditions.

Informative Note: Validation of the computational model with physical measurements during design can improve the accuracy of the computational model and the *zone air distribution effectiveness* of the system. Field measurements could also be performed post building occupancy to verify *zone air distribution effectiveness*.

(This is a normative appendix and is part of the standard.)

NORMATIVE APPENDIX D

AIR DENSITY CORRECTION FACTOR

This appendix presents an alternative procedure for determining the air density correction factor (E_ρ). This analytical method can be used instead of Table 6-5. The volumetric airflow rates in this standard are based on a dry-air density of 0.075 lb/ft³ (1.2 kg/m³) at a barometric pressure of 1 atm (101.3 kPa) and an air temperature of 70°F (21°C).

D1. AIR DENSITY CORRECTION FACTOR

The air density correction factor (E_ρ) shall be calculated in accordance with Sections D1.1 through D1.5.

D1.1 Design Condition. The air density correction factor (E_ρ) shall be calculated at the design condition with the lowest coincident air density.

D1.2 Elevation Adjustment. The elevation adjustment factor (C_Z) shall be calculated in accordance with Equation D-1a (I-P) or D-1b (SI).

$$C_Z = 1 / (1 - Z \times 6.8754 \times 10^{-6})^{5.2559} \quad (\text{D-1a})$$

$$C_Z = 1 / (1 - Z \times 2.25577 \times 10^{-5})^{5.2559} \quad (\text{D-1b})$$

where Z is the outdoor air intake elevation above sea level, ft (m).

D1.3 Temperature Adjustment. The temperature adjustment factor (C_T) shall be calculated in accordance with Equation D-2a (I-P) or D-2b (SI).

$$C_T = (T + 459.67) / 529.67 \quad (\text{D-2a})$$

$$C_T = (T + 273.15) / 294.15 \quad (\text{D-2b})$$

where T is the outdoor air temperature at design condition, °F (°C).

D1.3.1 The temperature adjustment factor (C_T) may be taken as 1.0 where the outdoor air temperature at design conditions is less than 104°F (40°C).

D1.4 Moisture Adjustment. The moisture adjustment factor shall be calculated in accordance with Equation D-3.

$$C_W = (1 + W) / (1 + 1.6078 \times W) \quad (\text{D-3})$$

where W is the humidity ratio at design conditions, lb moisture per lb dry air (kg moisture per kg dry air).

D1.4.1 The moisture adjustment factor may be taken as 1.0 where the outdoor air humidity ratio at design conditions is less than 0.024 lb_w/lb_{da} (0.024 kg_w/kg_{da}).

D1.5 Air Density Correction. The air density correction factor (E_ρ) shall be calculated in accordance with Equation D-4.

$$E_\rho = C_Z \times C_T \times C_W \quad (\text{D-4})$$

D2. ALTERNATIVE CALCULATION METHODS

The use of other calculation methods, including computer simulation, to determine the air density correction factor shall be permitted.

D2.1 Design Condition. The air density shall be calculated at the design condition with the lowest coincident air density.

D2.2 Calculation Parameters. The calculation method shall include elevation above sea level, design air temperature, and design moisture content as factors contributing to the air density.

D2.2.1 Pressurized Outdoor Air Intakes. Where *outdoor air* is supplied to the ventilation system under pressure and the pressure is maintained at all times that the ventilation system is operating, the air density calculation may include the static pressure at the ventilation system outdoor air intake.

D2.3 Air Density Correction. The air density correction factor (E_ρ) shall be calculated in accordance with Equation D-5a (I-P) or D-5b (SI).

$$E_\rho = 0.075 / \rho \quad (\text{D-5a})$$

$$E_\rho = 1.2 / \rho \quad (\text{D-5b})$$

where ρ is the minimum outdoor air density at design conditions, lb_{da}/ft³ (kg_{da}/m³).

D2.3.1 The air density correction factor (E_ρ) may be taken as 1.0 where the air density at design conditions is greater than 0.072 lb_{da}/ft³ (1.1 kg_{da}/m³).

(This is a normative appendix and is part of the standard.)

NORMATIVE APPENDIX E VENTILATION RATES FOR OUTPATIENT FACILITIES NOT COVERED BY ASHRAE/ASHE STANDARD 170

E1. GENERAL

This appendix presents minimum ventilation rates in *breathing zones* for spaces in outpatient facilities to which the *authority having jurisdiction* deemed ANSI/ASHRAE/ASHE Standard 170²⁵ not applicable.

E2. VENTILATION RATES

For zones in outpatient facilities not covered by ASHRAE/ASHE Standard 170, *ventilation zone* parameters shall be determined in accordance with Sections 6.2.1.1 through 6.2.1.4 using the rates in Table E-1.

Table E-1 Minimum Ventilation Rates in Breathing Zone

Occupancy Category	People Outdoor		Area Outdoor		Default Values	Air Class	OS (6.2.6.2)
	Air Rate (R_p)		Air Rate (R_a)		Occupant Density		
	cfm/ person	L/s· person	cfm/ft ²	L/s·m ²	#/1000 ft ² or #/100 m ²		
Outpatient Health Care Facilities ^{a,b}							
Birthing room	10	5	0.18	0.9	15	2	
<i>Class 1 imaging rooms</i>	7.5	3.8	0.12	0.6	5	1	
Dental operator	10	5	0.18	0.9	20	1	
General examination room	7.5	3.8	0.12	0.6	20	1	
Other dental treatment areas	5	2.5	0.06	0.3	5	1	
Physical therapy exercise area	20	10	0.18	0.9	7	2	
Physical therapy individual room	10	5	0.12	0.6	20	2	
Physical therapeutic pool area	—	—	0.48	2.4	—	3	
Prosthetics and orthotics room	10	5	0.18	0.9	20	2	
Psychiatric consultation room	5	2.5	0.06	0.3	20	1	
Psychiatric examination room	5	2.5	0.06	0.3	20	1	
Psychiatric group room	5	2.5	0.06	0.3	50	1	
Psychiatric seclusion room	10	5	0.12	0.6	5	1	
Speech therapy room	5	2.5	0.06	0.3	20	1	
Urgent care examination room	7.5	3.8	0.12	0.6	20	2	
Urgent care observation room	5	2.5	0.06	0.3	20	2	
Urgent care treatment room	7.5	3.8	0.12	0.6	20	2	
Urgent care triage room	10	5	0.18	0.9	20	3	

a. The requirements of this table provide for *acceptable IAQ*. The requirements of this table do not address the airborne transmission of viruses, bacteria, and other infectious contagions.

b. These rates are intended only for outpatient dental clinics where the amount of nitrous oxide is limited. They are not intended for dental operatories in institutional buildings where nitrous oxide is piped.

(This appendix is not part of this standard. It is merely informative and does not contain requirements necessary for conformance to the standard. It has not been processed according to the ANSI requirements for a standard and may contain material that has not been subject to public review or a consensus process. Unresolved objectors on informative material are not offered the right to appeal at ASHRAE or ANSI.)

INFORMATIVE APPENDIX F

INFORMATION ON SELECTED NATIONAL STANDARDS AND GUIDELINES FOR PM₁₀, PM_{2.5}, AND OZONE

Section 4 requires that the status of compliance with ambient air quality standards be determined for the geographical area of the building site. Table F-1 is a representative table presenting the NAAQS information for the United States. Links to detailed information on the ambient air quality standards and *contaminant* levels for other select counties and regions are as follows:

- U.S. NAAQS:^{P10} www.epa.gov/green-book and www.epa.gov/criteria-air-pollutants/naaqs-table
- Canadian Ambient Air Quality Standards: www.ccme.ca/en/air-quality-report#slide-7
- Hong Kong Air Quality Objectives: www.epd.gov.hk/epd/english/environmentinhk/air/air_quality_objectives/air_quality_objectives.html
- Singapore Air quality Targets: www.nea.gov.sg/our-services/pollution-control/air-pollution/air-quality
- European Commission Air Quality Standards: ec.europa.eu/environment/air/quality/standards.htm
- Brazil Air Quality Standards: www.transportpolicy.net/index.php?title=Brazil:_Air_Quality_Standards
- World Health Organization (WHO) Air Quality Guideline Values: [www.who.int/news-room/fact-sheets/detail/ambient-\(outdoor\)-air-quality-and-health](http://www.who.int/news-room/fact-sheets/detail/ambient-(outdoor)-air-quality-and-health)

The Clean Air Act (www.epa.gov/clean-air-act-overview), which was last amended in 1990, requires the United States Environmental Protection Agency (U.S. EPA) to set NAAQS (40 CFR part 50) for pollutants considered harmful to public health and the environment. The Clean Air Act identifies two types of NAAQS. *Primary standards* provide public health protection, including protecting the health of sensitive populations, such as asthmatics, children, and the elderly. *Secondary standards* provide public welfare protection, including protection against decreased visibility and damage to animals, crops, vegetation, and buildings.

U.S. EPA has set NAAQS for six principal pollutants, which are called “criteria” air pollutants (www.epa.gov/criteria-air-pollutants). Periodically, the standards are reviewed and may be revised. The current standards are listed in Table F-1. Units of measure for the standards are parts per million (ppm) by volume, parts per billion (ppb) by volume, and micrograms per cubic metre of air ($\mu\text{g}/\text{m}^3$).

Table F-1 NAAQS for the United States
(www.epa.gov/criteria-air-pollutants/naaqs-table)

Pollutant		Primary/ Secondary	Averaging Time	Level	Form
Carbon Monoxide (CO) www.epa.gov/co-pollution/table-historical-carbon-monoxide-co-national-ambient-air-quality-standards-naaqs		Primary	Eight (8) hours	9 ppm	Not to be exceeded more than once per year
			One (1) hour	35 ppm	
Lead (Pb) www.epa.gov/lead-air-pollution/table-historical-lead-pb-national-ambient-air-quality-standards-naaqs		Primary and secondary	Rolling three (3) month average	0.15 µg/m ³ (Note 1)	Not to be exceeded
Nitrogen Dioxide (NO ₂) www.epa.gov/no2-pollution/table-historical-nitrogen-dioxide-national-ambient-air-quality-standards-naaqs		Primary	One (1) hour	100 ppb	Ninety-eighth (98th) percentile of one-hour daily maximum <i>concentrations</i> , averaged over three years
		Primary and secondary	One (1) year	53 ppb (Note 2)	Annual mean
Ozone (O ₃) www.epa.gov/ozone-pollution/table-historical-ozone-national-ambient-air-quality-standards-naaqs		Primary and secondary	Eight (8) hours	0.070 ppm (Note 3)	Annual fourth-highest daily maximum eight-hour <i>concentration</i> , averaged over three years
Particulate Matter (PM) www.epa.gov/pm-pollution/table-historical-particulate-matter-pm-national-ambient-air-quality-standards-naaqs	PM _{2.5}	Primary	One (1) year	9.0 µg/m ³	Annual mean, averaged over three years
		Secondary	One (1) year	15.0 µg/m ³	Annual mean, averaged over three years
		Primary and secondary	Twenty-four (24) hours	35 µg/m ³	Ninety-eight (98th) percentile, averaged over three years
	PM ₁₀	Primary and secondary	Twenty-four (24) hours	150 µg/m ³	Not to be exceeded more than once per year on average over three years
Sulfur Dioxide (SO ₂) www.epa.gov/so2-pollution/table-historical-sulfur-dioxide-national-ambient-air-quality-standards-naaqs		Primary	One (1) hour	75 ppb (Note 4)	Ninety-ninth (99th) percentile of one-hour daily maximum <i>concentrations</i> , averaged over three years
		Secondary	Three (3) hours	0.5 ppm	Not to be exceeded more than once per year

Notes:

1. In areas designated “nonattainment” for the Pb standards prior to the promulgation of the current (2008) standards, and for which implementation plans to attain or maintain the current (2008) standards have not been submitted and approved, the previous standards (1.5 µg/m³ as a calendar quarter average) also remain in effect.
2. The level of the annual NO₂ standard is 0.053 ppm. It is shown here in terms of ppb for the purpose of clearer comparison to the one-hour standard level.
3. Final rule signed October 1, 2015, and effective December 28, 2015. The previous (2008) O₃ standards additionally remain in effect in some areas. Revocation of the previous (2008) O₃ standards and transitioning to the current (2015) standards will be addressed in the implementation rule for the current standards.
4. The previous SO₂ standards (0.14 ppm 24-hour and 0.03 ppm annual) will additionally remain in effect in certain areas: (a) any area for which it is not yet one year since the effective date of designation under the current (2010) standards, and (b) any area for which an implementation plan providing for attainment of the current (2010) standard has not been submitted and approved and that is designated “nonattainment” under the previous SO₂ standards or is not meeting the requirements of an SIP call under the previous SO₂ standards (40 CFR 50.4[3]). An SIP call is an EPA action requiring a state to resubmit all or part of its State Implementation Plan to demonstrate attainment of the required NAAQS.

(This appendix is not part of this standard. It is merely informative and does not contain requirements necessary for conformance to the standard. It has not been processed according to the ANSI requirements for a standard and may contain material that has not been subject to public review or a consensus process. Unresolved objections on informative material are not offered the right to appeal at ASHRAE or ANSI.)

INFORMATIVE APPENDIX G

ACCEPTABLE MASS-BALANCE EQUATIONS FOR USE WITH THE IAQ PROCEDURE

When applying the Indoor Air Quality Procedure (IAQP) from Section 6.3, mass-balance analysis may be employed to determine ventilation requirements.

Table G-1 presents mass-balance equations for analysis of single-zone systems. Figures G-1 and G-2 show representative single-zone systems. A filter may be located in the recirculated airstream (location A) or in the supply (mixed) airstream (location B). The equations do not account for sources within the HVAC system that may occur, such as filter off-gassing, energy recovery carryover of specific gases, or generation of particles or compounds.

VAV single-zone systems reduce the circulation rate when the thermal load is lower than the design load. This is accounted for by a flow reduction fraction (F_r).

A mass-balance equation for each *design compound* or *PM2.5* may be written and used to determine the required outdoor airflow or the breathing zone resultant *concentration* for the various system arrangements. Six permutations for air-handling and single-zone air distribution systems are described in Table G-1. The mass-balance equations for computing the required outdoor airflow and the breathing-zone *contaminant concentration* at steady-state conditions for each single-zone system are presented in Table G-1.

If the allowable breathing zone design target is specified, the equations in Table G-1 may be solved for the zone outdoor airflow rate (V_{oz}). When the zone outdoor airflow rate is specified, the equations may be solved for the resulting *breathing zone design compound* or *PM2.5 concentration*.

While the calculation methods in this appendix are based on single-zone systems and steady-state analysis, calculation methods that account for multiple-zone and transient effects are also available (see Dols and Polidoro^{P13} in Informative Appendix P). An IAQP calculator that accounts for single-zone and multiple-zone systems, along with a user's guide, can be downloaded at www.ashrae.org/pub9373452.

Table G-1 Required Zone Outdoor Airflow or Space Breathing Zone Contaminant Concentration with Recirculation and Filtration for Single-Zone Systems

Required Recirculation Rate			Required Zone Outdoor Airflow (V_{oz} in Section 6)	Space Breathing Zone Contaminant Concentration
Filter Location	Flow	Outdoor Airflow		
None	VAV	100%	$V_{oz} = \frac{N}{E_z F_r (C_{bz} - C_o)}$	$C_{bz} = C_o + \frac{N}{E_z F_r V_{oz}}$
A	Constant	Constant	$V_{oz} = \frac{N - E_z R V_r E_f C_{bz}}{E_z (C_{bz} - C_o)}$	$C_{bz} = \frac{N + E_z V_{oz} C_o}{E_z (V_{oz} + R V_r E_f)}$
A	VAV	Constant	$V_{oz} = \frac{N - E_z F_r R V_r E_f C_{bz}}{E_z (C_{bz} - C_o)}$	$C_{bz} = \frac{N + E_z V_{oz} C_o}{E_z (V_{oz} + F_r R V_r E_f)}$
B	Constant	Constant	$V_{oz} = \frac{N - E_z R V_r E_f C_{bz}}{E_z [C_{bz} - (1 - E_f)(C_o)]}$	$C_{bz} = \frac{N + E_z V_{oz} (1 - E_f) C_o}{E_z (V_{oz} + R V_r E_f)}$
B	VAV	100%	$V_{oz} = \frac{N}{E_z F_r [C_{bz} - (1 - E_f)(C_o)]}$	$C_{bz} = \frac{N + E_z F_r V_{oz} (1 - E_f) C_o}{E_z F_r V_{oz}}$
B	VAV	Constant	$V_{oz} = \frac{N - E_z F_r R V_r E_f C_{bz}}{E_z [C_{bz} - (1 - E_f)(C_o)]}$	$C_{bz} = \frac{N + E_z V_{oz} (1 - E_f) C_o}{E_z (V_{oz} + F_r R V_r E_f)}$

Table G-2 Required Zone Outdoor Airflow or Space Breathing Zone Contaminant Concentration with Recirculation and Filtration for Single-Zone Systems (Continued)

Symbol or Subscript	Definition
A, B	filter location
V	volumetric flow
C	contaminant concentration
E_z	zone air distribution effectiveness
E_f	filter efficiency
F_r	design flow reduction fraction factor
N	contaminant generation rate
R	recirculation flow factor
Subscript: o	outdoor
Subscript: r	return
Subscript: b	breathing
Subscript: z	zone

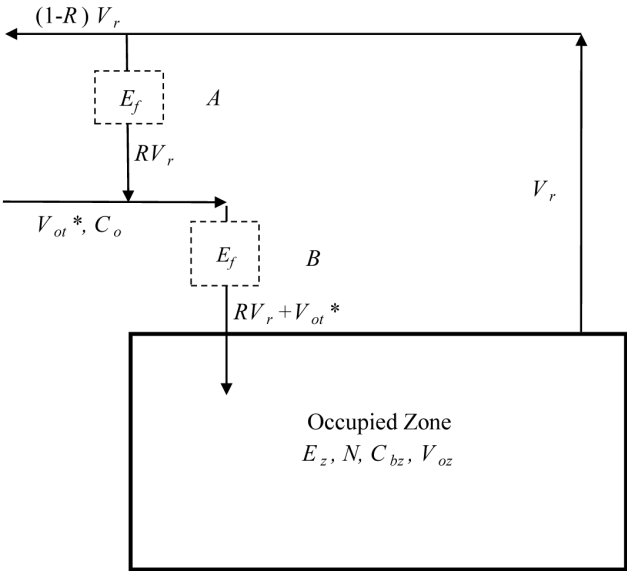


Figure G-1 Ventilation system schematic—constant-volume system with no infiltration/exfiltration. (* $V_{ot} = V_{oz}$ for single-zone systems.)

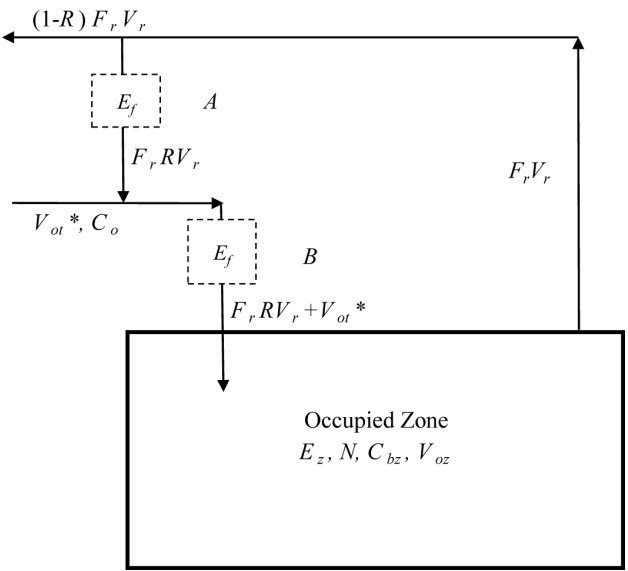


Figure G-2 Ventilation system schematic—variable-air-volume system with no infiltration/exfiltration. (* $V_{ot} = V_{oz}$ for single-zone systems.)

Copyright American Society of Heating, Refrigerating and Air-Conditioning Engineers, Inc.

(This appendix is not part of this standard. It is merely informative and does not contain requirements necessary for conformance to the standard. It has not been processed according to the ANSI requirements for a standard and may contain material that has not been subject to public review or a consensus process. Unresolved objectors on informative material are not offered the right to appeal at ASHRAE or ANSI.)

INFORMATIVE APPENDIX H

SIMPLIFIED VENTILATION RATE CALCULATION FOR MULTIPLE-ZONE RECIRCULATING SYSTEMS SERVING ONLY SPECIFIED OCCUPANCY CATEGORIES IN EXISTING BUILDINGS

H1. USE OF THIS APPENDIX

This appendix is intended to be used to assess ventilation rates in existing buildings for third-party building evaluation programs such as ASHRAE Building EQ (bEQ), Leadership in Energy and Environmental Design for Existing Buildings: Operations and Maintenance (LEED EBOM), Energy Star, etc. Zone minimum primary airflow is included as guidance in evaluating and adjusting minimum box settings. This informative appendix is not intended to be used as the Basis of Design or for regulatory applications.

H2. OUTDOOR AIR INTAKE

For multiple-zone recirculating systems serving only occupancy categories listed in Table H-1, the target outdoor air intake flow (V_{target}) is determined in accordance with Equation H-1. For all other systems, V_{target} is set equal to V_{ot} in accordance with Section 6.2.4.4. If the minimum outdoor air intake flows measured at the system level meet or exceed V_{target} , then the system meets the criteria of this informative appendix.

$$V_{target} = \sum_{all\ zones} A_z \times R_s \quad (H-1)$$

where

A_z = zone floor area, the net occupiable floor area of the *ventilation zone*, ft² (m²)

R_s = outdoor airflow rate required per unit area as determined from Table H-1

H3. ZONE MINIMUM PRIMARY AIRFLOW

For each zone, the minimum primary airflow (V_{pz-min}) is determined in accordance with Equation H-2.

$$V_{pz-min} = A_z \times R_{pz} \quad (H-2)$$

where

R_{pz} = minimum primary airflow rate required per unit area as determined from Table H-1. This is the minimum zone airflow required for *ventilation* purposes.

Table H-1 Minimum Outdoor and Primary Airflow Rates

Occupancy Category	Zone Minimum Airflow			
	Outdoor Airflow Rate R_s		Minimum Primary Airflow Rate, R_{pz}	
	cfm/ft ²	L/s·m ²	cfm/ft ²	L/s·m ²
Educational Facilities				
Classrooms (ages 5 to 8)	0.65	3.25	1.12	5.60
Classrooms (ages 9+)	0.82	4.10	1.41	7.05
Computer lab	0.65	3.25	1.12	5.60
Media center	0.65	3.25	1.12	5.60
Music/theater/dance	0.72	3.60	1.24	6.20
Multiuse assembly	1.42	7.10	2.45	12.25
General				
Break rooms	0.43	2.4	0.74	4.13
Conference/meeting	0.44	2.20	0.76	3.80
Corridors	0.11	0.55	0.19	0.95
Office Buildings				
Main entry lobbies	0.19	0.95	0.33	1.65
Occupiable storage rooms for dry materials	0.12	0.60	0.21	1.05
Office space	0.15	0.75	0.26	1.30
Reception areas	0.37	1.85	0.64	3.20
Telephone/data entry	0.63	3.15	1.09	5.45
Public Assembly Spaces				
Libraries	0.30	1.50	0.52	2.60

(This appendix is not part of this standard. It is merely informative and does not contain requirements necessary for conformance to the standard. It has not been processed according to the ANSI requirements for a standard and may contain material that has not been subject to public review or a consensus process. Unresolved objectors on informative material are not offered the right to appeal at ASHRAE or ANSI.)

INFORMATIVE APPENDIX I APPLICATION

This appendix contains application and compliance suggestions that are intended to assist users and enforcement agencies in applying this standard.

Although the standard may be applied to both new and existing buildings, its provisions are not intended to be applied retroactively when it is used as a mandatory regulation or code.

For the most part, Standard 62.1 is specifically written for new buildings because some of its requirements assume that other requirements within the standard have been met. In the case of existing buildings, retroactive application and compliance with all the requirements of this standard may not be practical. However, the principles established in this standard may be applied to most existing commercial and institutional buildings. Some existing buildings may achieve *acceptable indoor air quality* (IAQ) despite not meeting the requirements of Standard 62.1 due to, for example, good maintenance and capital improvement procedures; building materials that, by virtue of their age, have very low *contaminant* emission rates; and many other factors.

11. APPLICATION

11.1 New Buildings. All sections and normative appendices should apply to new buildings falling within the scope of this standard.

11.2 Existing Buildings. The standard should be applied to existing buildings at least in the circumstances described in the following subsections.

11.2.1 Additions to Existing Buildings. All additions to existing buildings should meet the requirements of this standard as if the addition were a new building. An exception may be made when an existing ventilation system is extended to serve the addition. In this case, the existing system components, such as fans and cooling and heating equipment, need not meet the requirements of this standard. However, the extended existing system should remain in compliance with ventilation codes and standards that were in effect at the time it was permitted for construction.

11.2.2 Repairs. Repairing (making operational) existing equipment or other building components does not require the building or any of its components to retroactively comply with this standard.

11.2.3 Replacement. Any component of a building that is removed and replaced should meet the applicable requirements of Section 5, “Systems and Equipment,” for that component. An exception may be made in cases when replacing a component of like size and kind, provided all requirements of codes and standards used at the time of original system design and installation are met. For example, replacement of an air-conditioning unit with one of similar capacity would not require retroactive compliance with ventilation rates and other requirements of this standard. Unaltered components do not need to be retroactively brought into compliance except when there are substantial alterations (as defined below).

11.2.4 Substantial Alterations. If a building is substantially altered, the requirements of this standard should be met as if the building were new. A building would be considered substantially altered if the cost of the revisions exceeds 50% of the building’s fair market value, excluding the cost of compliance with this standard.

11.2.5 Change in Use. If the space application category, as listed in Table 6-1, changes—such as from office to retail—the minimum ventilation rates required by Section 6, “Procedures,” should be met for that space.

11.2.6 Contaminants. Ventilation requirements of this standard are based on chemical, physical, and biological *contaminants* that can affect air quality.

11.2.7 Thermal Comfort. Control of thermal comfort is not required by this standard. Requirements for thermal comfort are contained in ASHRAE Standard 55.^{P14} Note that there are strong correlations between peoples’ perception of IAQ and their perception of thermal comfort.

11.2.8 Limitations. *Acceptable IAQ* might not be achieved in all buildings meeting the requirements of this standard for one or more of the following reasons:

- a. Because of the diversity of sources and *contaminants* in *indoor air*

- b. Because of the many other factors that might affect occupant perception and acceptance of IAQ, such as air temperature, humidity, noise, lighting, and psychological stress
- c. Because of the range of susceptibility in the population
- d. Because *outdoor air* brought into the building might be unacceptable or might not be adequately cleaned

(The following section provides suggested model code language.)

APPLICATION AND COMPLIANCE

Application

New Buildings. All sections and normative appendices apply to new buildings falling within the scope of this standard.

Existing Buildings

Additions to Existing Buildings. All additions to existing buildings within the scope of this standard shall meet the requirements of all sections and normative appendices.

Exception: When an existing ventilation system is extended to serve an addition, the existing system components, such as fans and cooling and heating equipment, need not meet the requirements of this standard. However, the extended existing system must remain in compliance with ventilation codes and standards that were in effect at the time it was permitted for construction.

Repairs. Repairing (making operational) existing equipment or other building components shall be allowed without requiring the building or any of its components to comply with this standard.

Replacement. Any component of a building that is removed and replaced shall meet the applicable requirements of Section 5, “Systems and Equipment,” of this standard for that component. Unaltered components are not required to be brought into compliance except as required due to a change in use.

Exception: Replacement of a building component or individual piece of equipment with a component of like size and kind, provided that all requirements of codes effective at the time of original system design and installation are met. For example, replacement of an air-conditioning unit with one of similar capacity would not require that the ventilation rate requirements and other requirements of this standard be met.

Substantial Alterations. If a building is substantially altered, all sections and normative appendices of this standard shall be met as if the building were new. A building shall be considered substantially altered if the cost of the revisions exceeds 50% of the building’s fair market value, excluding the cost of compliance with all sections and normative appendices of this standard.

Change in Use. If the space occupancy category as listed in Table 6-1 or Table 6-2 changes, such as from office to retail, the minimum ventilation rates required by Section 6, “Procedures,” shall be met for that space.

Compliance

Demonstrating that *acceptable IAQ* has been achieved, such as by measuring *contaminant concentrations* or surveying occupants, is not required by this standard except where required by the IAQ Procedure.

(This appendix is not part of this standard. It is merely informative and does not contain requirements necessary for conformance to the standard. It has not been processed according to the ANSI requirements for a standard and may contain material that has not been subject to public review or a consensus process. Unresolved objectors on informative material are not offered the right to appeal at ASHRAE or ANSI.)

INFORMATIVE APPENDIX J DOCUMENTATION

This appendix summarizes the requirements for documentation contained in the body of the standard using a series of templates that summarize the design criteria used and assumptions made to comply with this standard. One way to comply with the documentation requirements of the standard is to complete these templates as appropriate during the project design process.

J1. OUTDOOR AIR QUALITY

Section 4.3 of this standard requires an investigation of the outdoor air quality in the vicinity of the project site. This template offers a means of documenting the results of both the regional and local investigations and the conclusions reached concerning the acceptability of the outdoor air quality for *ventilation*.

J2. BUILDING VENTILATION DESIGN CRITERIA

This template provides a means of documenting significant design criteria for the overall building. Only the last column, in accordance with Section 5.10.3, is specifically required by the standard. Use of the other columns is motivated by the general documentation requirement described in Section 6.6.

J3. VENTILATION RATE PROCEDURE

Section 6.2 permits the use of this prescription-based procedure to design ventilation systems. This template documents the assumptions made when using this procedure as required by Sections 5.14.4 and 6.6.

J4. INDOOR AIR QUALITY PROCEDURE

Section 6.3 permits the use of this performance-based procedure to design ventilation systems. The first template documents the design criteria and assumptions made when using this procedure and justification of the design approach, as required by Section 6.3.2. This template is also provided in the IAQP calculator (see Informative Appendix G). The second template provides documentation for filtration and air cleaning using mechanical fibrous filters and/or sorbents, as required by Section 6.3.4. The third template covers the documentation for other types of filtration and *air-cleaning systems*, excluding mechanical fibrous filters and/or sorbents, as required by Section 6.3.4.

Regional Outdoor Air Quality Pollutants	Attainment or Nonattainment According to the United States Environmental Protection Agency (U.S. EPA)
Particulates (<i>PM</i> _{2.5})	(Yes/No)
Particulates (<i>PM</i> ₁₀)	(Yes/No)
Carbon monoxide—1 hour/8 hours	(Yes/No)
Ozone	(Yes/No)
Nitrogen dioxide	(Yes/No)
Lead	(Yes/No)
Sulfur dioxide	(Yes/No)
Local Outdoor Air Quality Survey	Date: _____ Time: _____
(a) Area surveyed	(Brief description of the site)
(b) Nearby facilities	(Brief description type of facilities—industrial, commercial, hospitality, etc.)
(c) <i>Odors</i> or irritants	(List and describe)
(d) Visible plumes	(List and describe)
(e) Nearby sources of vehicle exhaust	(List and describe)
(f) Prevailing winds	(Direction)
(g) Other observations	
(h) Conclusions	(Remarks concerning the acceptability of the outdoor air quality)

Building Ventilation Design Criteria

Designed according to ASHRAE Standard 62.1-[year of standard]

Total Building Outdoor Air Intake	Total Building Exhaust Air (see Section 5.18)	Outdoor Air Cleaning Required (See Section 6.1.4)		Filter/Air Cleaner Rating and Location	Indoor Air Dew Point (Section 5.12)		Air Balancing (See Sections 5.10.3 and 7.2.2) and Date Last Performed
		Particulate Matter	Ozone		Peak Outdoor DP at Dehumidification Design Condition	Calculated Space DP at Concurrent Outdoor Condition	
(cfm)	(cfm)	(Yes/No)	(Yes/No)	(MERV or ISO rating, upstream of coil)	(Dew point)	(Dew point)	(NEBB, AABC, etc., January 1, 2022)

Space Identification	Space Type	Occupant Density	Rate/Person	Rate/SF	Zone Air Distribution Effectiveness	System Ventilation Efficiency	Class of Air
(List number or name of each <i>ventilation zone</i> , such as office number or name, retail space name, or classroom number.)	(List occupancy category of the space from Table 6-1, such as Office Space, Retail Sales, Classroom Ages 5 to 8, etc.)	(People/ft ² or m ²)	(cfm or L/s)	(cfm or L/s)	(Table 6-4)	(Section 6.2.4.2; Normative Appendix A)	(Table 6-1, 6-2, or 6-3; include justification for classification if not in these tables)

1. IAQ Procedure Assumptions

Contaminant of Concern	Contaminant Source	Contaminant Strength	Contaminant Target Concentration			Perceived IAQ	Design Approach
			Limit	Exposure Period	Cognizant Authority Reference		
(Identify and list)	(Identify and list)	(Determine and list)	(List)	(List)	(List)	(Percentage of satisfied building occupants)	(Select from Section 6.3.6 and include justification.)

2. Documentation for Filtration and Air-Cleaning Systems Based on Mechanical Fibrous Filters and/or Sorbents

Air Cleaning Test Method

Type of Air Cleaning	Approved Test Methods	Air-Cleaner Efficiency Test Method (check all that apply)
Particulate matter filters	MERV per ASHRAE Standard 52.2 ¹³ or ISO 16890-1 ¹⁴	☐ ASHRAE Standard 52.2 ¹³ ☐ ISO 16890-1 ¹⁴
Gas-phase air cleaners	Efficiency per ASHRAE Standard 145.2 ²⁹ or ISO 10121-2 ³⁰	☐ ASHRAE Standard 145.2 ²⁹ ☐ ISO 10121-2 ³⁰

3. Documentation for Filtration and Air-Cleaning Systems, Excluding Mechanical Fibrous Filters and/or Sorbents

Air Cleaning Test Method

Type of Air Cleaning	Approved Test Methods	Air Cleaner Efficiency Test Method (check all that apply)
Other air cleaners	Efficiency per ASHRAE standards, other national consensus standard approved by AHJ, or custom efficiency test approved by AHJ	☐ ASHRAE Standard 52.2 ¹³ ☐ ISO 16890-1 ¹⁴ ☐ ASHRAE Standard 145.2 ²⁹ ☐ ISO 10121-2 ³⁰ ☐ National consensus standard approved by the local AHJ ☐ Custom efficiency test approved by the local AHJ

3.1 This section only applies if a national consensus standard is used to determine air cleaning efficiency.

National consensus standard(s) used: _____

Approval: AHJ _____ Approver name _____ Approval date _____

3.2 This section only applies if a custom efficiency test is used to determine air cleaning efficiency.

Testing Requirements for Each of the Design Compounds (DCs) and PM2.5	Approved by AHJ
Conducted by third-party lab	☐ Yes (compliant) ☐ No (noncompliant)
Test of the background <i>concentration</i> without the air cleaning in operation	☐ Yes (compliant) ☐ No (noncompliant)
Test of the output <i>concentration</i> with the air cleaning in operation	☐ Yes (compliant) ☐ No (noncompliant)
Test conducted under air-cleaning operating conditions that match the IAQP design operating conditions (Include fan voltage, flow rate, and other settings that are consistent with the manufacturer's operating specifications.)	☐ Yes (compliant) ☐ No (noncompliant)
Test conducted using the relevant laboratory methods for analysis and quantification specified in Tables 7-1 and 7-2	☐ Yes (compliant) ☐ No (noncompliant)

Compound	Allowed Test Methods
VOCs except formaldehyde, acetaldehyde and acetone	EPA IP-1; ³⁶ EPA TO-17; ³⁷ ISO 16000-6; ³⁸ ISO 16017-1; ³⁹ ISO 16017-2 ⁴⁰
Formaldehyde	ASTM D5197; ⁴¹ EPA IP-6; ⁴² EPA TO-11A; ISO 16000-3; ⁴³ or testing method that is compliant with the California Air Resources Board's (CARB) § 93120
Acetaldehyde and acetone	ASTM D5197; EPA IP-6; EPA TO-11A; ⁴⁴ EPA TO-17; ISO 16000-3
Carbon monoxide	EPA IP-3 ⁴⁵ ; ISO 4224 ⁴⁶

	Ozone	PM2.5	Carbon Monoxide	Formaldehyde ^a
Accuracy (±)	5 ppb	Greater of 5 µg /m ³ or 20% of reading	Greater of 3 ppm or 20% of reading	2% full scale within calibrated linearity range
Resolution (±)	1 ppb	5 µg/m ³	1 ppm	0.1 ppb

a. Include the "calibrated linearity range" in all reports.

Ozone Generating Devices—UL 2998¹⁷ certification for air-cleaning devices that generate ozone. ☐ Yes (compliant)
☐ No (noncompliant)

Approval: AHJ _____ Approver Name: _____ Approval Date: _____

(This appendix is not part of this standard. It is merely informative and does not contain requirements necessary for conformance to the standard. It has not been processed according to the ANSI requirements for a standard and may contain material that has not been subject to public review or a consensus process. Unresolved objectors on informative material are not offered the right to appeal at ASHRAE or ANSI.)

INFORMATIVE APPENDIX K RATE RATIONALE

Table K-1 provides description and rationale for the Ventilation Rate Procedure (VRP) rates in Table 6-1, “Minimum Ventilation Rates in Breathing Zone.” This information may be helpful to designers and other practitioners.

Table K-1 Rate Rationale (see Table 6-1)

Occupancy Category	Description/Rationale	People Outdoor Air Rate, cfm/ person	People Outdoor Air Rate, L/s/person	Area Outdoor Air Rate, cfm/ft ²	Area Outdoor Air Rate, L/s·m ²	Air Class
Correctional Facilities						
Booking/waiting	Occupant activity varies between sedentary and moderate walking. Occupants are generally more vocal. Occupants may not be as well-groomed as typical occupants. Occupant stress levels are generally high. All of which result in higher metabolic rates. There are no significant space-related <i>contaminants</i> .	7.5	3.8	0.06	0.3	2
Cell	Occupant activity is primarily sedentary (seated or sleeping). There are typically higher levels of space-related <i>contaminants</i> due to presence of a water closet, sink, and stored clothing. The presence of a water closet is the primary reason why this space has an Air Class of 2.	5	2.5	0.12	0.6	2
Day room	Occupant activity is primarily sedentary (seated, watching television). There are no significant space-related <i>contaminants</i> .	5	2.5	0.06	0.3	1
Guard stations	Occupant activity is primarily sedentary (seated). There are no significant space-related <i>contaminants</i> .	5	2.5	0.06	0.3	1
Educational Facilities						
Art <i>classroom</i>	Occupant activity is moderate. There is considerable aerobic activity in addition to the occupants being very vocal. Also, the occupants are primarily children with higher metabolic rates. There are significant space-related <i>contaminants</i> , including open paints, glues, and cleaning agents. The presence of these open <i>contaminants</i> result in this space being classified as Air Class 2.	10	5	0.18	0.9	2
<i>Classrooms</i> (ages 5 through 8)	Occupant activity is primarily sedentary (seated or light walking). However, occupants are generally more vocal. Also, the occupants are primarily children with higher metabolic rates and often more bioeffluents. There are some significant space-related <i>contaminants</i> , typically stored arts-and-crafts supplies and cleaning agents.	10	5	0.12	0.6	1
<i>Classrooms</i> (age 9+)	Occupant activity is primarily sedentary (seated or light walking). However, occupants are generally more vocal. Also, the occupants are primarily children with higher metabolic rates and often more bioeffluents. There are some significant space-related <i>contaminants</i> , typically stored arts-and-crafts supplies and cleaning agents.	10	5	0.12	0.6	1
Computer lab	Occupant activity is primarily sedentary (seated or light walking). However, occupants are generally more vocal. Also, the occupants can be children/young adults with higher metabolic rates. There are some significant space-related <i>contaminants</i> , typically toner cartridges and paper.	10	5	0.12	0.6	1

Table K-1 Rate Rationale (see Table 6-1) (Continued)

Occupancy Category	Description/Rationale	People Outdoor Air Rate, cfm/ person	People Outdoor Air Rate, L/s/person	Area Outdoor Air Rate, cfm/ft ²	Area Outdoor Air Rate, L/s·m ²	Air Class
Corridors (ages 5+)	Persons passing through the corridor are considered to be transitory and thus not occupants, however brief periods of high occupant density are anticipated in educational facilities. In educational facilities personal lockers are also likely to be present and contain <i>contaminants</i> .	—	—	0.12	0.6	1
Daycare (through age 4)	Occupant activity is moderate. There is considerable aerobic activity in addition to the occupants being very vocal. Also, the occupants are primarily young children with higher metabolic rates. There are significant space-related <i>contaminants</i> , including diapers, arts-and-crafts supplies, and cleaning agents. These <i>contaminants</i> , particularly the diapers, result in this space being classified as Air Class 2.	10	5	0.18	0.9	2
Lecture hall (fixed seats)	Occupant activity is primarily sedentary (seated or light walking). However, occupants are generally more vocal, resulting in higher metabolic rates. There are no significant space-related <i>contaminants</i> .	7.5	3.8	0.06	0.3	1
Media center	Occupant activity is primarily sedentary (seated or light walking). However, occupants are generally more vocal. Also, the occupants are primarily children/young adults with higher metabolic rates and often more bioeffluents. There are some significant space-related <i>contaminants</i> , typically toner cartridges and paper (both loose leaf and bound).	10	5	0.12	0.6	1
Multiuse assembly	Occupant activity is primarily sedentary (seated or light walking). However, occupants are generally more vocal, resulting in higher metabolic rates. There are no significant space-related <i>contaminants</i> .	7.5	3.8	0.06	0.3	1
Music/theater/dance	Occupant activity is high. There is considerable aerobic activity in addition to the occupants being very vocal. There are no significant space-related <i>contaminants</i> .	10	5	0.06	0.3	1
Science laboratories	Occupant activity is moderate. There is considerable aerobic activity in addition to the occupants being very vocal. Also, the occupants are primarily children with higher metabolic rates. There are significant space-related <i>contaminants</i> , including open chemicals and cleaning agents. The presence of these open <i>contaminants</i> result in this space being classified as Air Class 2. OSHA regulated exposure limits must be maintained, ensuring Class 2 defined air is present. This condition is covered in Table 6-2.	10	5	0.18	0.9	2
University/college laboratories	Occupant activity is moderate. There is considerable aerobic activity in addition to the occupants being very vocal. Also, the occupants have higher metabolic rates. There are significant space-related <i>contaminants</i> , including open chemicals and cleaning agents. The presence of these open <i>contaminants</i> results in this space being classified as Air Class 2. OSHA regulated exposure limits must be maintained, ensuring Class 2 defined air is present. This condition is covered in Table 6-2.	10	5	0.18	0.9	2
Wood/metal shop	Occupant activity is moderate. There is considerable aerobic activity in addition to the occupants being very vocal. Also, the occupants can be children/young adults with higher metabolic rates. There are significant space-related <i>contaminants</i> , including sawdust, oils, metal shavings, and chemical agents. The presence of these open <i>contaminants</i> result in this space being classified as Air Class 2.	10	5	0.18	0.9	2
Food and Beverage Service						
Bars, cocktail lounges	Occupant activity is moderate (standing, talking, eating/drinking, waiting tables). The presence of large quantities of open drinks and prepared foods creates higher levels of space-related <i>contaminants</i> . The associated food and drink <i>odors</i> result in this space being classified as Air Class 2.	7.5	3.8	0.18	0.9	2
Cafeteria/ fast-food dining	Occupant activity is moderate (standing, talking, eating, cleaning tables). The presence of large quantities of unpackaged, prepared foods creates higher levels of space-related <i>contaminants</i> . The associated food <i>odors</i> result in this space being classified as Air Class 2.	7.5	3.8	0.18	0.9	2

Table K-1 Rate Rationale (see Table 6-1) (Continued)

Occupancy Category	Description/Rationale	People Outdoor Air Rate, cfm/ person	People Outdoor Air Rate, L/s/person	Area Outdoor Air Rate, cfm/ft ²	Area Outdoor Air Rate, L/s·m ²	Air Class
Kitchen	Occupant activity is very active (walking, talking, eating, food preparation and cooking). The presence of large quantities of unpackaged, cooking prepared foods creates higher levels of space-related <i>contaminants</i> . The associated food <i>odors</i> results in this space being classified as Air Class 2.	7.5	3.8	0.12	0.6	2
Restaurant dining rooms	Occupant activity is moderate (standing, talking, eating, waiting tables). The presence of large quantities of unpackaged, prepared foods creates higher levels of space-related <i>contaminants</i> . The associated food <i>odors</i> result in this space being classified as Air Class 2.	7.5	3.8	0.18	0.9	2
General						
Break rooms	Occupant activity is primarily sedentary (seated). There are limited space-related <i>contaminants</i> .	5	2.5	0.06	0.3	
Coffee stations	Occupant activity is primarily sedentary. There are limited space-related <i>contaminants</i> .	5	2.5	0.06	0.3	
Conference/meeting	Occupant activity is primarily sedentary (seated). There are no significant space-related <i>contaminants</i> .	5	2.5	0.06	0.3	1
Corridors	Persons passing through the corridor are considered to be transitory and thus not occupants. There are no significant space-related <i>contaminants</i> .	—	—	0.06	0.3	1
Occupiable storage rooms for liquids or gels	Occupant activity is primarily sedentary (seated) The <i>concentration</i> of stored products increases the level of space-related <i>contaminants</i> . Current <i>ventilation</i> rate is consistent with other minimal/transient occupancy environments.	5	2.5	0.12	0.6	2
Hotels, Motels, Resorts, Dormitories						
Barracks sleeping areas	Occupant activity is primarily sedentary (sleeping). There are no significant space-related <i>contaminants</i> .	5	2.5	0.06	0.3	1
Bedroom/living room	Occupant activity is primarily sedentary (seated or sleeping). There are no significant space-related <i>contaminants</i> .	5	2.5	0.06	0.3	1
Laundry rooms, central	Occupant activity is primarily moderate. There are often usual space-related <i>contaminants</i> related to cleaning.	5	2.5	0.12	0.6	2
Laundry rooms within dwelling units	Occupant activity is primarily moderate. There are often usual space-related <i>contaminants</i> related to cleaning.	5	2.5	0.12	0.6	1
Lobbies/prefunction	Occupant activity is primarily standing and light walking. However, occupants are generally more vocal, resulting in higher metabolic rates. There are no significant space-related <i>contaminants</i> .	7.5	3.8	0.06	0.3	1
Multipurpose assembly	Occupant activity is primarily sedentary (seated or light walking). There are no significant space-related <i>contaminants</i> .	5	2.5	0.06	0.3	1
Miscellaneous Spaces						
Banks or bank lobbies	Occupant activity is primarily standing and light walking. However, occupants are generally more vocal, resulting in higher metabolic rates. There are no significant space-related <i>contaminants</i> .	7.5	3.8	0.06	0.3	
Bank vaults/safe deposit	Occupant activity is light, typically standing. There are no significant space-related <i>contaminants</i> .	5	2.5	0.06	0.3	2
Computer (not printing)	Occupant activity is primarily sedentary (seated). There are no significant space-related <i>contaminants</i> .	5	2.5	0.06	0.3	1

Table K-1 Rate Rationale (see Table 6-1) (Continued)

Occupancy Category	Description/Rationale	People Outdoor Air Rate, cfm/ person	People Outdoor Air Rate, L/s/person	Area Outdoor Air Rate, cfm/ft ²	Area Outdoor Air Rate, L/s·m ²	Air Class
Freezer and refrigerated spaces (<50°F [10°C])	Refrigerated warehouse spaces are significantly different from conventional warehouses in a number of ways. The low temperatures will slow the emission of <i>contaminants</i> , such as VOCs, from the materials stored in the space; the characteristics of the items being stored will be different; and the amount of time spent in the space by occupants may be shorter (particularly for spaces kept at subfreezing temperatures).	10	5	0	0	2
General manufacturing (excludes heavy industrial and processes using chemicals)	Occupant activity is moderate (standing, walking, assembly). Moderate levels of space-related <i>contaminants</i> are expected. The unknown nature of the <i>contaminants</i> leads to a category of Air Class 3.	10	5	0.18	0.9	3
Pharmacy (prep. area)	Occupant activity is primarily light work and standing. There are space-related <i>contaminants</i> , including open containers of liquid medicines. The presence of these open containers results in this space being classified as Air Class 2.	5	2.5	0.18	0.9	2
Photo studios	Occupant activity is primarily standing and light work. There are large quantities of chemicals, many of them open, resulting in higher levels of space-related <i>contaminants</i> .	5	2.5	0.12	0.6	1
Shipping/receiving	Persons moving materials have a higher level of activity. The flow of products increases the level of space-related <i>contaminants</i> in addition to the typical use of forklifts.	10	5	0.12	0.6	1
Sorting, packing, light assembly	Occupant activity is moderate (standing, walking, assembly). There may be moderate levels of space-related <i>contaminants</i> .	7.5	3.8	0.12	0.6	2
Telephone closets	This should be handled as unoccupied space.					
Transportation waiting	Occupant activity is primarily standing and moderate-to-heavy walking. There are no significant space-related <i>contaminants</i> .	7.5	3.8	0.06	0.3	1
Warehouses	Occupant activity is moderate (standing, walking, assembly). There may be moderate levels of space-related <i>contaminants</i> .	10	5	0.06	0.3	2
Office Buildings						
Main entry lobbies	Occupant activity is primarily transitory light walking. There are few anticipated space-related <i>contaminants</i> .	5	2.5	0.06	0.3	1
Occupiable storage rooms for dry materials	Occupant activity is primarily sedentary (seated). The <i>concentration</i> of stored products increases the level of space-related <i>contaminants</i> ; however, dry material emissions are expected to be low.	5	2.5	0.06	0.3	1
Office space	Occupant activity is primarily sedentary (seated). There are no significant space-related <i>contaminants</i> .	5	2.5	0.06	0.3	1
Reception areas	Occupant activity is primarily sedentary (seated). There are no significant space-related <i>contaminants</i> .	5	2.5	0.06	0.3	1
Telephone/data entry	Occupant activity is primarily sedentary (seated). There are no significant space-related <i>contaminants</i> .	5	2.5	0.06	0.3	1
Public Assembly Spaces						
Auditorium seating area	Occupant activity is primarily sedentary (seated). There are no significant space-related <i>contaminants</i> .	5	2.5	0.06	0.3	1
Courtrooms	Occupant activity is primarily sedentary (seated). There are no significant space-related <i>contaminants</i> .	5	2.5	0.06	0.3	1
Legislative chambers	Occupant activity is primarily sedentary (seated). There are no significant space-related <i>contaminants</i> .	5	2.5	0.06	0.3	1

Table K-1 Rate Rationale (see Table 6-1) (Continued)

Occupancy Category	Description/Rationale	People Outdoor Air Rate, cfm/ person	People Outdoor Air Rate, L/s/person	Area Outdoor Air Rate, cfm/ft ²	Area Outdoor Air Rate, L/s·m ²	Air Class
Libraries	Occupant activity is primarily sedentary (seated or light walking). The large quantities of books create higher levels of space-related <i>contaminants</i> (dust and <i>odors</i>).	5	2.5	0.12	0.6	1
Lobbies	Occupant activity is primarily sedentary (seated or light walking). There are no significant space-related <i>contaminants</i> .	5	2.5	0.06	0.3	1
Museums (children's)	Occupant activity is moderate. There is considerable aerobic activity in addition to the occupants being very vocal. Also, the occupants are typically young children with higher metabolic rates. There are typically some significant space-related <i>contaminants</i> , such as food and drink.	7.5	3.8	0.12	0.6	1
Museums/galleries	Occupant activity is primarily standing and light walking. However, occupants are generally more vocal, resulting in higher metabolic rates. There are no significant space-related <i>contaminants</i> .	7.5	3.8	0.06	0.3	1
Places of religious worship	Occupant activity is primarily sedentary (seated). There are no significant space-related <i>contaminants</i> .	5	2.5	0.06	0.3	1
Residential						
Common corridors	Persons passing through the corridor are considered to be transitory and thus not occupants. There are no significant space-related <i>contaminants</i> .	—	—	0.06	0.3	—
Retail						
Sales (except as below)	Occupant activity is moderate. There is considerable occupant movement, including carrying packages and being more vocally active. The presence of new merchandise creates higher levels of space-related <i>contaminants</i> . This is primarily the reason for the space being classified as Air Class 2.	7.5	3.8	0.12	0.6	2
Barber shop	Occupant activity is primarily sedentary, with moderate work being performed by the staff. Occupants are generally more vocal, resulting in higher metabolic rates. There are some significant space-related <i>contaminants</i> (shampoos, disinfecting agents, high levels of human hair). However, these are directly related to the occupancy rather than the floor area. This is the primary reason why this space is classified as Air Class 2.	7.5	3.8	0.06	0.3	2
Beauty and nail salons	Occupant activity is primarily sedentary, with moderate work being performed by the staff. Occupants are generally more vocal, resulting in higher metabolic rates. There are some significant space-related <i>contaminants</i> (shampoos, disinfecting agents, high levels of hair).	20	10	0.12	0.6	2
Coin-operated laundries	Occupant activity is primarily moderate-to-heavy walking and may include carrying packages. There are some significant space-related <i>contaminants</i> (detergents, disinfecting agents, soiled laundry). However, these are directly related to the occupancy rather than the floor area. This is the primary reason why this space is classified as Air Class 2.	7.5	3.8	0.12	0.6	2
Mall common areas	Occupant activity is primarily moderate to heavy walking and may include carrying packages. Occupants are generally more vocal, resulting in higher metabolic rates. There are no significant space-related <i>contaminants</i> .	7.5	3.8	0.06	0.3	1
Pet shops (animal areas)	Occupant activity is moderate (standing, talking, stooping, walking, and carrying packages). The <i>concentration</i> of animals of various species in containment creates higher levels of space-related <i>contaminants</i> . This <i>concentration</i> of animals, and the fact that they are kept in open containment, results in this space being classified as Air Class 2.	7.5	3.8	0.18	0.9	2
Supermarket	Occupant activity is primarily moderate to heavy walking and may include carrying packages. There are no significant space-related <i>contaminants</i> .	7.5	3.8	0.06	0.3	1

Table K-1 Rate Rationale (see Table 6-1) (Continued)

Occupancy Category	Description/Rationale	People Outdoor Air Rate, cfm/ person	People Outdoor Air Rate, L/s/person	Area Outdoor Air Rate, cfm/ft ²	Area Outdoor Air Rate, L/s·m ²	Air Class
Sports and Entertainment						
Bowling alley (seating)	Occupant activity is moderate (seated, standing, walking, talking, drinking). The presence of open food and drink creates moderately high levels of significant <i>contaminants</i> .	10	5	0.12	0.6	1
Disco/dance floors	Occupant activity is high. There is considerable aerobic activity. There are often considerable quantities of open drink, creating high levels of space-related <i>contaminants</i> related to the people using the space.	20	10	0.06	0.3	1
Gambling casinos	Occupant activity is moderate (seated, standing, walking, talking, drinking). The presence of open food and drink creates moderately high levels of significant <i>contaminants</i> .	7.5	3.8	0.18	0.9	1
Game arcades	Occupant activity is moderate (seated, standing, walking, talking, drinking). The presence of open food and drink creates moderately high levels of significant <i>contaminants</i> .	7.5	3.8	0.18	0.9	1
Gym, stadium (play area)	Occupant activity is high. There is considerable aerobic activity. There are no significant space-related <i>contaminants</i> . Occupancy is variable, and the high area <i>outdoor air</i> rate compensates for the varying occupancy and local source. CO ₂ -based <i>demand controlled ventilation</i> in these spaces should consider that the volume per person in these spaces is typically large, which means that CO ₂ <i>concentration</i> changes will have longer than usual lag times behind occupancy changes.	20	10	0.18	0.9	2
Health club/ aerobics room	Occupant activity is high. There is considerable aerobic activity in addition to the occupants being very vocal. There are significant space-related <i>contaminants</i> related to the people using the space.	20	10	0.06	0.3	2
Health club/ weight rooms	Occupant activity is high. There is considerable aerobic activity in addition to the occupants being very vocal. There are significant space-related <i>contaminants</i> related to the people using the space.	20	10	0.06	0.3	2
Spectator areas	Occupant activity is moderate. While the occupants may be primarily seated, there is considerable vocal activity, as well as standing, cheering, and walking to concessions, etc. There are often considerable quantities of open food and drink, creating high levels of space-related <i>contaminants</i> .	7.5	3.8	0.06	0.3	1
Sports arena (play area)	Occupant activity is high. There is considerable aerobic activity. Occupancy is variable, and the high area <i>outdoor air</i> rate compensates for the varying occupancy and local sources. The presence of playing surface cleaning/resurfacing equipment results in significantly high levels of space-related <i>contaminants</i> .	20	10	0.18	0.9	2
Stages, studios	Occupant activity is moderate. While the occupants may be primarily seated, there is considerable vocal activity, as well as standing, cheering, and walking to concessions, etc. The stage props result in higher levels of space-related <i>contaminants</i> . <i>Contaminant</i> level is not high enough to justify an Air Class 2.	10	5	0.06	0.3	1
Swimming (pool and deck)	While the occupant activity (swimming) is high, it is primarily anaerobic. Occupancy is variable, and the high area outdoor air rate compensates for the varying occupancy and local source. Also, the bioeffluents, such as sweat are discharged into the water rather than the air. For these reasons, there is no occupancy-related outdoor air rate. The high level of chemicals in the pool water that are absorbed into the air as the pool water evaporates create exceptionally high levels of space-related <i>contaminants</i> . The presence of these chemicals, and their noxious <i>odor</i> , result in the space being classified as Air Class 2.	—	—	0.48	2.4	2

(This appendix is not part of this standard. It is merely informative and does not contain requirements necessary for conformance to the standard. It has not been processed according to the ANSI requirements for a standard and may contain material that has not been subject to public review or a consensus process. Unresolved objections on informative material are not offered the right to appeal at ASHRAE or ANSI.)

INFORMATIVE APPENDIX L INFORMATION ON NATURAL VENTILATION

L1. OUTDOOR AIR QUALITY DATA

Outdoor air quality data may be considered valid if it is demonstrated that the data are both physically and spatially representative.

L1.1 Physically representative data accurately reflect the air quality conditions at the monitoring station from which they are derived. Data are considered physically representative if they are obtained from

- a. reports of historical levels of air pollutants published by the relevant local, regional, or federal entity with statutory responsibility for collecting and reporting air quality information in accordance with applicable air quality regulations, or
- b. an on-site monitoring campaign that is verifiably comparable to local, regional, or federal guidelines and methods for demonstration of compliance with applicable air quality regulations.

L1.2 Spatially representative data are collected from a monitoring site that may differ from the proposed project location but is informative of the air quality conditions at the proposed project location. Data may be considered spatially representative if they are

- a. the same as those used by the entity charged with demonstrating regulatory compliance for the geographic region that includes the proposed project location, or
- b. derived from an on-site monitoring campaign that also meets the requirement stated by the criteria in Section L1.1(b).

L2. NATURAL VENTILATION RATE

When calculating the ventilation rate, specific path(s) of the intended airflow passage must first be determined along with flow directions. The two driving forces for natural ventilation are buoyancy and wind, which can work cooperatively or competitively based on the environmental conditions of wind speed, direction, indoor/outdoor air/surface temperatures, as well as the intentional airflow path and mechanisms.

- a. In the case of engineered *natural ventilation* that results in multiple flow scenarios, each force must be examined and considered separately.
- b. Specific pressure-based calculation of natural ventilation flow rate is documented in *ASHRAE Handbook—Fundamentals*,^{P3} Chapter 16, Section 10:
 1. Buoyancy-induced airflow can be calculated following Equation 46.
 2. Wind-driven airflow can be calculated following Equation 45.
 3. The overall pressure (driven by both wind and stack effect) converted to resulting pressure difference between openings can found in Equation 44.

For obtaining wind-driven pressure, several methods are available:

- a. *ASHRAE Handbook—Fundamentals*,^{P3} Chapter 24, provides a method to convert wind speed and direction into pressure coefficients that can be used to determine wind-driven pressure.
- b. CIBSE AM10,^{P15} Chapter 4, provides a method to account for wind-driven *ventilation* and outlines specific challenges to it in Section 4.4.1.
- c. If the building has undergone wind tunnel test for structural stress, the same test can provide detailed pressure coefficients.
- d. Outdoor airflow simulation (such as computational-fluid-dynamics-based simulation) can be used to obtain the specific flow condition at the intended openings.

For intended openings that are large, such as an open atrium or open balcony, and/or when the flow path is not well defined, such as when only single or single-side openings are available, the pressure-based method can be invalid, and outdoor-indoor linked simulation should be used.

L3. PRESCRIPTIVE PATH A CALCULATIONS

L3.1 Ventilation Intensity. Spaces have been defined by a ventilation intensity, which represents the amount of flow rate needed per Equation 6-1, divided by the floor area of the space. Its units are (L/s)/m² of floor area or cfm/ft² of floor area.

Table L-1 Ventilation Intensity Brackets

Bracket	(L/s)·m ²	cfm/ft ²	Commonly Encountered Space Typologies Bracket
1	0.0 to 1.0	0.0 to 0.2	Office, living room, main entry lobby
2	1.0 to 2.0	0.2 to 0.4	Reception area, general manufacturing, kitchen, lobby
3	2.0 to 3.0	0.4 to 0.6	Classroom, daycare
4	3.0 to 4.0	0.6 to 0.8	Restaurant dining room, places of religious worship
5	4.0 to 5.5	0.6 to 1.1	Auditorium, health club/aerobics room, bar, gambling

Not addressed: Lecture Hall and spectator areas (6 [L/s]/m²) and disco/dance floors (10.3 [L/s]/m²)

$$\text{Ventilation Intensity} = \frac{V_{bz}}{A_z} = \frac{R_p \times P_z + R_a \times A_z}{A_z} \quad (\text{L-1})$$

The ventilation intensity brackets in Table L-1 are used.

L3.2 Single Openings. The flow through a single sharp opening due to bidirectional buoyancy-driven flow (V_{bd_sharp}) (see Etheridge and Sandberg^{P16}) is expressed as follows:

$$V_{so_sharp} = 0.21 \times A_w \times \sqrt{g H_s \frac{\Delta T}{T_{ref}}} \quad (\text{L-2})$$

where

- A_w = free unobstructed area of the window, or *openable area*
- ΔT = temperature difference between indoors and outdoors. Given the conservative nature of a prescriptive path, a temperature difference of 1°C (1.8°F) is assumed for these calculations. In reality, this temperature will depend on the internal gains in the space and will likely be higher than 1°C (1.8°F), leading to higher airflows (and a smaller window area requirement).
- H_s = vertical dimension of the opening
- g = gravity constant
- T_{ref} = reference temperature in Kelvin (or Rankine), typically equal to T_{in} , T_{out} or an expected average. A reference temperature of 21°C (70°F, 294K) was assumed for these calculations.

A safety factor is incorporated assuming that an awning window is used. Awning (or top-hinged) windows are among the most common windows used for *natural ventilation* and, because of their uneven vertical area distribution, are more inefficient than a sliding window (sharp opening) at driving flow. An efficiency (ϵ_v) of around 83% (value used in these calculations) when compared to sliding windows is inferred from

$$V_{so} = V_{so_sharp} \times \epsilon_w \quad (\text{L-3})$$

Assuming a height-to-width ratio for the window of $R_{H/W}$ ($R = H/W$), the window area can be rewritten as

$$A_w = \frac{H_s^2}{R_{H/W}} \quad (\text{L-4})$$

The required *openable area* as a fraction of the zone's floor area is therefore calculated by equating the bidirectional buoyancy-driven flow through a single awning opening (V_{so}) to the goal flow rate (V_{bz}) obtained from Table 6-1.

$$V_{so} = V_{bz} \quad (\text{L-5})$$

And solving for window area,

$$\frac{A_w}{A_z} = \left(\frac{V_{bz}}{0.21 \times 0.83 \times R_{H/W}^4 \times \sqrt{g \frac{\Delta T}{T_{ref}}}} \right)^{4/5} \times \frac{1}{A_z} \times 100 \quad (\text{L-6})$$

L3.3 Vertically Spaced Openings. The flow rate (V_{vs}) through vertically spaced openings of areas A_s (the smallest sum of opening areas, either upper openings or lower openings) and A_l (the largest sum of opening areas, either upper openings or lower openings) is obtained using the following equation:

$$V_{vs} = A_{eff} \times C_d \times \sqrt{2g\Delta H \frac{\Delta T}{T_{ref}}} \quad (L-7)$$

where

A_{eff} = effective window area, defined as

$$A_{eff} = \frac{1}{\sqrt{\frac{1}{A_s^2} + \frac{1}{A_t^2}}} = \frac{A_s}{\sqrt{1+R^2}} = \frac{A_w}{\sqrt{1+R^2} \times \left(1 + \frac{1}{R}\right)} \quad (L-8)$$

A_w = total sum of all opening areas

$$A_w = A_s + A_t \quad (L-9)$$

R = area ratio between A_s and A_t

$$R = \frac{A_s}{A_t} \quad (L-10)$$

ΔH = shortest vertical distance between the center of the lowest openings and that of the upper openings.

All other constants are the same as in the single opening scenario.

The required *openable area* as a fraction of the zone's floor area is therefore calculated by equating the flow through two sets of vertically spaces openings (V_{vs}) to the goal flow rate (V_{bz}) obtained from Table 6-1.

$$V_{vs} = V_{bz} \quad (L-11)$$

Solving for window area:

$$\frac{A_w}{A_z} = \frac{V_{bz}}{C_d \times \sqrt{2g\Delta H \frac{\Delta T}{T_{ref}}}} \times \sqrt{1+R^2} \times \left(1 + \frac{1}{R}\right) \times \frac{1}{A_z} \times 100 \quad (L-12)$$

L4. CONTROL AND ACCESSIBILITY (MIXED-MODE VENTILATION)

Mixed-mode *ventilation* is a hybrid system used to maintain *indoor air* quality (IAQ) and internal thermal temperatures year-round using both natural and mechanical ventilation systems.

- Natural ventilation systems use natural forces such as wind and thermal buoyancy to ventilate and cool spaces.
- Mechanical ventilation systems use mechanical systems with fans to supply and *exhaust air* from a space, provide humidity control, and, if required, filter possible *contaminants*.

By preferentially using *natural ventilation* when outdoor air conditions are suitable, energy costs and carbon emissions can be minimized. Sensors are used to identify when *natural ventilation* is less effective at providing suitable indoor temperatures, humidity levels, and *contaminant* levels, and indicate that a transition to *mechanical ventilation* should occur. The transition between modes can be manual or automatic, as dictated by the needs of the owner/occupants. The use of each mode when appropriate will ensure year-round *acceptable IAQ*.

(This appendix is not part of this standard. It is merely informative and does not contain requirements necessary for conformance to the standard. It has not been processed according to the ANSI requirements for a standard and may contain material that has not been subject to public review or a consensus process. Unresolved objections on informative material are not offered the right to appeal at ASHRAE or ANSI.)

INFORMATIVE APPENDIX M COMPLIANCE

This appendix contains compliance suggestions that are intended to assist users and enforcement agencies in applying this standard.

M1. SECTION 4

- ☐ Is documentation of outdoor air quality included as required in Section 4.3?

M2. SECTION 5

- ☐ Are air balancing provisions included in design documentation as required in Section 5.10?
- ☐ If the system is a plenum system, are provisions for providing minimum breathing zone airflow specified?
- ☐ Do exhaust ducts comply with the requirements of Section 5.15?
- ☐ Are ventilation systems controls specified as per Section 5.19?
- ☐ Do specifications include resistance to mold and erosion for airstream surfaces per Section 5.11?
- ☐ Are separation distances between outdoor air intakes and sources listed and in compliance with Section 5.4?
- ☐ Is there any noncombustion equipment that requires exhaust (Section 5.17)?
- ☐ Is combustion air provided for fuel-burning appliances (Section 5.16)?
- ☐ Are appropriate filters specified upstream of cooling coils or wetted surfaces (Section 5.5)?
- ☐ Are dehumidification capability and building *exfiltration* calculations provided (Section 5.12)?
- ☐ Do specifications for drain pans comply with requirements of Section 5.7?
- ☐ Are coils specified per the requirements of Section 5.6?
- ☐ If present, do humidifiers and water spray systems comply with the requirements of Section 5.8?
- ☐ Is access provided for inspection, cleaning, and maintenance of all ventilation equipment and air distribution equipment (Section 5.21)?
- ☐ Is moisture management (Section 5.1) included in building envelope design, including specifically,
 - ☐ weather barrier;
 - ☐ vapor retarder;
 - ☐ sealing exterior joints, seams, and penetrations; and
 - ☐ insulation on pipes, ducts, or other surfaces whose temperatures are expected to fall below dew point of surrounding air?
- ☐ If there is an attached parking garage, do airflow control measures comply with requirements of Section 5.2?
- ☐ Is recirculation from spaces containing Class 2 air limited to spaces with the same purpose and with the same pollutants following requirements of Section 5.14.3.2?
- ☐ Is air from spaces containing Class 3 air contained and not transferred to any other space (Section 5.14.3.3)?
- ☐ Is all air from spaces containing Class 4 air exhausted directly to the outdoors (Section 5.14.3.4)?
- ☐ If *ETS* is expected to be present, does the design comply with all separation requirements of Section 5.3?

M3. SECTION 6 VENTILATION RATE PROCEDURE

- ☐ Are occupancy categories consistent with the space design documents?
- ☐ Are there any *unusual sources* of *contaminants* or compounds? If yes, *ventilation* must be added per Section 6.3.5.

M3.1 Filtration

- ☐ If PM10 standard is exceeded as reported in Section 4, is required filtration per Section 6.1.4.1 provided?
- ☐ If PM2.5 standard is exceeded as reported in Section 4, is required filtration per Section 6.1.4.2 provided?
- ☐ If ozone standard is exceeded, and the area is Serious, Severe15, Severe17, or Extreme, filtration per Section 6.1.4.3 is required unless an exception is documented.

M3.2 Ventilation Rates. Check compliance with the outdoor air intake flow (V_{ot}) using the following process.

- ☐ Calculate $V_{otdefault}$ using Equation M-1 using the combined default rate (R_c) from Informative Appendix N and the occupiable area (A_z) of each zone.

$$V_{otdefault} = \sum_{all\ zones} R_c \times A_z \quad (M-1)$$

- ☐ Calculate additional *ventilation air* required by Section 6.2.1.1.2. Additional *ventilation air* is $V_{otadditional}$.
- ☐ Calculate V_{otmax} using Equation M-2.

$$V_{otmax} = V_{otdefault} + V_{otadditional} \quad (M-2)$$

- ☐ Calculate V_{otmin} using Equation M-3.

$$V_{otmin} = V_{otmax} \times 0.75 \quad (M-3)$$

- ☐ Designed system outdoor air intake flow (V_{ot}) should fall between V_{otmin} and V_{otmax} .
- ☐ Values of V_{ot} for multiple-zone recirculating VAV systems should be close to V_{otmax} .
- ☐ Values of V_{ot} for 100% (dedicated) outdoor air systems providing tempered air should be equivalent to V_{otmin} .
- ☐ Values of V_{ot} for other systems should fall between these values.
- ☐ If dynamic reset is included as a part of the design, does it comply with all requirements of Section 6.2.6?

Exceptions to M3.2:

1. Minimum outdoor airflow for multiple-zone recirculating systems designed using Normative Appendix A could be below V_{otmin} . A calculation spreadsheet should be provided to confirm that E_v for the system is greater than 0.75.
2. Minimum outdoor airflow for systems designed using Normative Appendix C could be below V_{otmin} . Calculation assumptions of any modeling criteria and results should be provided to confirm that E_z values are greater than 1.0.

M4. SECTION 6 INDOOR AIR QUALITY PROCEDURE

For the IAQP:

- ☐ Do the design documents provide evaluation of the following?
 - ☐ Compounds included in the design (Section 6.3.1)
 - ☐ List includes all *contaminants* of concern
 - ☐ Indoor sources and emissions rates for each compound
 - ☐ Outdoor sources and expected *concentrations* for each compound
 - ☐ Exposure periods and *concentration* limits
 - ☐ Evaluation of mixtures
 - ☐ Specification of perceived IAQ acceptability
 - ☐ Calculation of resultant *concentrations* from the design by mass balance
- ☐ Do specifications include test methods?
- ☐ Do specifications require that the subjective evaluation process be performed in the completed building?
 - ☐ If a substantially similar zone is used for subjective evaluation, are previous test results, conditions, and system design provided to verify that the zone is substantially similar?
- ☐ If applicable, are appropriate specifications for dynamic reset monitoring and controls included?

M5. SECTION 6 NATURAL VENTILATION PROCEDURE

Natural ventilation systems follow either the prescriptive or the engineered system compliance path.

For the prescriptive compliance path:

- ☐ Is a mechanical system compliant with either Section 6.2 or 6.3 included?
 - ☐ If no, does design comply with Exceptions 1 or 2 of Section 6.4.1?
- ☐ Do maximum distances from openings comply with Sections 6.4.1.2, Section 6.4.1.3, or Section 6.4.1.4?
- ☐ Do opening sizes comply with the requirements of Section 6.4.2?
 - ☐ Is net *free area* of openings specified?
 - ☐ Are sill-to-head heights specified?
 - ☐ Are aggregate widths specified?
- ☐ Are controls *readily accessible*?

For the engineered compliance path:

- ☐ Do the design documents provide evaluation of the following?
 - ☐ Hourly environmental conditions, including, but not limited to, outdoor air dry-bulb temperature; dew-point temperature; outdoor *concentration of contaminants* of concern (including but not limited to *PM_{2.5}*, *PM₁₀*, and ozone), where data are available; wind speed and direction; and internal heat gains during expected hours of operation of *natural ventilation*.
 - ☐ The effect of pressure losses along airflow paths of natural ventilation airflow on the resulting flow rates, including, but not limited to, inlet vents, air transfer grills, ventilation stacks, and outlet vents.
 - ☐ Qualification of natural ventilation airflow rates of identified airflow paths accounting for wind and thermally induced driving pressures.
 - ☐ *Outdoor air* is provided in sufficient quantities to ensure pollutants and *odors* of indoor origin do not result in unacceptable IAQ as established under Section 6.2.1.1 and/or 6.3.
 - ☐ *Outdoor air* introduced into the space through natural ventilation system openings does not result in unacceptable IAQ according to Sections 6.1.4.1 through 6.1.4.4.
 - ☐ Effective interior air barriers and insulation are provided that separate naturally ventilated spaces from mechanically cooled spaces, ensuring that high-dew-point *outdoor air* does not come into contact with mechanically cooled surfaces.
- ☐ Are controls *readily accessible*?

M6. SECTION 6 EXHAUST

Exhaust ventilation systems follow either the prescriptive or the performance compliance path.

For the prescriptive compliance path:

- ☐ Does airflow comply with requirements of Tables 6-1 and 6-3?
 - ☐ If no for any space, does it qualify as an exception?
- ☐ Have source strengths been evaluated as required in Section 6.5.1.3?

For the performance compliance path:

- ☐ Do the design documents provide evaluation of the following?
 - ☐ Compounds of interest for the design
 - ☐ Indoor sources and emissions rates for each compound
 - ☐ Outdoor sources and emissions rates brought in by *ventilation air*
 - ☐ Exposure periods and *concentration* limits
 - ☐ Evaluation of mixtures
 - ☐ Calculation of resultant *concentrations* from the design
- ☐ If applicable, are appropriate specifications for dynamic reset monitoring and controls included?

M7. VENTILATION FOR EXISTING BUILDINGS

This section provides guidance for determining compliance with the standard for existing buildings. Many sustainability and energy programs require that ventilation rates for systems comply with ASHRAE Stan-

dard 62.1; however, the methods for determining compliance vary widely. This appendix is intended to provide a standardized approach and clear guidance for practitioners who work with existing buildings.

A ventilation system in an existing building may be deemed to comply with the current version of Standard 62.1 if the system complies with all the sections in this appendix. The building may be deemed to comply if all systems in the building comply with all the sections in this appendix (Sections M7.1, Section M7.2, and M7.3).

M7.1 Filtration. Filtration complies with Sections M7.1.1 and Section M7.1.2.

M7.1.1 Filtration Before Coils. Filtration complies with Section 5.5, which requires a minimum of MERV-8 or ISO ePM10 filtration upstream of cooling coils serving occupied spaces.

M7.1.2 Filtration of Outdoor Air. Filtration complies with Section 6.1.4, which requires treating *outdoor air* with appropriate filters or air cleaners if local air quality conditions do not meet national air quality standards for PM10, PM2.5, ozone, or other outdoor *contaminants*.

M7.2 Outdoor Airflow. The following process may be used to determine if outdoor airflow rates comply with the VRP of the current standard. Occupied areas may be determined by measurement, dimensioned floor plans, or from the building manager's data. Occupancy categories should reflect how spaces are currently used, which may differ from the original design intent. Check design criteria and assumptions documented according to Section 6.6, if available.

M7.2.1 Measure System Outdoor Airflow. Measure system outdoor airflow. Measurements may be made directly or by installed flow measurement devices in the system that are calibrated. This rate is $V_{otmeasured}$.

M7.2.2 Determine System Type and Design Outdoor Airflow. Determine the system type and then follow the guidance in the appropriate section to determine the design outdoor airflow, $V_{otdesign}$.

M7.2.2.1 Single Zone Systems. Determine $V_{otdesign}$ using Section 6.2.2. Section 6.2.2 provides Equation 6-3, which sets the system outdoor air intake flow equal to the zone outdoor airflow. Zone outdoor airflows are calculated in Equation 6-2 using the *zone air distribution effectiveness* (E_z) listed in Table 6-4.

M7.2.2.2 100% Outdoor Air Systems. Determine $V_{otdesign}$ using Section 6.2.3. Section 6.2.3 provides Equation 6-4, which sums all zone outdoor airflow requirements to determine the system outdoor airflow requirement. Zone outdoor airflows are calculated in Equation 6-2 using the *zone air distribution effectiveness* (E_z) listed in Table 6-4.

M7.2.2.3 Multiple Zone Recirculating Systems. Determine $V_{otdesign}$ using any process listed in this section. Calculations are ordered from simplest to most complex. The simpler methods typically result in more conservative (higher) ventilation airflows.

M7.2.2.3.1 Appendix G. Determine $V_{otdesign}$ using Informative Appendix H, which provides a table of minimum outdoor and primary airflow rates for specified occupancy categories in existing buildings: some educational facilities, conference rooms, corridors, office spaces, and libraries.

M7.2.2.3.2 Simplified Procedure. Determine $V_{otdesign}$ using Section 6.2.4 and the Simplified Procedure in Section 6.2.4.3, which provides a simplified method to determine system ventilation efficiency (E_v) based on occupant distribution. System ventilation efficiency values can vary from 0.22 to 0.75.

M7.2.2.3.3 Alternative Procedure. Determine $V_{otdesign}$ using Section 6.2.4 and the Alternative Procedure in Normative Appendix A, which provides credit for secondary recirculation but requires detailed system and occupancy information at the zone level. The system ventilation efficiency is determined by the zone ventilation efficiency of the "critical zone." The alternative procedure provides credit for secondary recirculation.

M7.2.3 Determine Outdoor Airflow Compliance. If $V_{otmeasured} > V_{otdesign}$, the system complies with the outdoor airflow requirements of the VRP.

M7.3 Controls. Confirm that ventilation system controls comply with requirements of Section 5.19, which requires that minimum ventilation airflows be maintained under all load conditions through manual or automatic controls.

(This appendix is not part of this standard. It is merely informative and does not contain requirements necessary for conformance to the standard. It has not been processed according to the ANSI requirements for a standard and may contain material that has not been subject to public review or a consensus process. Unresolved objections on informative material are not offered the right to appeal at ASHRAE or ANSI.)

INFORMATIVE APPENDIX N VENTILATION RATE CHECK TABLE

Table N-1 is not for design purposes. It is intended to provide check values. Default rate per unit area is based on a multiple-zone system with default occupancy and default E_v that equals 0.75. This is the default E_v in the simplified rate when $D > 0.60$.

Table N-1 Check Table for the Ventilation Rate Procedure (VRP)

Occupancy Category	Combined Outdoor Air Rate (R_c)	
	cfm/ft ²	L/s·m ²
Animal Facilities		
Animal exam room (veterinary office)	0.43	2.13
Animal imaging (MRI/CT/PET)	0.51	2.53
Animal operating rooms	0.51	2.53
Animal postoperative recovery room	0.51	2.53
Animal preparation rooms	0.51	2.53
Animal procedure room	0.51	2.53
Animal surgery scrub	0.51	2.53
Large-animal holding room	0.51	2.53
Necropsy	0.51	2.53
Small-animal-cage room (static cages)	0.51	2.53
Small-animal-cage room (ventilated cages)	0.51	2.53
Correctional Facilities		
Booking/waiting	0.58	2.93
Cell	0.33	1.63
Dayroom	0.28	1.40
Guard stations	0.18	0.90
Educational Facilities		
Art classroom	0.51	2.53
Classrooms (ages 5 through 8)	0.49	2.47
Classrooms (ages 9+)	0.63	3.13
Computer lab	0.49	2.47
Daycare sickroom	0.57	2.87
Daycare (through age 4)	0.57	2.87
Lecture hall (fixed seats)	1.58	8.00
Libraries	0.23	1.13
Media center	0.49	2.47
Multiuse assembly	1.08	5.47
Music/theater/dance	0.55	2.73

Table N-1 Check Table for the Ventilation Rate Procedure (VRP) (Continued)

Occupancy Category	Combined Outdoor Air Rate (R_c)	
	cfm/ft ²	L/s·m ²
Science laboratories	0.57	2.87
University/college laboratories	0.57	2.87
Wood/metal shop	0.51	2.53
Food and Beverage Service		
Bars, cocktail lounges	1.24	6.27
Cafeteria/fast-food dining	1.24	6.27
Kitchen (cooking)	0.36	1.81
Restaurant dining rooms	0.94	4.75
General		
Break rooms	0.25	1.23
Coffee stations	0.21	1.07
Conference/meeting	0.41	2.07
Corridors	0.08	0.40
Occupiable storage rooms for liquids or gels	0.17	0.87
Hotels, Motels, Resorts, Dormitories		
Barracks sleeping areas	0.21	1.07
Bedroom/living room	0.15	0.73
Laundry rooms (central)	0.23	1.13
Laundry rooms within <i>dwelling units</i>	0.23	1.13
Lobbies/prefunction	0.38	1.92
Multipurpose assembly	0.88	4.40
Miscellaneous Spaces		
Banks or bank lobbies	0.23	1.16
Bank vaults/safe deposit	0.11	0.57
Computer (not printing)	0.11	0.53
Freezer and refrigerated spaces (<50°F [10°C])	0.03	0.13
General manufacturing (excludes heavy industrial and processes using chemicals)	0.33	1.67
Pharmacy (prep area)	0.31	1.53
Photo studios	0.23	1.13
Shipping/receiving	0.19	0.93
Sorting, packing, light assembly	0.23	1.15
Telephone closets	0.00	0.00
Transportation waiting	1.08	5.47
Warehouses	0.09	0.47
Office Buildings		
Main entry lobbies	0.15	0.73
Occupiable storage rooms for dry materials	0.09	0.47

Table N-1 Check Table for the Ventilation Rate Procedure (VRP) (Continued)

Occupancy Category	Combined Outdoor Air Rate (R_c)	
	cfm/ft ²	L/s· m ²
Office space	0.11	0.57
Reception areas	0.28	1.40
Telephone/data entry	0.48	2.40
Outpatient Health Care Facilities		
Birthing room	0.44	2.20
<i>Class I imaging rooms</i>	0.19	0.97
Dental operator	0.51	2.53
General examination room	0.36	1.81
Other dental treatment areas	0.11	0.57
Physical therapy exercise area	0.43	2.13
Physical therapy individual room	0.35	1.73
Physical therapeutic pool area	0.64	3.20
Prosthetics and orthotics room	0.51	2.53
Psychiatric consultation room	0.21	1.07
Psychiatric examination room	0.21	1.07
Psychiatric group room	0.41	2.07
Psychiatric seclusion room	0.15	0.73
Urgent care examination room	0.36	1.81
Urgent care observation room	0.21	1.07
Urgent care treatment room	0.44	2.21
Urgent care triage room	0.51	2.53
Speech therapy room	0.21	1.07
Public Assembly Spaces		
Auditorium seating area	1.08	5.40
Courtrooms	0.55	2.73
Legislative chambers	0.41	2.07
Libraries	0.23	1.13
Lobbies	1.08	5.40
Museums (children's)	0.56	2.83
Museums/galleries	0.48	2.43
Places of religious worship	0.88	4.40
Retail		
Sales (except as below)	0.31	1.56
Barbershop	0.33	1.67
Beauty and nail salons	0.83	4.13
Coin-operated laundries	0.36	1.81
Mall common areas	0.48	2.43

Table N-1 Check Table for the Ventilation Rate Procedure (VRP) (Continued)

Occupancy Category	Combined Outdoor Air Rate (R_c)	
	cfm/ft ²	L/s· m ²
Pet shops (animal areas)	0.34	1.71
Supermarket	0.16	0.81
Sports and Entertainment		
Bowling alley (seating)	0.69	3.47
Disco/dance floors	2.75	13.73
Gambling casinos	1.44	7.28
Game arcades	0.44	2.21
Gym, sports arena (play area)	0.43	2.13
Health club/aerobics room	1.15	5.73
Health club/weight rooms	0.35	1.73
Spectator areas	1.58	8.00
Stages, studios	1.01	5.07
Swimming (pool and deck)	0.64	3.20

(This appendix is not part of this standard. It is merely informative and does not contain requirements necessary for conformance to the standard. It has not been processed according to the ANSI requirements for a standard and may contain material that has not been subject to public review or a consensus process. Unresolved objectors on informative material are not offered the right to appeal at ASHRAE or ANSI.)

INFORMATIVE APPENDIX O INDOOR AIR QUALITY PROCEDURE

01. SUMMARY OF SELECTED AIR QUALITY GUIDELINES

If the Indoor Air Quality Procedure (IAQP) is used, acceptable indoor *concentration* limits are needed for *design compounds* (DCs) and particles. When using this procedure, these *concentration* limits need to be referenced from a *cognizant authority* as defined in the standard. At present, no single organization develops acceptable *concentration* limits for all substances in *indoor air*, nor are limits available for all potential DCs or particles.

Cognizant authorities, such as the United States Environmental Protection Agency (U.S. EPA), California EPA, and the Committee for Health Related Evaluation of Building Products (AgBB) publish *concentration* limits for compounds, many of which may be present in the indoor environment. Compounds included in the IAQP design need to be included if data were judged sufficient to indicate a compound was likely to be found in buildings at *concentrations* that were a substantial fraction of the proposed design target. The goal is not to include every possible compound that may appear in *indoor air* but rather sufficient numbers of diverse compounds, such that control of the compounds is anticipated to result in air quality that meets the standard's definition of "acceptable."

A summary of considerations is presented below:

- Is a compound expected to be present in *indoor air* with reasonable frequency at *concentrations* relevant to (but not necessarily above) the design target? Specifically, the design outdoor airflow rate (V_{oz}) and design features will be controlled by the compounds with the highest emission rates and lowest targets (taking mixtures into account); thus, compounds with low *concentrations* and high targets will have little or no impact on the calculated V_{oz} .
- Is there a design target that has been proposed by a *cognizant authority*?
- Does it seem reasonable to expect that product emissions rates may be available for the proposed compound?
- Is there an established sampling and analytical method for the proposed compound?

Occupational exposure limits (e.g., permissible exposure limits and threshold limit values) are not appropriate as design limits (DLs), as they are not established for *acceptable IAQ* or for typical commercial buildings. In general, they were developed for industrial applications evaluating effects of substances on healthy adult male workers.

02. GUIDELINE FOR EMISSION RATES

Several published peer-reviewed papers provide a reference for design teams to use to compile reasonable DC emission rates. A nonexhaustive list of peer-reviewed papers is shown in Section O4. In addition, multiple established certification programs include empirical measures of emission rates for construction materials as well as finishes, furniture, and equipment intended for indoor use. These include third-party programs as well as industry trade association programs and programs in support of government regulations (e.g., the AgBB evaluation scheme used in Germany and parts of Europe, Blue Angel, BIFMA, Green Label, France A+, CDPH Standard Method for testing and evaluation of VOC emissions [CDPH Section 01350], Greenguard, SCS Indoor Advantage Gold, and Floorscore). Engineers may use the emission rates for the specific materials that a designer is including or considering for use. The IAQP (Section 6.3) requires that emission rates must consider DCs emitted by occupants and their activities, by materials, and by specific sources within the occupied spaces and introduced into the building with *outdoor air*.

03. SUBJECTIVE EVALUATION

Section 7.3.2 requires that an occupant survey be conducted. Many subjective evaluation approaches have been used with varying degrees of success. The following is an example of an evaluation approach that focuses on adapted occupants:

- After the building is completed and substantially occupied, provide all occupants with an electronic or written survey of questions, including, "Do you perceive the air quality in your environment to be acceptable or unacceptable?"

- b. Anonymous surveys with neutrally framed questions provide the best responses.
- c. When conducting an evaluation of adapted occupants, respondents must record their perception of zone air quality after 30 minutes of residency in the occupied zone.
- d. All occupants should be surveyed, if possible. Otherwise, at least 50% of typical occupancy, or 300, whichever is less, should be randomly selected.
- e. A minimum 30% response rate from those surveyed is desirable. Each zone must be surveyed per requirements of Section 6.3. The subjective evaluation validates the acceptability of *indoor air* if 80% of respondents in the area do not express dissatisfaction. The Center for the Built Environment at UC Berkeley has developed a survey that includes IAQ questions and may be a useful template.

04. BIBLIOGRAPHY

- Ahn, J.H., J.E. Szulejko, K.H. Kim, Y.H. Kim, and B.W. Kim. 2014. Odor and VOC emissions from pan frying of mackerel at three stages: Raw, well-done, and charred. *International Journal of Environmental Research and Public Health* 11(11):11753–71.
- Apte, M.G. 2013. Final Report: Balancing energy conservation and occupant needs in ventilation rate standards for Big Box stores in California: Predicted indoor air quality and energy consumption using a matrix of ventilation scenarios. LBNL Paper LBNL-5551E, Lawrence Berkeley National Laboratory, University of California, Berkeley.
- Assadi, A.A., J. Palau, A. Bouzaza, J. Penya-Roja, V. Martinez-Soriac, and D. Wolbert. 2014. Abatement of 3-methylbutanal and trimethylamine with combined plasma and photocatalysis in a continuous planar reactor. *Journal of Photochemistry and Photobiology A: Chemistry* 282:1–8.
- Barro, R., J. Regueiro, M. Llompert, C. Garcia-Jares. 2009. Analysis of industrial contaminants in indoor air: Part 1. Volatile organic compounds, carbonyl compounds, polycyclic aromatic hydrocarbons and polychlorinated biphenyls. *Journal of Chromatography A* 1216(3):540–66.
- Batterman, S, F.C. Su, S. Li, B. Mukherjee, and C. Jia. 2014. Personal exposure to mixtures of volatile organic compounds: modeling and further analysis of the RIOPA data. Research report (Health Effects Institute). 181:3–63.
- Bi, X.H., G.Y. Sheng, Y.L. Feng, J.M. Fu, and J.X. Xie. 2005. Gas- and particulate-phase specific tracer and toxic organic compounds in environmental tobacco smoke. *Chemosphere* 61(10):1512–22.
- Borapheh, P., and P. Thiravetyan. 2015. Removal of trimethylamine (fishy odor) by C-3 and CAM plants. *Environmental Science and Pollution Research* 22(15):11543–57.
- Bradman, A., F. Gaspar, R. Castorina, J. Williams, T. Hoang, and P.L. Jenkins. 2017. Supporting Information: Formaldehyde and acetaldehyde exposure and risk characterization in California early childhood education environments. *Indoor Air* 27(1):104–113.
- CDPH. 2006. Long-term building air measurements for volatile organic compounds including aldehydes at a California five-building sustainable office complex, (Vol. 1/2 and 2/2). Report #CA/DHS/EHLB/R-172. California Department of Public Health, Sacramento, CA.
- Chan, W.R., S. Cohn, M. Sidheswaran, D.P. Sullivan, and W.J. Fisk. 2015. Contaminant levels, source strengths, and ventilation rates in California retail stores. *Indoor Air* 25(4):381–92.
- Choo, C.P., and J. Jalaludin. 2015. An overview of indoor air quality and its impact on respiratory health among Malaysian school-aged children. *Reviews on Environmental Health* 30(1):9–18.
- Claeson, A.S., M. Sandstrom, A.L. Sunesson. 2007. Volatile organic compounds (VOCs) emitted from materials collected from buildings affected by microorganisms. *Journal of Environmental Monitoring* 9(3):240–5.
- Cometto-Muniz, J.E., and M.H. Abraham. 2015. Compilation and analysis of types and concentrations of air-borne chemicals measured in various indoor and outdoor human environments. *Chemosphere* 127:70–86.
- Cometto-Muniz, J.E., and W.S. Cain. 1992. Sensory irritation. Relation to indoor air pollution. *Annals of the New York Academy of Sciences* 641:137–51.
- Dodson, R.E., J.I. Levy, E.A. Houseman, J.D. Spengler, and D.H. Bennett. 2009. Evaluating methods for predicting indoor residential volatile organic compound concentration distributions. *Journal of Exposure Science and Environmental Epidemiology* 19(7):682–93.
- Dutton, S.M. 2014. Evaluation of the indoor air quality procedure for use in retail buildings. LBNL Paper LBNL-6079E, Lawrence Berkeley National Laboratory, University of California, Berkeley, CA.
- Fenske, J.D., and S.E. Paulson. 1999. Human breath emissions of VOCs. *Journal of the Air and Waste Management Association* 49(5):594–8.
- Fischer, M.L., D. Littlejohn, M.M. Lunden, N.J. Brown. 2003. Automated measurements of ammonia and nitric acid in indoor and outdoor air. *Environmental Science and Technology* 37(10):2114–9.
- Godwin, C, and S. Batterman. 2007. Indoor air quality in Michigan schools. *Indoor Air* 17(2):109–21.

- Harrison, R.M., J.M. Delgado-Saborit, S.J. Baker, N. Aquilina, C. Meddings, and S. Harrad. 2009. Measurement and modeling of exposure to selected air toxics for health effects studies and verification by biomarkers. Research report (Health Effects Institute) 143:3-96, Discussion 7–100.
- Hau, K.M., D.W. Connell, and B.J. Richardson. 2000. Use of partition models in setting health guidelines for volatile organic compounds. *Regulatory Toxicology and Pharmacology* 31(1):22–9.
- Hau, K.M., D.W. Connell, and B.J. Richardson. 2000. Use of partition models to evaluate guidelines for mixtures of volatile organic compounds. *Regulatory Toxicology and Pharmacology* 32(1):36–41.
- Hodgson, A.T., and H. Levin. 2003. Volatile organic compounds in indoor air: A review of concentrations measured in North America since 1990. Report No. LBNL-51715, Lawrence Berkeley National Laboratory, Berkeley, CA.
- Hotchi, T., A.T. Hodgson, and W.J. Fisk. 2006. Indoor air quality impacts of a peak load shedding strategy for a large retail building. LBL Contract No. 500-03-026, PIER Demand Response Research Center, Lawrence Berkeley National Laboratory, Berkeley, CA.
- Ilacqua, V., O. Hänninen, N. Kuenzli, and M.F. Jantunen. 2007. Intake fraction distributions for indoor VOC sources in five European cities. *Indoor Air* 17(5):372–83.
- Kinney, P.L., S.N. Chillrud, S. Ramstrom, J. Ross, and J.D. Spengler. 2002. Exposures to multiple air toxics in New York City. *Environmental Health Perspectives* 110:539–46.
- Lamorena, R.B., and W. Lee. 2008. Influence of ozone concentration and temperature on ultra-fine particle and gaseous volatile organic compound formations generated during the ozone-initiated reactions with emitted terpenes from a car air freshener. *Journal of Hazardous Materials* 158(2-3):471–7.
- Lee, S.C., S. Lam, and H.K. Fai. 2001. Characterization of VOCs, ozone, and PM10 emissions from office equipment in an environmental chamber. *Building and Environment* 36(7):837–42.
- Lindgren, T. 2010. A case of indoor air pollution of ammonia emitted from concrete in a newly built office in Beijing. *Building and Environment* 45(3):596–600.
- Lioy, P.J., Z. Fan, J. Zhang, P. Georgopoulos, S.W. Wang, and P. Ohman-Strickland. 2011. Personal and ambient exposures to air toxics in Camden, New Jersey. Research report (Health Effects Institute) (160):3–127, Discussion 9–51.
- Liu W, Zhang J, Zhang L, Turpin BJ, Weisel CP, Morandi MT, et al. Estimating contributions of indoor and outdoor sources to indoor carbonyl concentrations in three urban areas of the United States. *Atmospheric Environment*. 2006;40(12):2202-14.
- Nagda, N.L., and H.E. Rector. 2003. A critical review of reported air concentrations of organic compounds in aircraft cabins. *Indoor Air* 13(3):292–301.
- Nazaroff, W.W., and C.J. Weschler. 2004. Cleaning products and air fresheners: exposure to primary and secondary air pollutants. *Atmospheric Environment* 38(18):2841–65.
- Norback, D., and G. Wieslander. 2002. Biomarkers and chemosensory irritations. *International Archives Of Occupational And Environmental Health* 75(5):298–304.
- Ohura, T., T. Amagai, X. Shen, S. Li, P. Zhang, and L. Zhu. 2009. Comparative study on indoor air quality in Japan and China: Characteristics of residential indoor and outdoor VOCs. *Atmospheric Environment* 43(40):6352–9.
- Oikawa, D., W. Takeuchi, S. Murata, K. Takahashi, and Y. Sekine. 2012. Measurement of concentrations of thioglycolic acid, dithiodiglycolic acid and ammonia in indoor air of a beauty salon. *Journal of Occupational Health*. 54(5):370–5.
- Pant, P., S.K. Guttikunda, and R.E. Peltier. 2016. Exposure to particulate matter in India: A synthesis of findings and future directions. *Environmental Research* 147:480–96.
- Park, J.S., and K. Ikeda. 2006. Variations of formaldehyde and VOC levels during 3 years in new and older homes. *Indoor Air* 16(2):129–35.
- Repace, J.L., J.N. Hyde, and D. Brugge. 2006. Air pollution in Boston bars before and after a smoking ban. *BMC Public Health* 6:266.
- Roda, C., I. Kousignian, C. Guihenneuc-Jouyaux, C. Dassonville, I. Nicolis, and J. Just. 2011. Formaldehyde exposure and lower respiratory infections in infants: Findings from the PARIS cohort study. *Environmental Health Perspectives*. 119(11):1653–8.
- Rohr, A.C. 2013. The health significance of gas- and particle-phase terpene oxidation products: a review. *Environment international*. 60:145-62.
- Saha, C.K., G.Q. Zhang, P. Kai, and B. Bjerg. 2010. Effects of a partial pit ventilation system on indoor air quality and ammonia emission from a fattening pig room. *Biosystems Engineering* 105(3):279–87.

- Salonen, H.J., A.L. Pasanen, S.K. Lappalainen, H.M. Riuttala, T.M. Tuomi, and P.O. Pasanen. 2009. Airborne concentrations of volatile organic compounds, formaldehyde and ammonia in Finnish office buildings with suspected indoor air problems. *Journal of Occupational and Environmental Hygiene* 6(3):200–9.
- Santarsiero, A., S. Fuselli, A. Piermattei, R. Morlino, G. De Blasio, and M. De Felice. 2009. Investigation of indoor air volatile organic compounds concentration levels in dental settings and some related methodological issues. *Annali dell'Istituto superiore di sanita* 45(1):87–98.
- Sarigiannis, D.A., S.P. Karakitsios, A. Gotti, I.L. Liakos, and A. Katsoyiannis. 2011. Exposure to major volatile organic compounds and carbonyls in European indoor environments and associated health risk. *Environment International* 37(4):743–65.
- Sax, S.N., D.H. Bennett, S.N. Chillrud, P.L. Kinney, and J.D. Spengler. 2004. Differences in source emission rates of volatile organic compounds in inner-city residences of New York City and Los Angeles. *Journal of Exposure Analysis and Environmental Epidemiology* 14:S95–S109.
- Schripp, T., S. Langer, and T. Salthammer. 2012. Interaction of ozone with wooden building products, treated wood samples and exotic wood species. *Atmospheric Environment* 54:365–72.
- Siegel, J.A., J. Srebric, N. Crain, E. Nirlo, M. Zaatari, and A. Hoisington. 2012. Ventilation and indoor air quality in retail stores (RP-1596). *HVAC&R Research* 20(2).
- Smith, T.J., M.E. Davis, J.E. Hart, A. Blicharz, F. Laden, and E. Garshick. 2012. Potential air toxics hot spots in truck terminals and cabs. Research report (Health Effects Institute) (172):5–82.
- Tham, K.W., M.S. Zuraimi, and S.C. Sekhar. 2004. Emission modelling and validation of VOCs' source strengths in air-conditioned office premises. *Environment International*. 30(8):1075–88.
- Vainiotalo, S., V. Vaananen, and R. Vaaranrinta. 2008. Measurement of 16 volatile organic compounds in restaurant air contaminated with environmental tobacco smoke. *Environmental Research* 108(3):280–8.
- Wang, S., H.M. Ang, and M.O. Tade. 2007. Volatile organic compounds in indoor environment and photocatalytic oxidation: state of the art. *Environment International* 33(5):694–705.
- Wang, T.C. 1975. A study of bioeffluents in a college classroom. *ASHRAE transactions* 81(1):1–12.
- Weisel, C.P. 2002. Assessing exposure to air toxics relative to asthma. *Environmental Health Perspectives* 110 Suppl. 4:527–37.
- Winkle, M.R., and P.A. Scheff. 2001. Volatile organic compounds, polycyclic aromatic hydrocarbons and elements in the air of ten urban homes. *Indoor Air* 11(1):49–64.
- Wolkoff, P. 2013. Indoor air pollutants in office environments: Assessment of comfort, health, and performance. *International Journal of Hygiene and Environmental Health* 216(4):371–94.
- Wolkoff, P., C.K. Wilkins, P.A. Clausen, and G.D. Nielsen. 2006. Organic compounds in office environments—Sensory irritation, odor, measurements and the role of reactive chemistry. *Indoor Air* 16(1):7–19.
- Wu, X.M., M.G. Apte, R. Maddalena, and D.H. Bennett. 2011. Volatile organic compounds in small- and medium-sized commercial buildings in California. *Environmental Science and Technology* 45(20):9075–83.
- Zaatari, M., and J. Siegel. 2014. Particle characterization in retail environments: Concentrations, sources, and removal mechanisms. *Indoor Air* 24(4):350–61.
- Zhu, J., R. Newhook, L. Marro, and C.C. Chan. 2005. Selected volatile organic compounds in residential air in the city of Ottawa, Canada. *Environmental Science and Technology* 39(11):3964–71.

(This appendix is not part of this standard. It is merely informative and does not contain requirements necessary for conformance to the standard. It has not been processed according to the ANSI requirements for a standard and may contain material that has not been subject to public review or a consensus process. Unresolved objectors on informative material are not offered the right to appeal at ASHRAE or ANSI.)

INFORMATIVE APPENDIX P

INFORMATIVE REFERENCES

- P1. GPO. 29 CFR 1910.1200, *Hazard Communication Standard*. Washington, DC: U.S. Government Printing Office. West Conshohocken, PA: ASTM International.
- P2. DHHS. 2020. *Biosafety in Microbiological and Biomedical Laboratories*. Washington, DC: U.S. Department of Health and Human Services.
- P3. ASHRAE. 2021. *ASHRAE Handbook—Fundamentals*. Peachtree Corners, GA: ASHRAE.
- P4. ASHRAE. 2024. *Protecting Building Occupants from Smoke During Wildfire and Prescribed Burn Events*. Peachtree Corners, GA: ASHRAE.
- P5. ASHRAE. 2024. *High-Performance Sequences of Operation for HVAC Systems*. Peachtree Corners, GA: ASHRAE.
- P6. ACGIH. 2025. TLVs and BEIs—Threshold Limit Values for Chemical Substances and Physical Agents and Biological Exposure Indices (Section 6). Cincinnati, OH: American Conference of Governmental Industrial Hygienists.
- P7. EPA. 1995. EPA-454/B-95-003B, *User's Guide for the Industrial Source Complex (ISC3) Dispersion Models, Vol. 2: Description of Model Algorithms*. Washington, DC: U.S. Environmental Protection Agency.
- P8. ASHRAE. 2001. ASHRAE RP-1009, *Simplified Diffuser Boundary Conditions for Numerical Room Airflow Models*. Peachtree Corners, GA: ASHRAE.
- P9. ASHRAE. 2009. ASHRAE RP-1373, *Air Distribution Effectiveness with Stratified Air Distribution Systems*. Peachtree Corners, GA: ASHRAE.
- P10. GPO. 2024. 40 CFR 50, National Primary and Secondary Ambient Air Quality Standards. Washington, DC: U.S. Government Printing Office.
- P11. ASHRAE. 2001. ASHRAE RP-1009, *Simplified Diffuser Boundary Conditions for Numerical Room Airflow Models*. Peachtree Corners, GA: ASHRAE.
- P12. ASHRAE. 2009. ASHRAE RP-1373, *Air Distribution Effectiveness with Stratified Air Distribution Systems*. Peachtree Corners, GA: ASHRAE.
- P13. Dols, W.S., and B.J. Polidoro. 2020. CONTAM User Guide and Program Documentation, Version 3.4. NIST Technical Note 1887, Revision 1. Gaithersburg, MD: National Institute of Standards and Technology.
- P14. ASHRAE. 2023. ASHRAE Standard 55, *Thermal Environmental Conditions for Human Occupancy*. Peachtree Corners, GA: ASHRAE.
- P15. CIBSE. 2005. CIBSE AM10, *Natural Ventilation in Non-Domestic Buildings*. London, UK: Chartered Institution of Building Services Engineers.
- P16. Etheridge, D.W., and M. Sandberg. 1996. *Building Ventilation: Theory and Measurement*, Vol. 50. Hoboken, NJ: Wiley & Sons.

(This appendix is not part of this standard. It is merely informative and does not contain requirements necessary for conformance to the standard. It has not been processed according to the ANSI requirements for a standard and may contain material that has not been subject to public review or a consensus process. Unresolved objectors on informative material are not offered the right to appeal at ASHRAE or ANSI.)

INFORMATIVE APPENDIX Q
ADDENDA DESCRIPTION

ANSI/ASHRAE Standard 62.1-2025 incorporates ANSI/ASHRAE Standard 62.1-2022 and Addenda a, b, c, d, e, f, g, h, i, j, k, m, n, o, p, q, r, s, t, u, v, w, x, y, z, aa, ab, and ag to ANSI/ASHRAE Standard 62.1-2022. Table Q-1 lists each addendum and describes the way in which the standard is affected by the change. It also lists the ASHRAE and ANSI approval dates for each addendum.

Table Q-1 Addenda to ANSI/ASHRAE Standard 62.1-2022

Addendum	Section(s) Affected	Description of Changes*	Approval Dates
a	Table 6-1	Addendum a adds corridors (ages 5 plus) to Table 6-1, “Minimum Ventilation Rates in Breathing Zone.” These corridors differ from those in general and residential buildings because they have periodically high occupant density and may have lockers permanently installed.	ASHRAE October 31, 2023 ANSI October 31, 2023
b	Table 6-1; Table G-1; Table J-1; Table M-1	The rate rationales for both general and office break rooms are identical in Informative Appendix J, and the area outdoor rate is listed as 0.06 cfm/ft2 (0.3 L/s·m2) for both break rooms in that appendix. Addendum b removes the office building break room from Table 6-1, “Minimum Ventilation Rates in Breathing Zone” because it is inconsistent with the rationale in Informative Appendix J. Tables G-1, J-1, and M-1 are also revised for consistency.	ASHRAE October 31, 2023 ANSI October 31, 2023
c	Informative Appendix F; Informative Appendix P	Addendum c provides a calculator for mass balance equations used with the revised Indoor Air Quality Procedure. It also updates the reference to the CONTAM Users Manual for mass balance calculations.	ASHRAE October 31, 2023 ANSI October 31, 2023
d	6.2.6.1.3.4	Addendum d modifies the accuracy range of CO ₂ sensors for demand controlled ventilation to reflect the capabilities of current commercial technology.	ASHRAE March 31, 2025 ANSI March 31, 2025
e	New 5.13	Addendum e adds a relative humidity limit of 65% when evaporative cooling systems are operating.	ASHRAE September 30, 2022 ANSI September 30, 2022
f	New 5.21	Addendum f requires emergency operation modes be provided on all ventilation equipment. The controls must include automatic reset to normal operations.	ASHRAE September 30, 2025 ANSI September 30, 2025
g	14	Addendum g provides additional templates to document air-cleaning systems used in compliance with the Indoor Air Quality Procedure in Section 6.3.	ASHRAE October 31, 2024 ANSI October 31, 2024

* These descriptions may not be complete and are provided for information only.

Table Q-1 Addenda to ANSI/ASHRAE Standard 62.1-2022 (Continued)

Addendum	Section(s) Affected	Description of Changes*	Approval Dates
h	8; Table 8-1; 9	Addendum h revises Table 8-1, separate from Standard 180, that is based on the requirements of Standard 62.1 and provides facility users with a list of inspection tasks that are important to maintaining acceptable IAQ. The results of the inspection task may have an impact on the preventive maintenance plan or could elevate the need for service or professional support.	ASHRAE September 30, 2025 ANSI September 30, 2025
i	Table 7-1; Table 7-2	Addendum i adds a set of alternative testing methods to Table 7-1 when testing formaldehyde, acetaldehyde, and acetone to include more affordable testing options, while maintaining testing standards and lowering the cost barriers to employing the IAQP in ventilation design.	ASHRAE October 31, 2024 ANSI October 31, 2024
j	Delete 6.2.1.1.3; Add New 6.2.1.3; New Table 6-5; 6.2.1.4; New Normative Appendix D	Addendum j changes the requirements for the ventilation zone outdoor airflow to include corrections for the density of the air at the ventilation system outdoor air intake. New Table 6-5 includes density correction factors calculated for elevation only, while new Normative Appendix D provides calculations for the density correction factor that include elevation, temperature, and humidity.	Standards Committee March 27, 2024 ANSI April 22, 2024
k	5.12	Addendum k clarifies the language of Section 5.12 to separate the requirements of the humidity limit and the control systems. An exception has been added for climates with low occurrences of dew-point temperatures above 68°F (20°C).	Standards Committee March 27, 2024 ANSI April 22, 2024
m	6.2.4.1.3	Addendum m allows the method published under Addendum f to Standard 170-2021, Ventilation of Health Care Facilities, to be used to calculate the design outdoor air intake flow if it is greater than the design outdoor air intake flow required by Standard 62.1.	ASHRAE September 30, 2022 ANSI September 30, 2022
n	6.3.4; 9	The Indoor Air Quality Procedure (IAQP) requires that a mass balance calculation be performed. Any mass balance that includes filtration or air cleaning requires a particle filtration efficiency or gaseous removal efficiency. Addendum n requires that the particle filtration efficiency or gaseous removal efficiency of filtration or air cleaning devices be tested to current standards. To ensure objectivity for test equipment suppliers, no specific design of the test apparatus is specified. Instead requirements of apparatus properties and validation tests are specified.	ASHRAE September 30, 2022 ANSI September 30, 2022
o	Informative Appendix I	Addendum o adds additional documentation recommendations to Informative Appendix I for the edition of the standard used in design, rating of filters and air cleaners, and date of the last air balancing. The inclusion of this information will improve operators' awareness of the design conditions of their ventilation systems.	ASHRAE April 30, 2024 ANSI April 30, 2024
p	3; 5.4.2; 9	Addendum p adds a definition for hurricane-prone regions in line with the 2021 International Building Code (IBC) definition, clarifies and improves the applicable options regarding rain entrainment requirements (Section 5.4.2, options b and c), and adds ANSI/AMCA Standard 550 to the normative references. The reference to AMCA Standard 550 is intended to more effectively align the code requirements in IMC Sections 401.5 and 501.3.2.	ASHRAE April 30, 2024 ANSI April 30, 2024
q	Table 6-5; Table 6-6	Addendum q updates Table 6-5 and 6-6 related to the IAQ Procedure with information obtained from users of testing procedures for design compounds. 1,1,1-trichloroethane has been removed and the design limit for phenol has been updated to 70 µg/m ³ consistent with the 2018 AgBB listing.	ASHRAE October 31, 2024 ANSI October 31, 2024

* These descriptions may not be complete and are provided for information only.

Table Q-1 Addenda to ANSI/ASHRAE Standard 62.1-2022 (Continued)

Addendum	Section(s) Affected	Description of Changes*	Approval Dates
r	Table 6-5; 9; Informative Appendix E	Addendum r updates the PM _{2.5} limit to align with the latest EPA limit of 9.0 µg/m ³ .	ASHRAE April 30, 2025 ANSI April 30, 2025
s	Table 7-1; 9	Addendum s expands the available testing methods for acetaldehyde and acetone by allowing these compounds to be tested using EPA TO-17. Also, this addendum updates EPA TO-11 to EPA TO-11A to reflect the latest revision of the method.	ASHRAE April 30, 2025 ANSI April 30, 2025
t	7.3.1	Addendum t changes the carbon monoxide limit from peak reading to an average reading over an 8 hour interval.	ASHRAE April 30, 2025 ANSI April 30, 2025
u	3.1; 6.2.1.1.7	Addendum u provides definitions for modular rooms and sleep pods and requires that ventilation be provided to these spaces based on the most similar occupancy category.	ASHRAE April 30, 2025 ANSI April 30, 2025
v	3.1; 5.4.1.4; 5.8.1; 5.14.2; 6.2.1.1.5; 6.5.1; Table 6-5; Table 7-1; 7.1.5; 9; Appendix B; Appendix P	Addendum v updates references to the most recent relevant version of the referenced standards. In some cases, the cognizant authority has been changed or clarified to indicate the correct agency and the relevant references updated in the text for consistency.	ASHRAE September 30, 2025 ANSI September 30, 2025
w	Table 6-1	Addendum w adjusts the default occupant densities given in Table 6-1 so as not to exceed the occupant density allowances in the 2024 International Building Code. This addendum also renames three occupancy categories for clarity.	ASHRAE September 30, 2025 ANSI September 30, 2025
x	6.5; 9; Table 6-2	Addendum x relocates requirements in the notes of Table 6-2 to the body of the standard. The performance compliance path is modified to be consistent with previous changes to the IAQ Procedure.	Standards Committee September 23, 2022 BOD October 14, 2022 ANSI November 8, 2022
y	6.2.6.1.5	Addendum y modifies Section 6.2.6.1.5, “Documentation,” to include the value of the occupant component ($R_p \times P_z$) and building component ($R_a \times A_z$) for each demand controlled ventilation (DCV) zone.	ASHRAE September 30, 2025 ANSI September 30, 2025
z	6.2.1.1.5; 6.2.1.1.6; Table 6-2	Addendum z provides an exemption from the exhaust requirements of Table 6-2 when a qualified professional establishes the minimum exhaust rates required. Furthermore, the minimum exhaust rate in science laboratories has been reduced to 0.35 cfm/ft ² , consistent with the commentary in ANSI Z9.5-2022 which indicates that laboratories can be safely operated at as low as 2 ACH under demand control sequences. Additionally, the “Educational science laboratories” occupancy category in Table 6-2 has been corrected to “Science laboratories” to be consistent with Table 6-1, Informative Appendix J, and Informative Appendix M.	ASHRAE September 30, 2025 ANSI September 30, 2025

* These descriptions may not be complete and are provided for information only.

Table Q-1 Addenda to ANSI/ASHRAE Standard 62.1-2022 (Continued)

Addendum	Section(s) Affected	Description of Changes*	Approval Dates
aa	New 6.3.4.1	Addendum aa introduces the requirement to document the end-of-useful-life efficiency (EEOL) for formaldehyde and a non-polar volatile organic compound. Furthermore, the EEOL must be used for formaldehyde calculations under the indoor air quality procedure.	ASHRAE September 30, 2025 ANSI September 30, 2025
ab	6.2.6.1; 6.2.6.2; Table 6-1	Addendum ab introduces CO ₂ based demand control ventilation control sequences informed by ASHRAE RP-1547. Furthermore it adds differential CO ₂ concentration limits above ambient to Table 6-1 for use with these control sequences.	ASHRAE October 31, 2023 ANSI October 31, 2023
ag	Normative Appendix B; Figure B-1 Deleted; New Figures B-1 and B-2; Informative Appendix P	Addendum ag replaces the calculation method in Normative Appendix B, “Separation of Exhaust Outlets and Outdoor Air Intakes,” with a new method based on ASHRAE Research Project (RP) 1635.	ASHRAE April 30, 2024 ANSI April 30, 2024

* These descriptions may not be complete and are provided for information only.

NOTE

Approved addenda, errata, or interpretations for this standard can be downloaded free of charge from the ASHRAE website at www.ashrae.org/technology.

POLICY STATEMENT DEFINING ASHRAE'S CONCERN FOR THE ENVIRONMENTAL IMPACT OF ITS ACTIVITIES

ASHRAE is concerned with the impact of its members' activities on both the indoor and outdoor environment. ASHRAE's members will strive to minimize any possible deleterious effect on the indoor and outdoor environment of the systems and components in their responsibility while maximizing the beneficial effects these systems provide, consistent with accepted Standards and the practical state of the art.

ASHRAE's short-range goal is to ensure that the systems and components within its scope do not impact the indoor and outdoor environment to a greater extent than specified by the Standards and Guidelines as established by itself and other responsible bodies.

As an ongoing goal, ASHRAE will, through its Standards Committee and extensive Technical Committee structure, continue to generate up-to-date Standards and Guidelines where appropriate and adopt, recommend, and promote those new and revised Standards developed by other responsible organizations.

Through its *Handbook*, appropriate chapters will contain up-to-date Standards and design considerations as the material is systematically revised.

ASHRAE will take the lead with respect to dissemination of environmental information of its primary interest and will seek out and disseminate information from other responsible organizations that is pertinent, as guides to updating Standards and Guidelines.

The effects of the design and selection of equipment and systems will be considered within the scope of the system's intended use and expected misuse. The disposal of hazardous materials, if any, will also be considered.

ASHRAE's primary concern for environmental impact will be at the site where equipment within ASHRAE's scope operates. However, energy source selection and the possible environmental impact due to the energy source and energy transportation will be considered where possible. Recommendations concerning energy source selection should be made by its members.

ASHRAE · 180 Technology Parkway · Peachtree Corners, GA 30092 · www.ashrae.org

About ASHRAE

Founded in 1894, ASHRAE is a global professional society committed to serve humanity by advancing the arts and sciences of heating, ventilation, air conditioning, refrigeration, and their allied fields.

As an industry leader in research, standards writing, publishing, certification, and continuing education, ASHRAE and its members are dedicated to promoting a healthy and sustainable built environment for all, through strategic partnerships with organizations in the HVAC&R community and across related industries.

To stay current with this and other ASHRAE Standards and Guidelines, visit www.ashrae.org/standards, and connect on LinkedIn, Facebook, Twitter, and YouTube.

Visit the ASHRAE Bookstore

ASHRAE offers its Standards and Guidelines in print, as immediately downloadable PDFs, and via ASHRAE Digital Collections, which provides online access with automatic updates as well as historical versions of publications. Selected Standards and Guidelines are also offered in redline versions that indicate the changes made between the active Standard or Guideline and its previous edition. For more information, visit the Standards and Guidelines section of the ASHRAE Bookstore at www.ashrae.org/bookstore.

IMPORTANT NOTICES ABOUT THIS STANDARD

To ensure that you have all of the approved addenda, errata, and interpretations for this Standard, visit www.ashrae.org/standards to download them free of charge.

Addenda, errata, and interpretations for ASHRAE Standards and Guidelines are no longer distributed with copies of the Standards and Guidelines. ASHRAE provides these addenda, errata, and interpretations only in electronic form to promote more sustainable use of resources.